

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD / DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

SCHEM, MLB, VENUS, X425G

EVT 01/12/2015

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70	Venus PCI-E	MARY_X425G	08/22/2014
71	Venus CORE/FB POWER	MARY_X425G	09/22/2014
72	Venus FRAME BUFFER I/F	J45G_AMD	06/30/2014
73	0V95 GPU / 1V35 FB Power Supply	ADITYA_X425G	09/16/2014
74	GDDR5 Frame Buffer A	MARY_X425G	09/22/2014
75	GDDR5 Frame Buffer B	MARY_X425G	09/22/2014
76	Venus HDMI/DP/GPIO	MARY_X425G	09/22/2014
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78	Venus DP PWR/GNDs	MARY_X425G	09/22/2014
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93	Thunderbolt Constraints	SIDLE_J45	12/10/2012
94	Camera Constraints	SIDLE_J45	12/10/2012
95	SMC Constraints	SIDLE_J45	12/10/2012
96	Project Specific Constraints	SIDLE_J45	12/10/2012
97	GPU (AMD VENUS) Constraints	J45G_AMD	07/01/2014

ALIASES RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-00383	1	SCHEM, MLB, VENUS, X425G	SCH	CRITICAL	
820-00163	1	PCBF, MLB, VENUS, X425G	PCB	CRITICAL	

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ABBREV=ABBREV
LAST_MODIFIED=Mon Jan 12 16:34:40 2015

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-00042	COMMON PARTS, MLB, VENUS, X425G	X425_COMMON
985-00050	DEV BOM, MLB, VENUS, X425G	X425_DEVEL:ENG
639-00682	PCBA, MLB, VENUS, CTO, 16GHYN, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:HYNIX_1600, FB_4G_HYNIX
639-00703	PCBA, MLB, VENUS, CTO, 16GMIC, VR-4GMIC, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:MICRON_1600, FB_4G_MICRON
639-00739	PCBA, MLB, VENUS, CTO, 16GHYN, VR-4GMIC, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:HYNIX_1600, FB_4G_MICRON
639-00740	PCBA, MLB, VENUS, CTO, 16GMIC, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:MICRON_1600, FB_4G_HYNIX
639-00798	PCBA, MLB, VENUS, BEST, 16GHYN, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:HYNIX_1600, FB_4G_HYNIX
639-00799	PCBA, MLB, VENUS, BEST, 16GMIC, VR-4GMIC, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:MICRON_1600, FB_4G_MICRON
639-00800	PCBA, MLB, VENUS, BEST, 16GHYN, VR-4GMIC, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:HYNIX_1600, FB_4G_MICRON
639-00801	PCBA, MLB, VENUS, BEST, 16GMIC, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, CPU_CRW:CTO, RAM:MICRON_1600, FB_4G_HYNIX
639-00803	PCBA, MLB, VENUS, NOCPU, 16GMIC, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, GFX_BOM, RAM:MICRON_1600, FB_4G_HYNIX
639-00974	PCBA, MLB, NOGPU, CTO, 16GMIC, VR-4GHYN, X425G	BASE_BOM, DEVEL_BOM, CPU_CRW:CTO, RAM:MICRON_1600, FB_4G_HYNIX

X425 BOM Groups

BOM GROUP	BOM OPTIONS
X425_COMMON	ALTERNATE, COMMON, X425_COMMON1, X425_COMMON2, X425_PROGPARTS, ACAPS:A2
X425_COMMON1	CPUMEM:S0, TBTHV:P15V, SKIP_5V3V3:AUDIBLE, CPUPEG:X8X4X4, S2_PWR:S0, SMC_SUSAC:SYES
X425_COMMON2	EDP:YES, XDP, SSD_PWR_EN:GPIO, CAM_WAKE:NO, SAMCONN, APCLKRQ:ISOL, CRW_SPRT,WLAN_SW:SIL
X425_PVT	BKLT:PROD, SENSOR_NONPROD:N
X425_PROGPARTS	SMC_PROG:BASE, BOOTROM_PROG:EVT, TBTRM:PROG, DPMUXMCU:PROG
X425_DEVEL:ENG	ALTERNATE, XDP_DEBUG, S0PGOOD_ISL, SENSOR_NONPROD:Y, SENSOR_NONPROD_R,BKLT:ENG, DBGLED, DPMUX_DEBUG, GPU_ROM:YES, SENSOR_GPU_NONPROD:Y
X425_DEVEL:DVT	ALTERNATE, XDP_DEBUG, BKLT:PROD, SENSOR_NONPROD:N, DBGLED
X425_DEVEL:PVT	XDP_DEBUG
GFX_BOM	VENUS:XTA
XDP_DEBUG	XDP_CONN, XDP_PCH

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00058	1	CRW, SR1ZX, PRQ, C8, 2.5, 47W, +3E, 1.2, 6M, BGA	U0500	CRITICAL	CPU_CRW: BEST
337S00059	1	CRW, SR1ZY, PRQ, C8, 2.8, 47W, +3E, 1.2, 6M, BGA	U0500	CRITICAL	CPU_CRW: CTO
337S4542	1	IC, QEW, LPT-M, HM87, C2, SR199, PRQ, FCBG	U1100	CRITICAL	
338S1247	1	IC, TBT, FR-4C, AB, PRQ, C10, SR1JC, FCBGA288	U2800	CRITICAL	
338S1264	1	IC, BCM1570BA2, S2 PCIE CMRA, 8XB, 20BF CPGA	U3900	CRITICAL	
333S0700	1	IC, SDRAM, 4GBIT, DDR3L-1600, GEMMA, 9GB FBGA	U4000	CRITICAL	
333S00032	32	IC, SDRAM, DDR3L-1600, 4GBIT, 78B FBGA	_____	CRITICAL	HYNIX_1600
333S0660	32	IC, SDRAM, 4GBIT, DDR3L-1600, V9BA, 78P, FBGA	_____	CRITICAL	MICRON_1600
337S00116	1	IC, GPU, VENUS XTAAX, Q5, 29X29MM, FCBGA962	U8400	CRITICAL	VENUS: XTA
333S00027	4	IC, GDDR5, 4GBIT, 6GBPS, 1.5V, 25MM, BGA130	U8800, U8850, U8900, U8950	CRITICAL	FB_4G_HYTX
333S0766	4	IC, GDDR5, 4GBIT, 6GBPS, 128MX32, 25MM, 176BGA	U8800, U8850, U8900, U8950	CRITICAL	FB_4G_MICRON

DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM:HYNIX_1600	HYNIX_1600, RAMCFG3:H, RAMCFG2:H, RAMCFG1:L, RAMCFG0:L
RAM:MICRON_1600	MICRON_1600, RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:L

COMMON/DEVEL BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00042	1	COMMON PARTS,MLB,VENUS,X425G	BASE	CRITICAL	BASE_BOM
985-00050	1	DEV,MLB,VENUS,X425G	DEVEL	CRITICAL	DEVEL_BOM

SYNC MASTER=CLEAN X305		SYNC DATE=05/30/2014	
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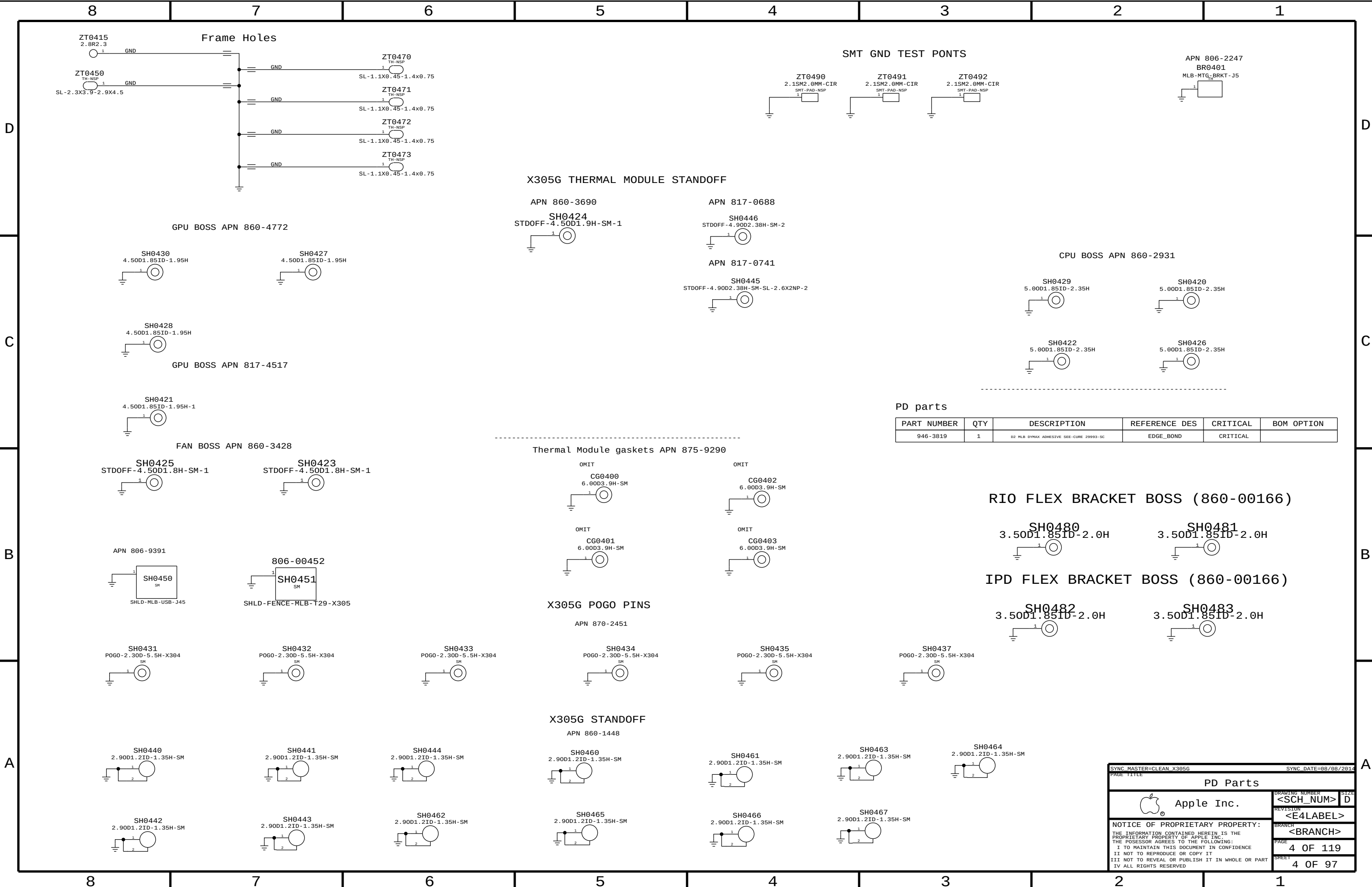
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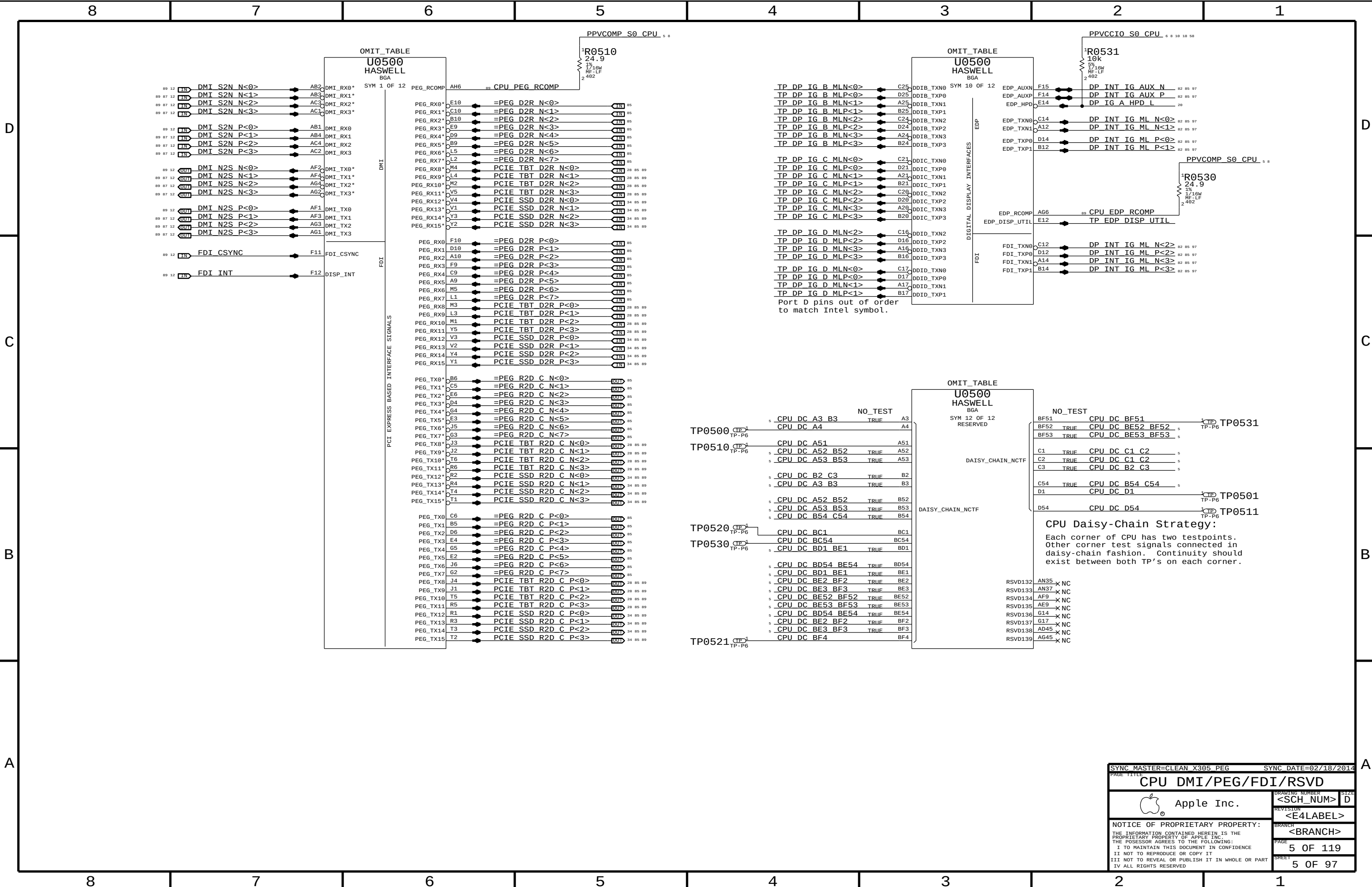
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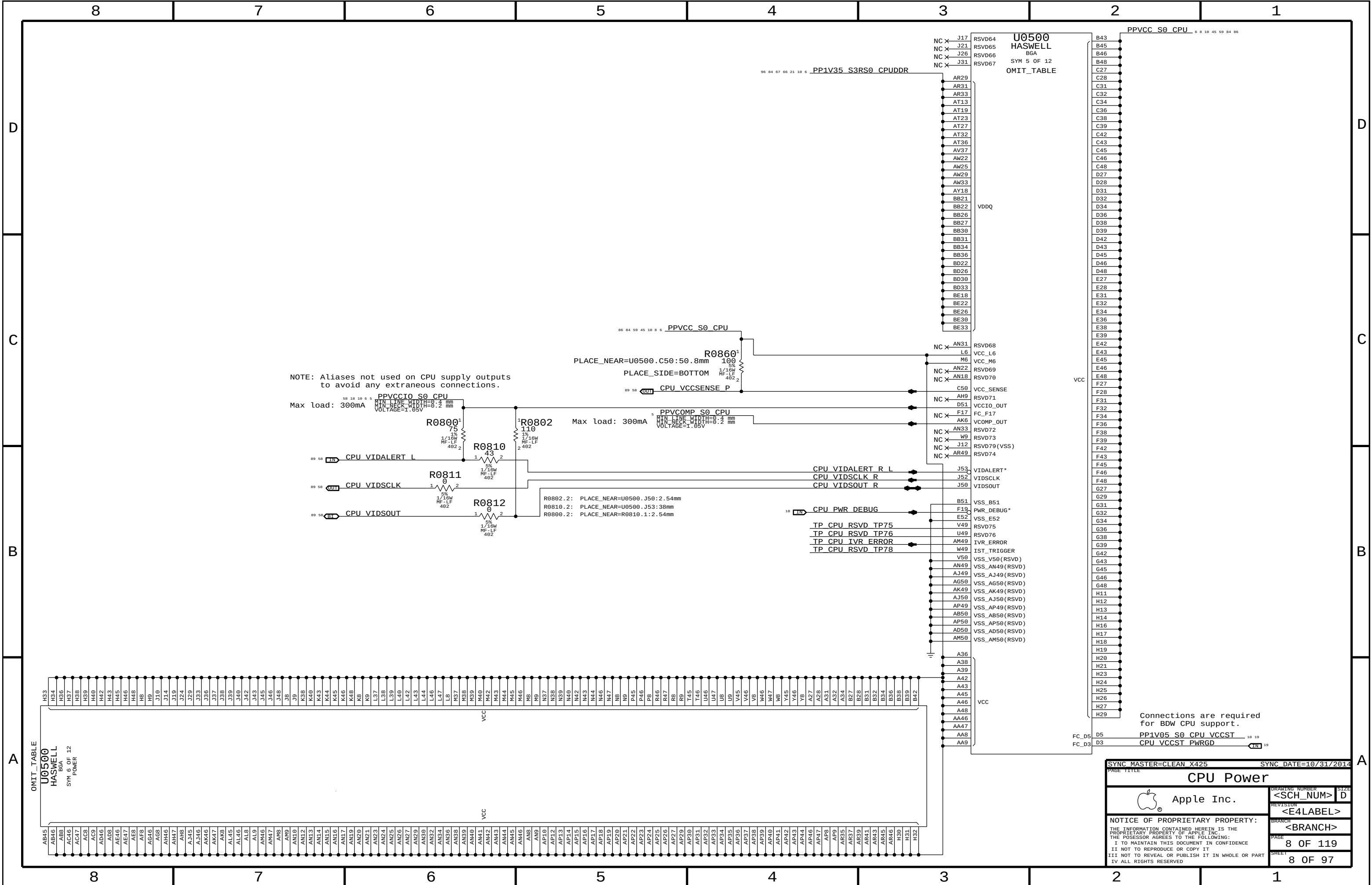
PD parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
946-3819	1	D2 MLB DYMAX ADHESIVE SEE-CURE 29993-SC	EDGE_BOND	CRITICAL	

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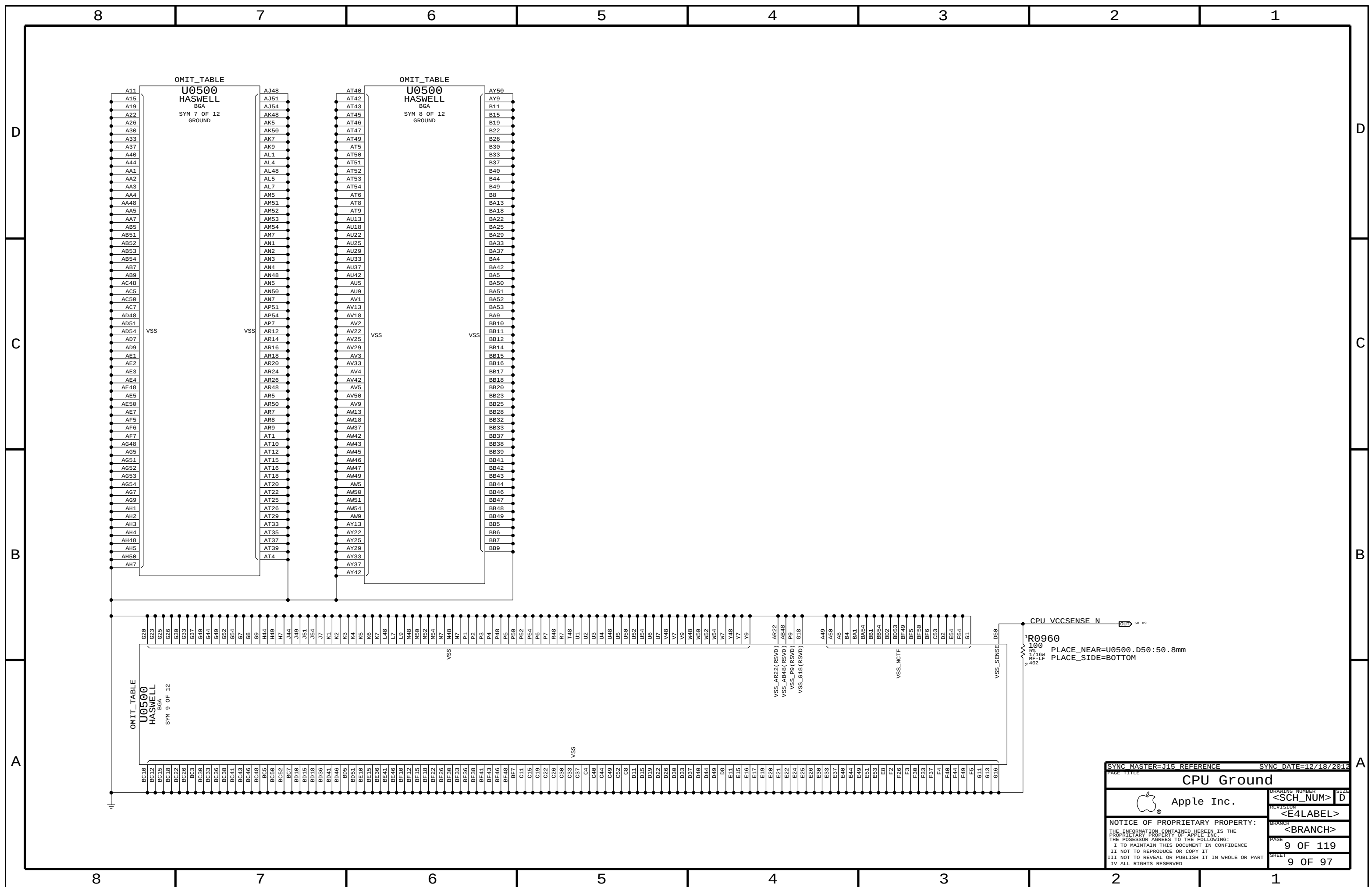
NOTE: Aliases not used on CPU supply outputs to avoid any extraneous connections.

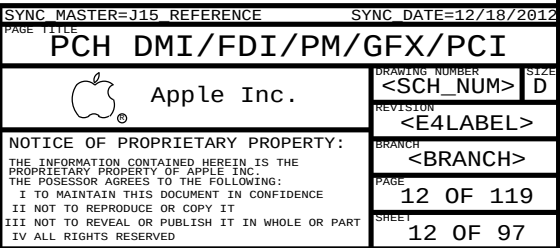
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Max load: 300mA

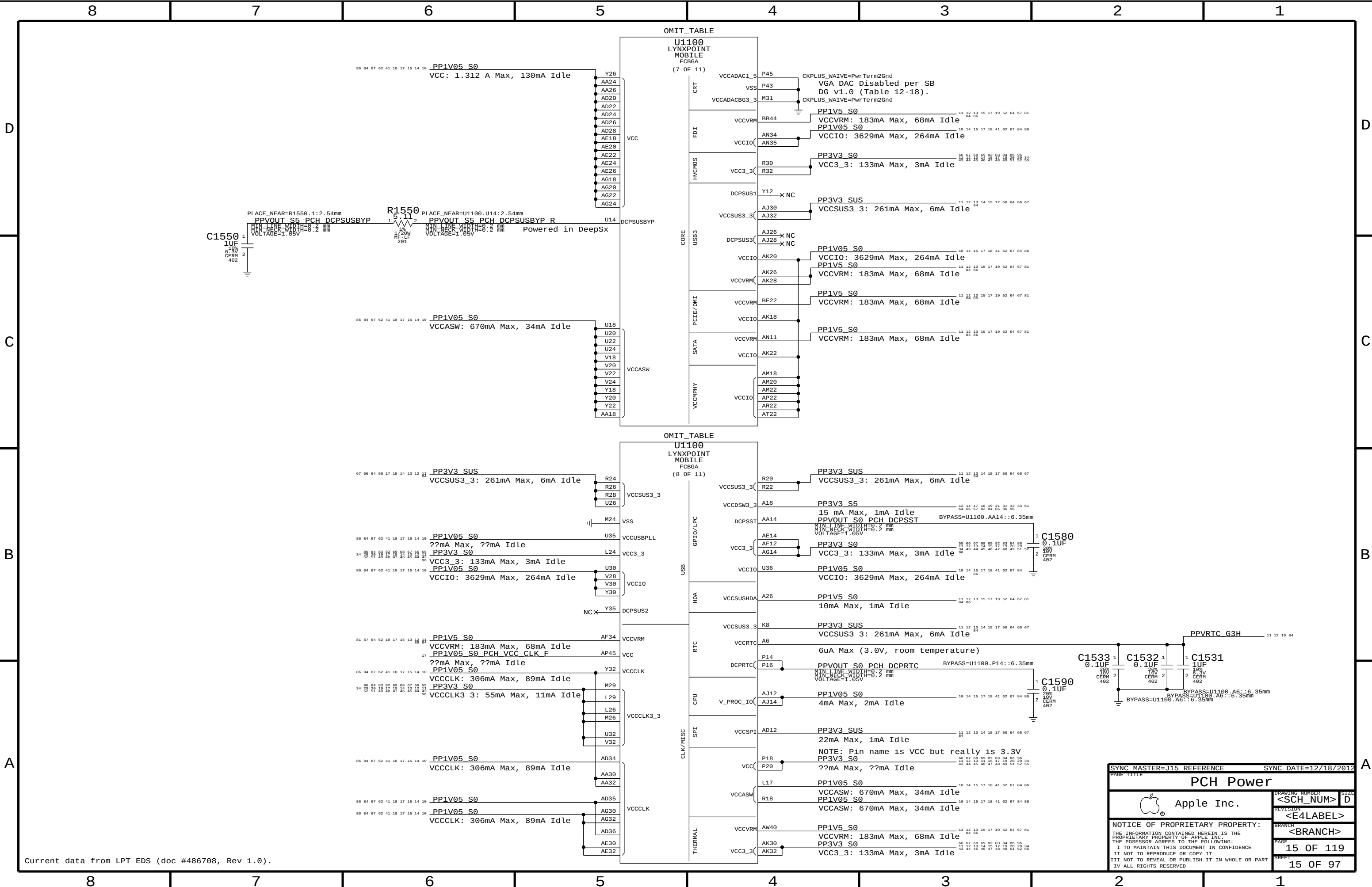
Connections are required for BDW CPU support.

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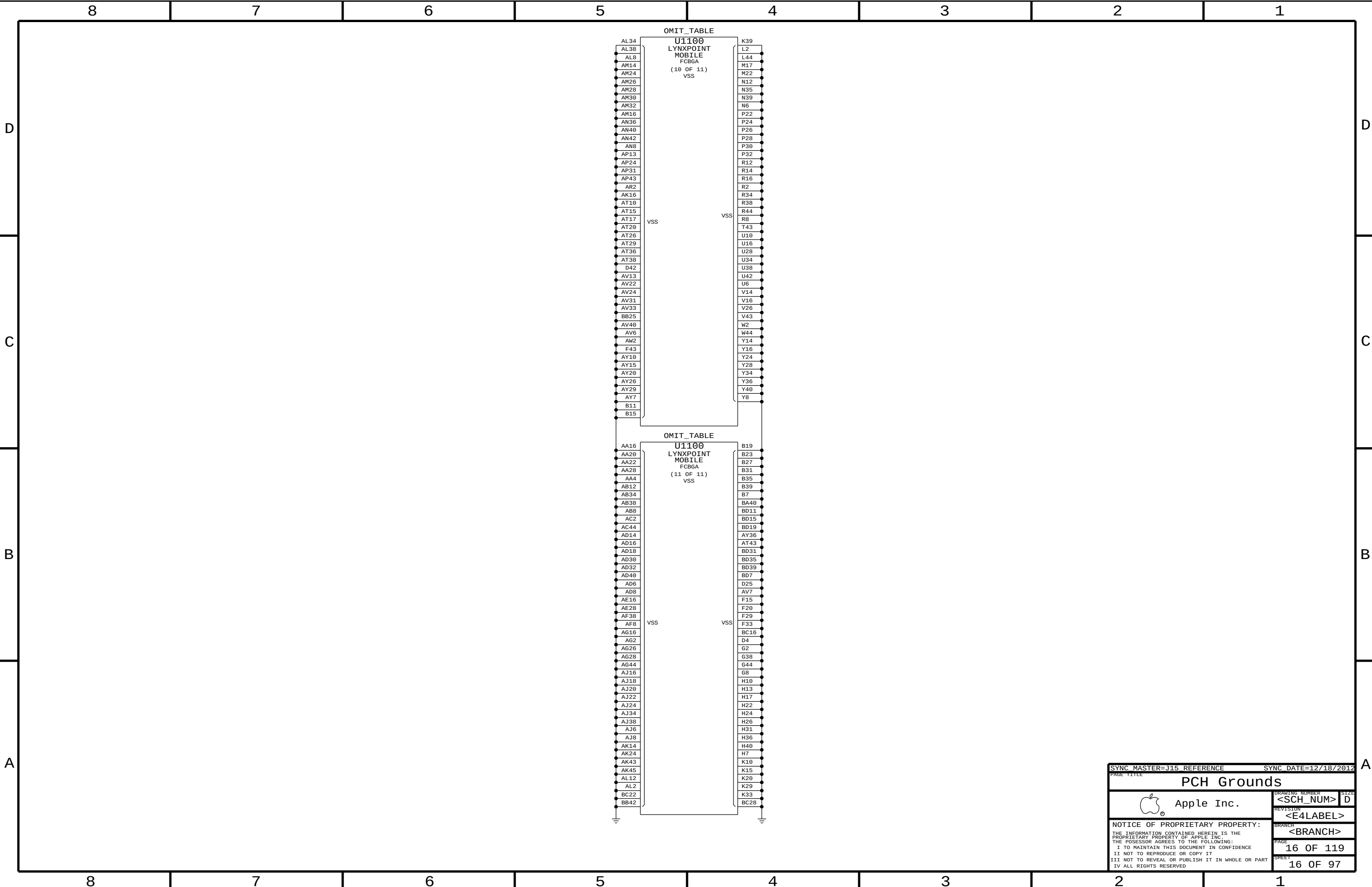






Current data from LPT EDS (doc #486708, Rev 1.0).

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PCH Grounds

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Extra BPM Testpoints

89 6	IN	XDP BPM L<2>	TP1802
89 6	IN	XDP BPM L<3>	TP1803
89 6	IN	XDP BPM L<4>	TP1804
89 6	IN	XDP BPM L<5>	TP1805
89 6	IN	XDP BPM L<6>	TP1806
89 6	IN	XDP BPM L<7>	TP1807

Merged (CPU/PCH) Micro2-XDP

NOTE: This is not the standard XDP pinout.
Use with 921-0133 Adapter Flex to support chipset debug.

TDI and TMS are terminated in CPU.

CPU JTAG Isolation

PCH/XDP Signals

		OMIT (All 10 R's)	
13	OUT	XDP DA0 USB EXTA OC L	R1890 SHORT 1 2 NONE NONE NONE 201
13	IN	XDP DA2 SSD PWR EN	R1895 SHORT 1 2 NONE NONE NONE 201
13	IN	XDP DA3 CAMERA PWR EN	R1893 SHORT 1 2 NONE NONE NONE 201
13	OUT	XDP DB0 USB EXTB OC L	R1894 SHORT 1 2 NONE NONE NONE 201
13	IN	XDP DB2 SD PWR EN	R1896 SHORT 1 2 NONE NONE NONE 201
13	OUT	XDP DB3 SDCONN STATE CHANGE L	R1897 SHORT 1 2 NONE NONE NONE 201
11	IN	XDP DC0 DP AUXCH ISOL L	R1872 SHORT 1 2 NONE NONE NONE 201
11	IN	XDP DC1 SATARDVR EN	MAKE_BASE=TRUE
14	IN	XDP DC2 ODD PWR EN L	MAKE_BASE=TRUE
14	IN	XDP DC3 JTAG ISP TCK	R1875 SHORT 1 2 NONE NONE NONE 201
14	OUT	XDP DD0 SSD PCIE SEL L	R1876 SHORT 1 2 NONE NONE NONE 201
20 18 14	OUT	XDP DD1 MLB RAMCFG1	MAKE_BASE=TRUE
20 18 11	OUT	XDP DD2 ENETSD CLKREQ L	MAKE_BASE=TRUE
11	OUT	XDP DD3 AP CLKREQ L	R1879 SHORT 1 2 NONE NONE NONE 201

Non-XDP Signals

27	IN	USB EXTA OC L
13 66	OUT	SSD PWR EN
13 20	OUT	CAMERA PWR EN PCH
13 86	IN	USB EXTB OC L
13 86	OUT	SD PWR EN
13 81 86	OUT	SDCONN STATE CHANGE L
11 85	OUT	DP AUXCH ISOL L
11 18	OUT	XDP DC1 SATARDVR EN
14 18	OUT	XDP DC2 ODD PWR EN L
14 20	OUT	JTAG ISP TCK
14	OUT	SSD PCIE SEL L
14 18 20	IN	XDP DD1 MLB RAMCFG1
11 18	IN	XDP DD2 ENETSD CLKREQ
11 33	IN	AP CLKREQ L

PCH/XDP Signal Isolation Notes:

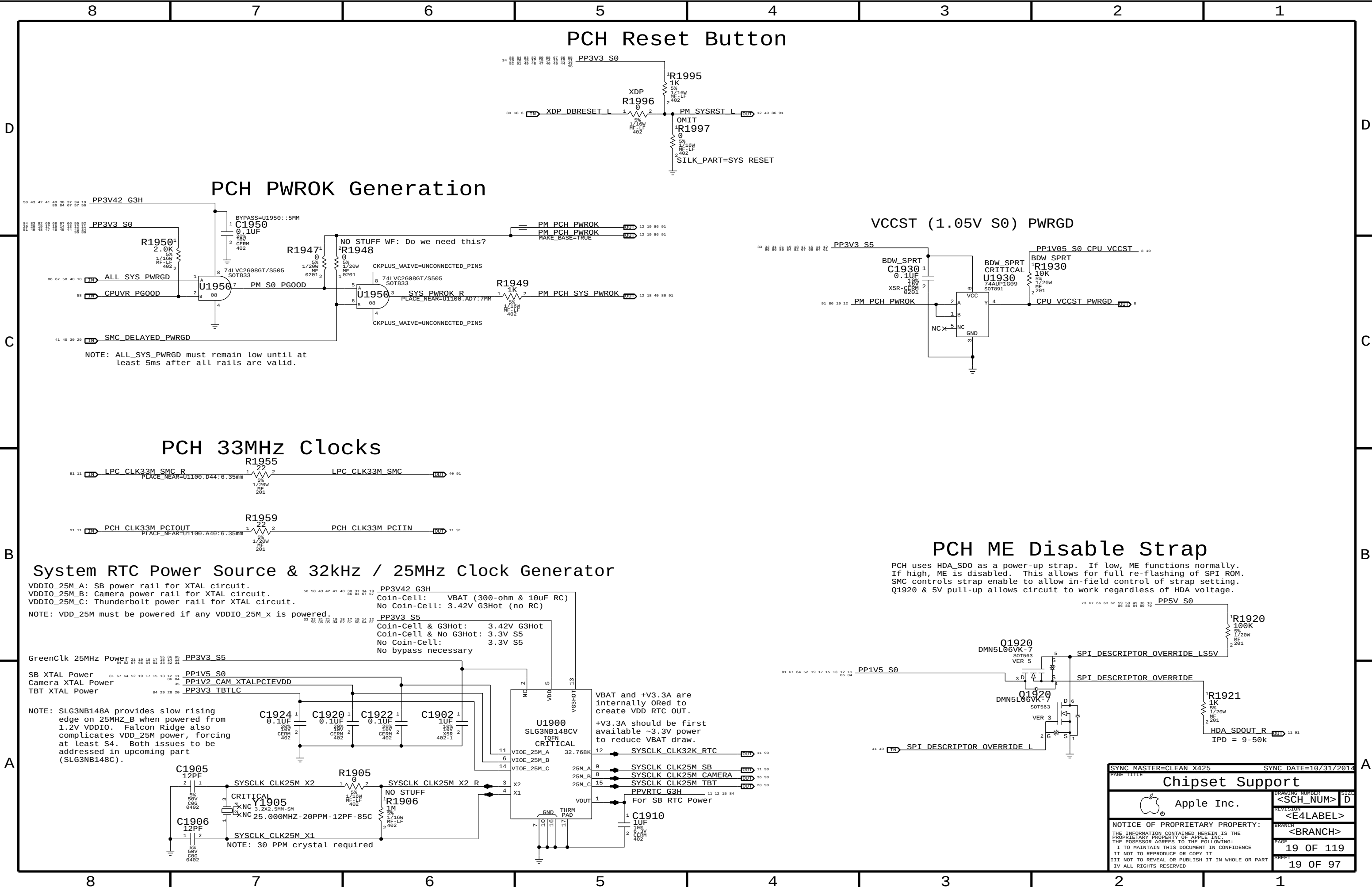
'Output' non-XDP signals require pulls.
'Output' PCH/XDP signals require pulls.

R187x and R189x should be placed where signal path needs to split between route from PCH to J1850 and path to non-XDP signal destination (to minimize stub).

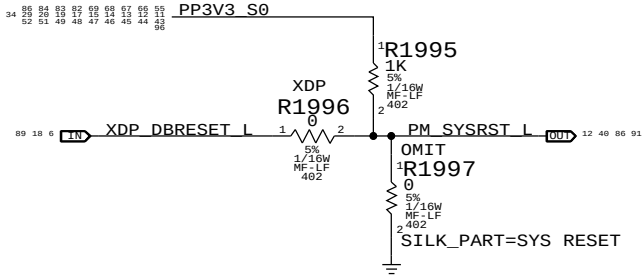
Unused PCH/XDP Signals

13	IN	XDP DA1 USB EXTC OC L	TP1810
13	IN	XDP DB1 USB EXTD OC L	TP1811
14	IN	XDP FC0 HDD PWR EN	TP1812
14	IN	XDP FC1 GPU GOOD	TP1813

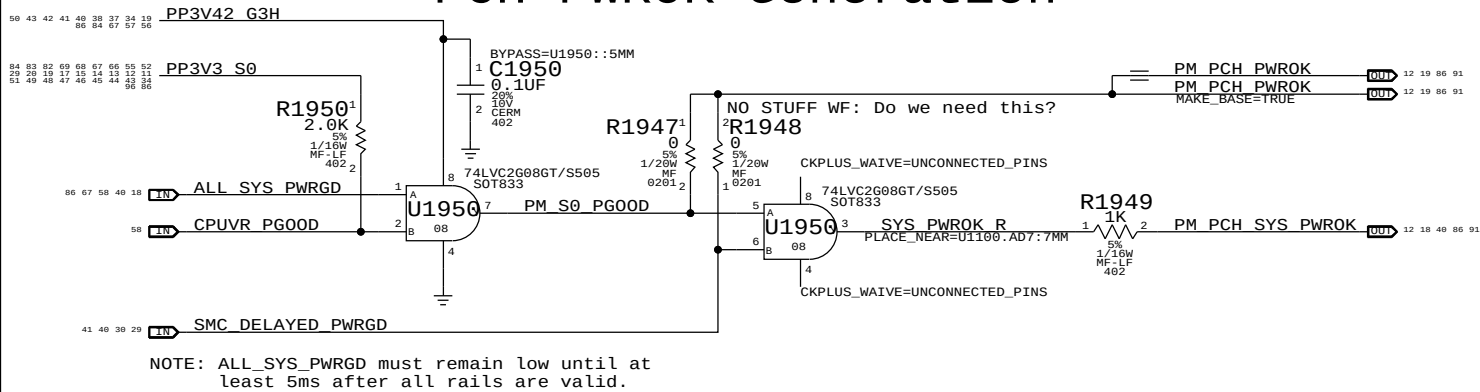
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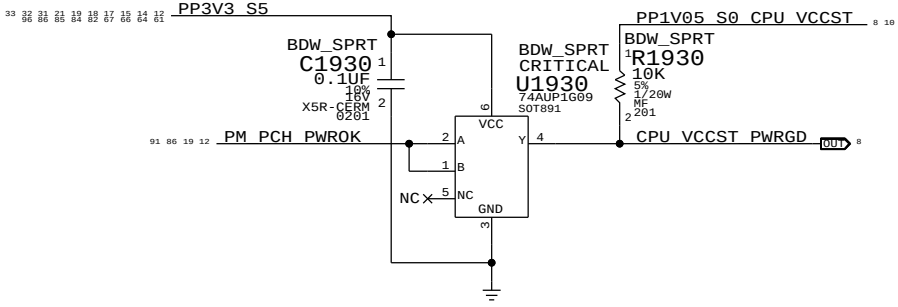
PCH Reset Button



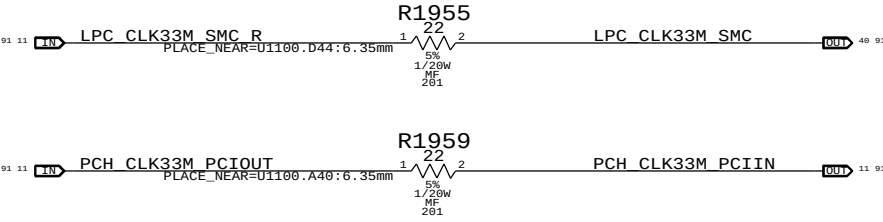
PCH PWROK Generation



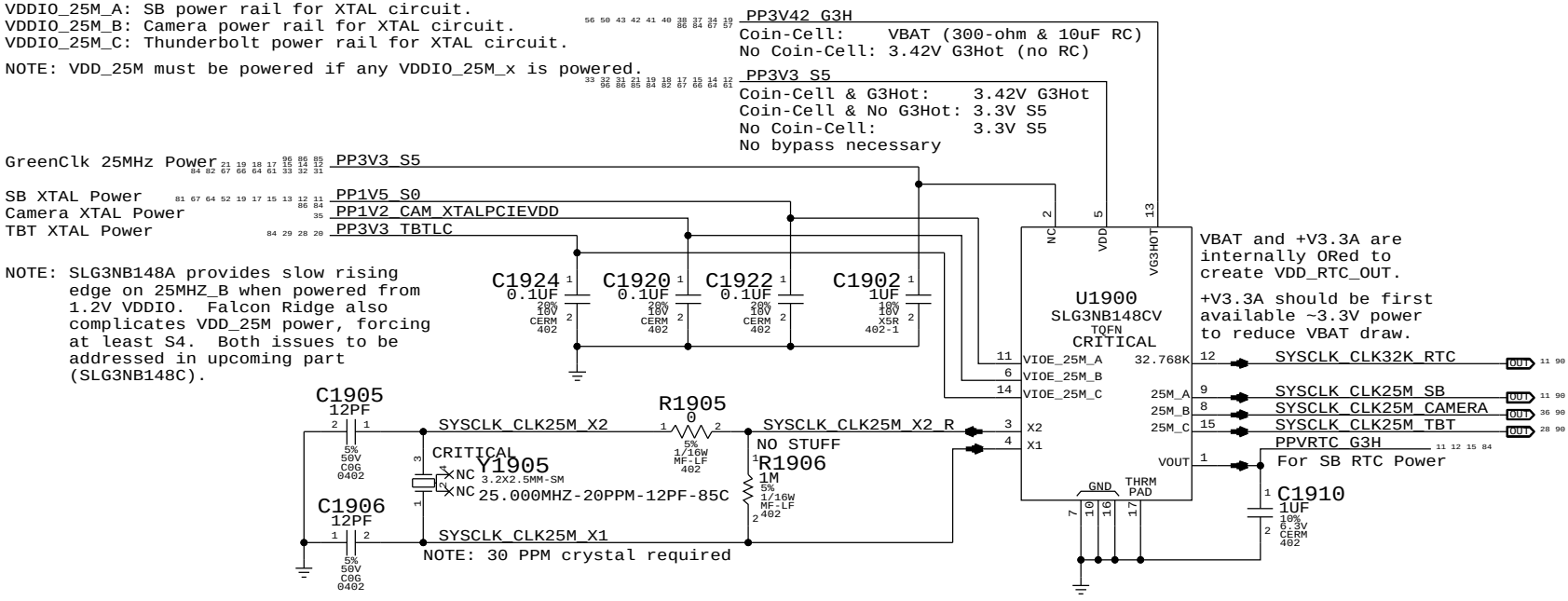
VCCST (1.05V S0) PWRGD



PCH 33MHz Clocks

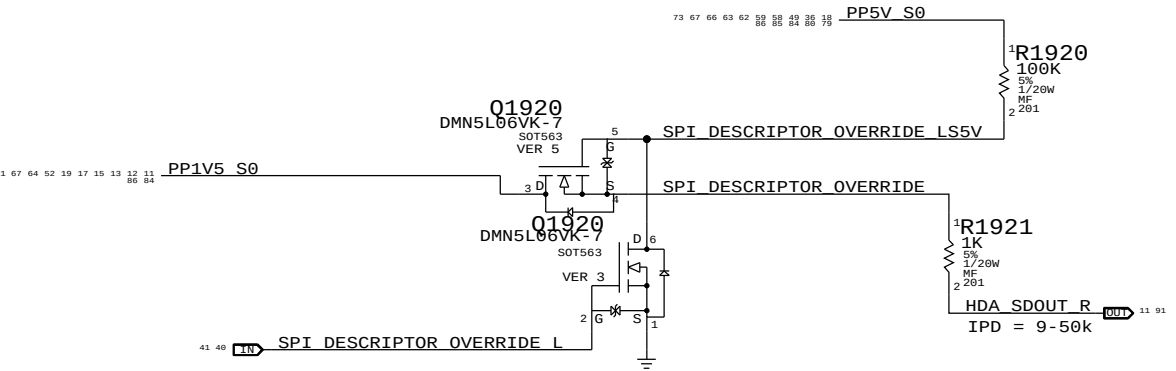


System RTC Power Source & 32kHz / 25MHz Clock Generator



PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.



SYNC MASTER=CLEAN X425		SYNC DATE=10/31/2014	
PAGE TITLE		Chipset Support	
Apple Inc.		DRAWING NUMBER	SIZE
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D



D

C

C

B

B

A

A

A



B

A

VRef Dividers

Always used, regardless of margining option.

VREFDQ A

VREFDQ B

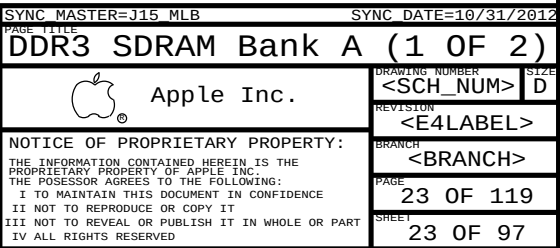
VREFCA

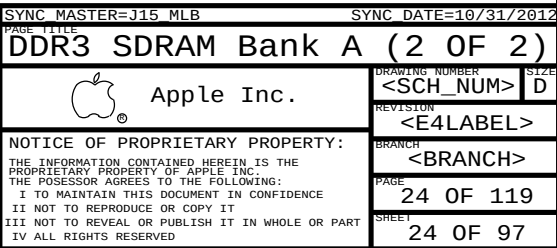
NOTE: CPU has single output for VREFCA. Connected to 4 DRAMs.

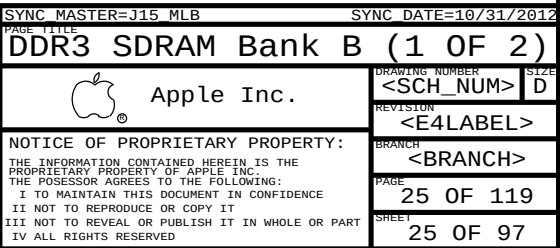
NOTE: CPU DAC output step sizes:

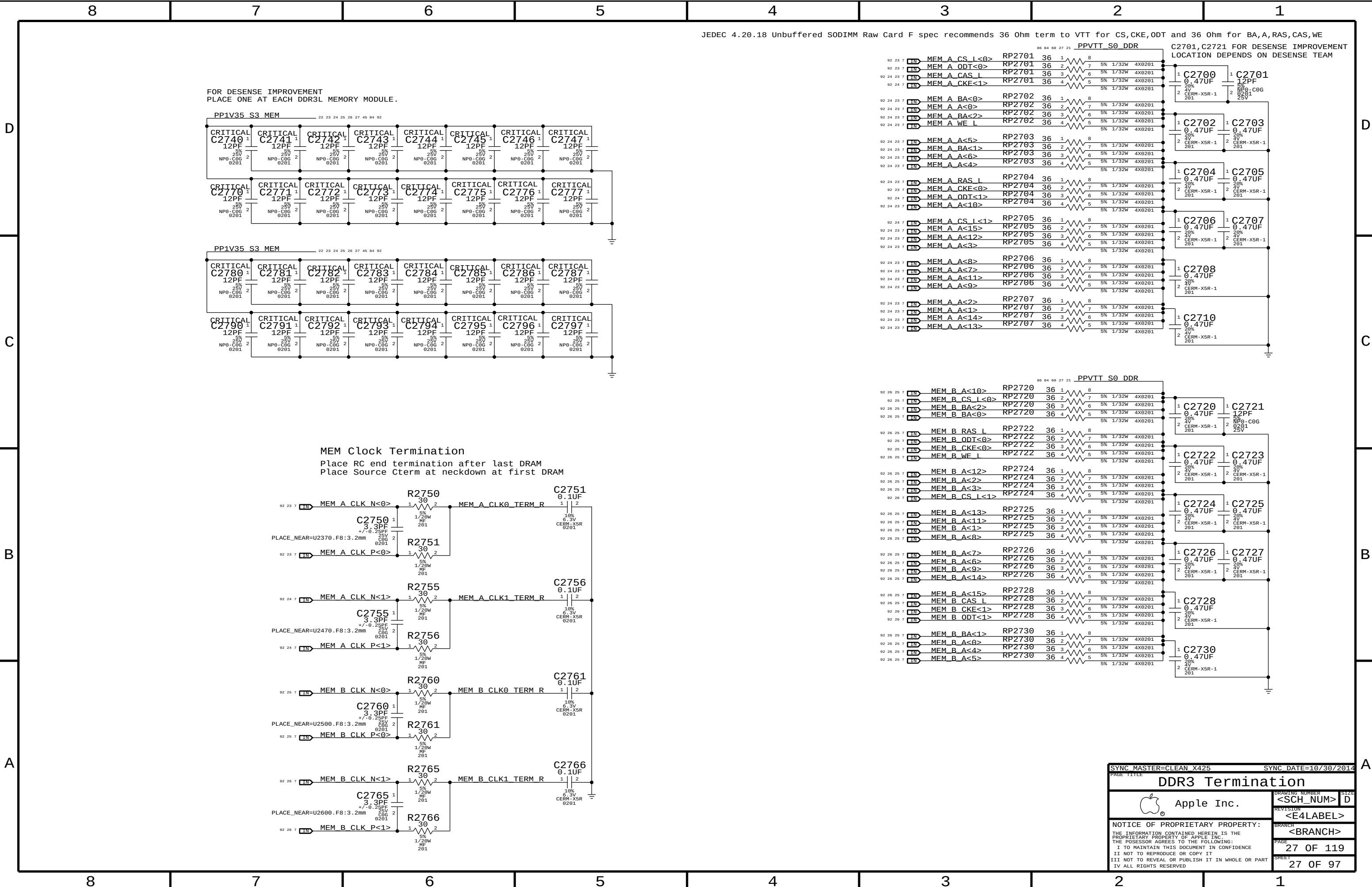
DDR3	(1.5V)	7.70mV	per step
DDR3L	(1.35V)	6.99mV	per step
LPDDR3	(1.2V)	?.??mV	per step

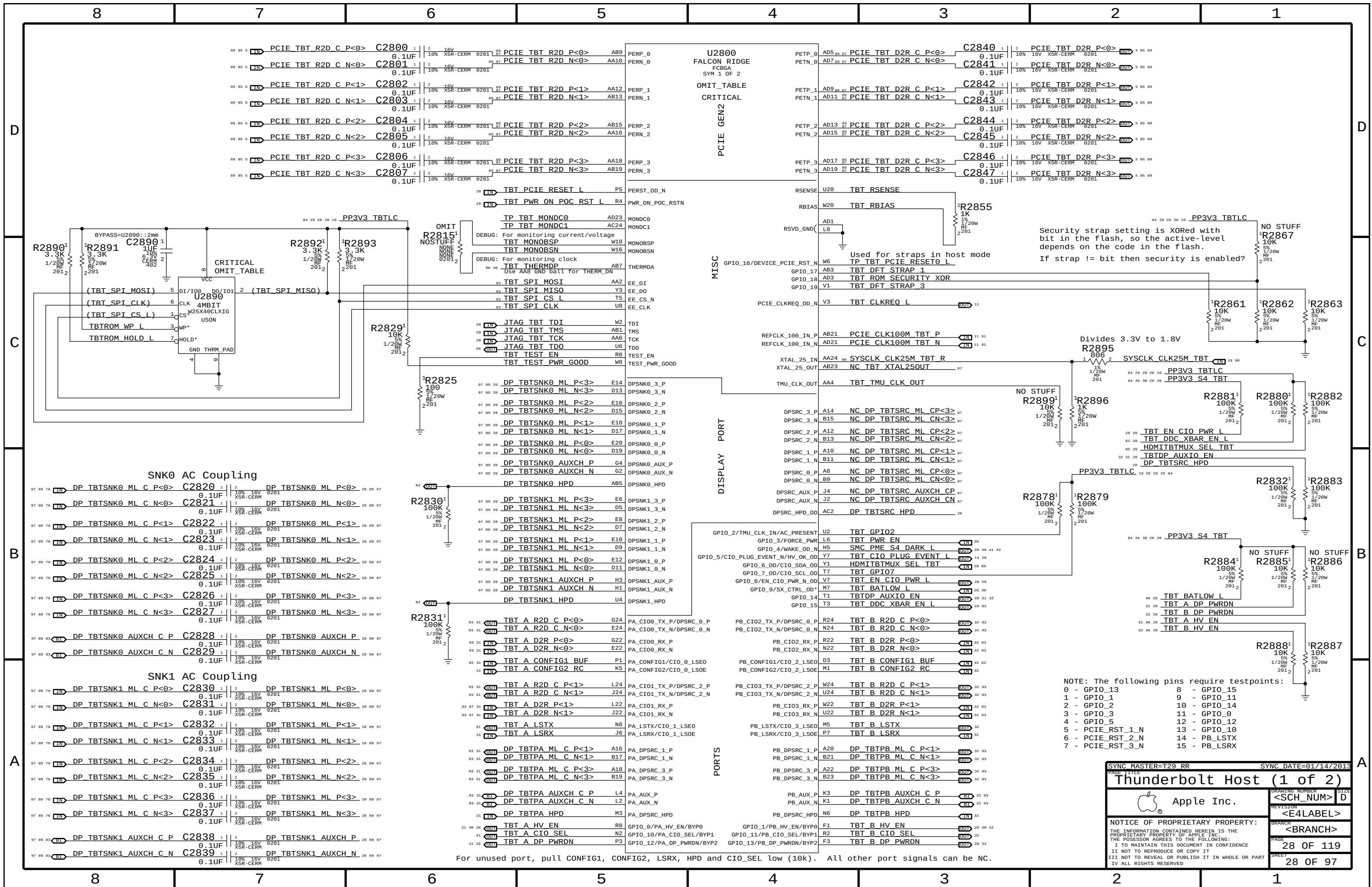
NOTE: CPU has single output for VREFCA.
Connected to 4 DRAMs.

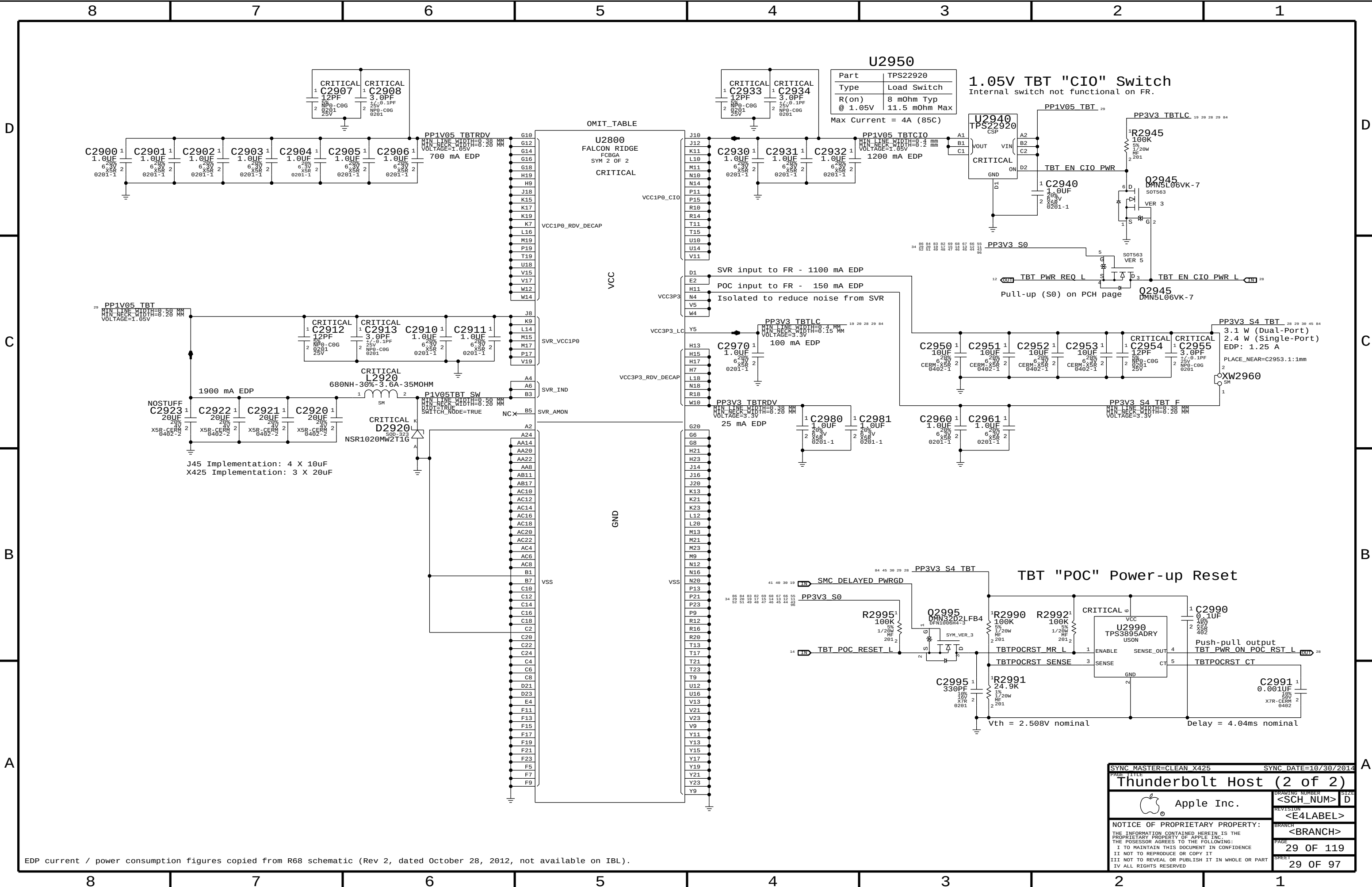












EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC MASTER=CLEAN X425

SYNC DATE=10/30/2014

Thunderbolt Host (2 of 2)

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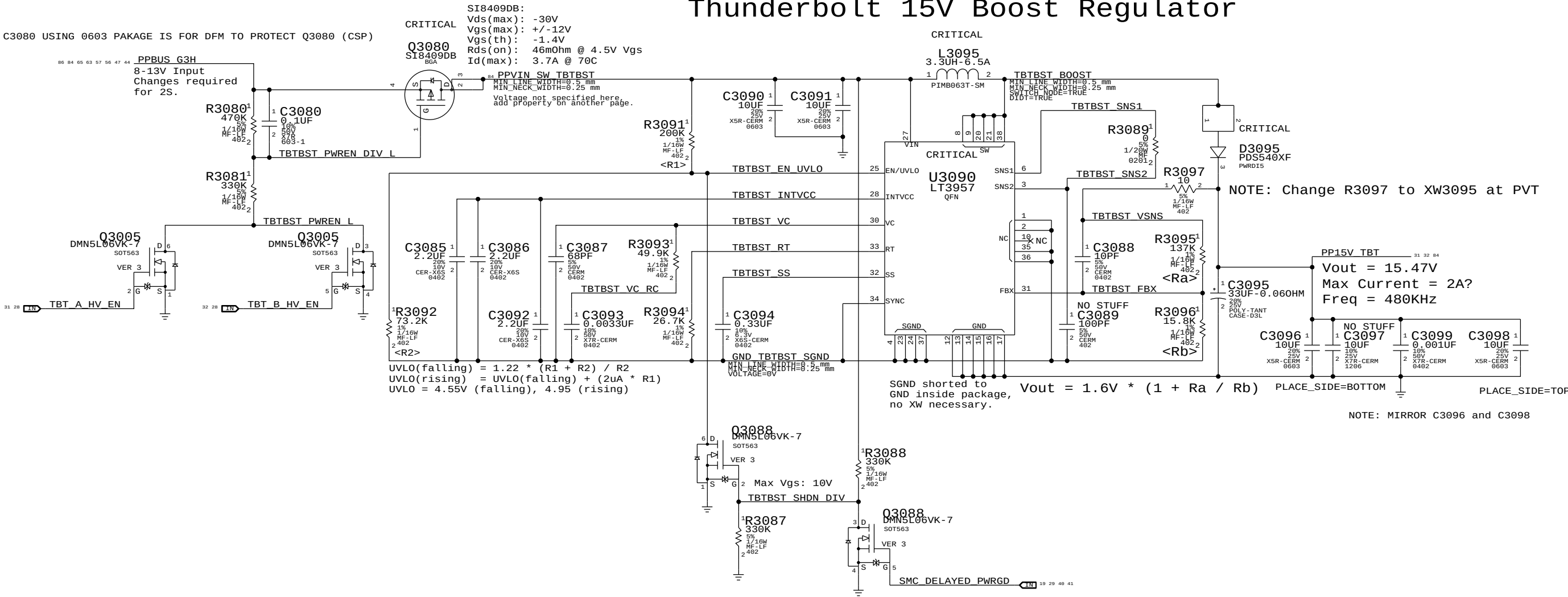
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SHEET
29 OF 97

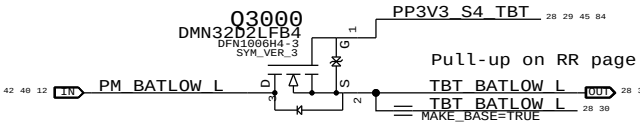
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
(NONE)

Thunderbolt 15V Boost Regulator



BATLOW# Isolation



SYNC MASTER=CLEAN X305

SYNC DATE=06/24/2014

Thunderbolt Mobile Support

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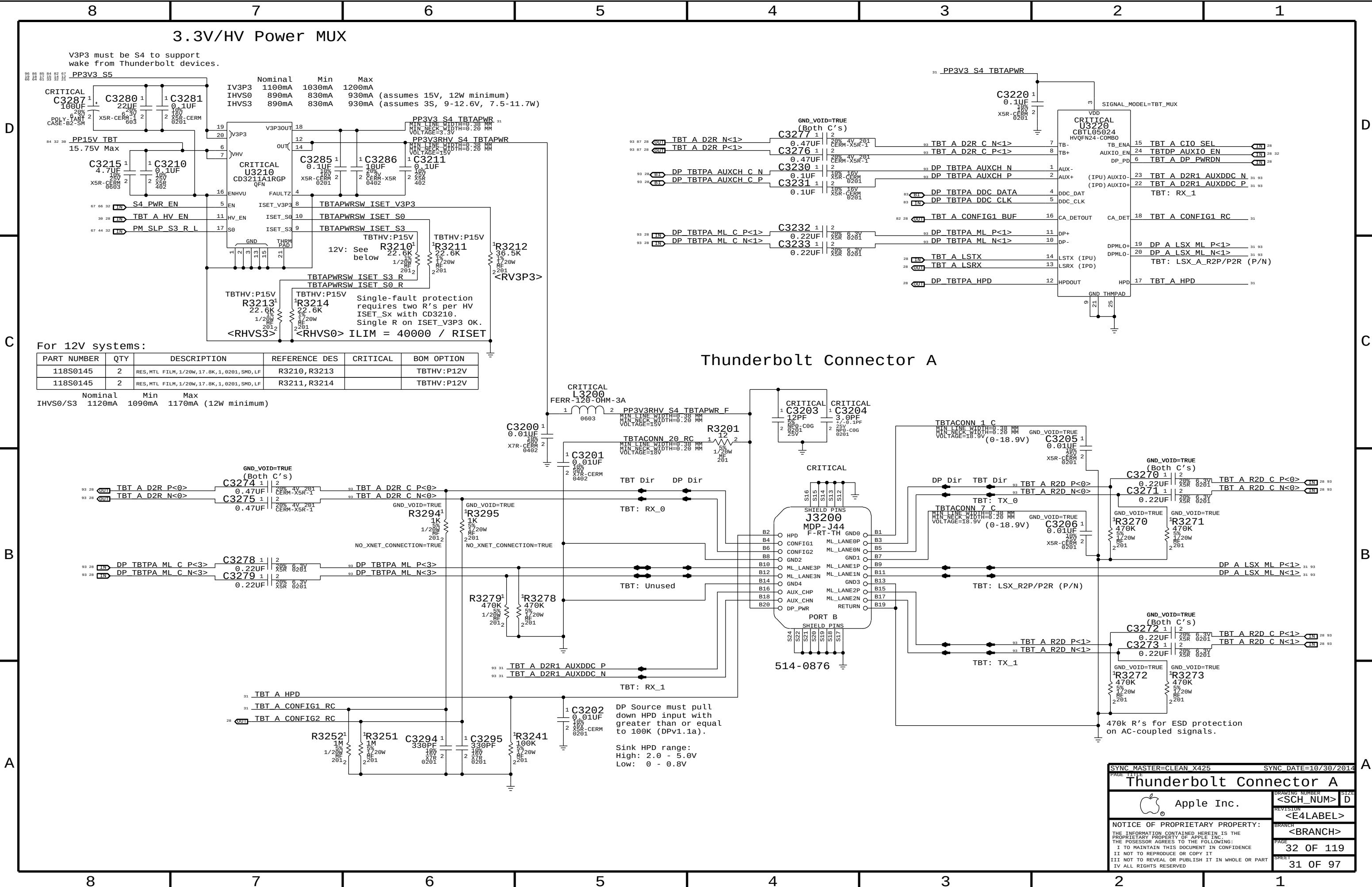
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For 12V systems:					
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF	R3210, R3213		TBTHV:P12V
118S0145	2	RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF	R3211, R3214		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)

SYNC MASTER=CLEAN X425

SYNC DATE=10/30/2014

Thunderbolt Connector A

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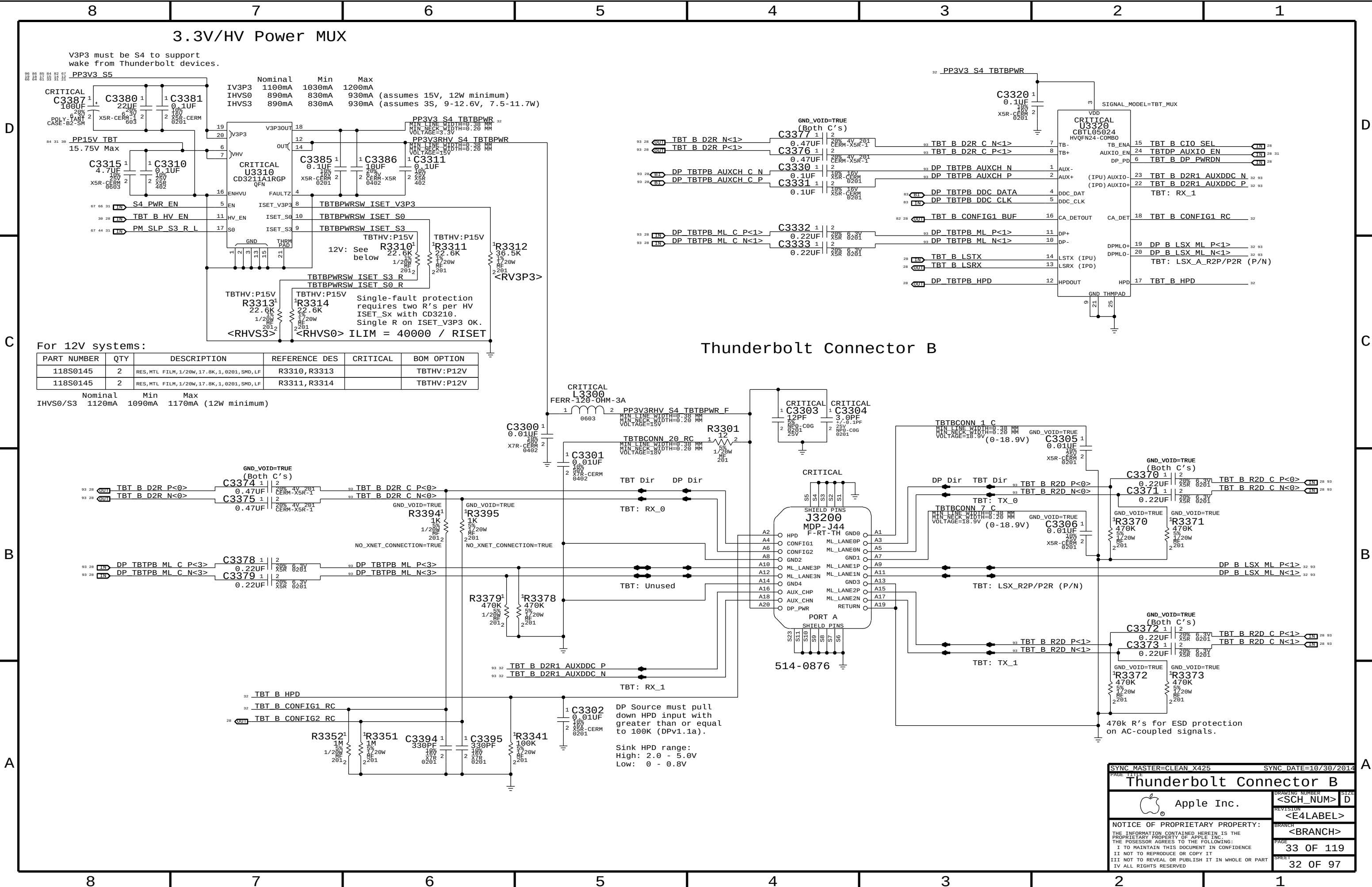
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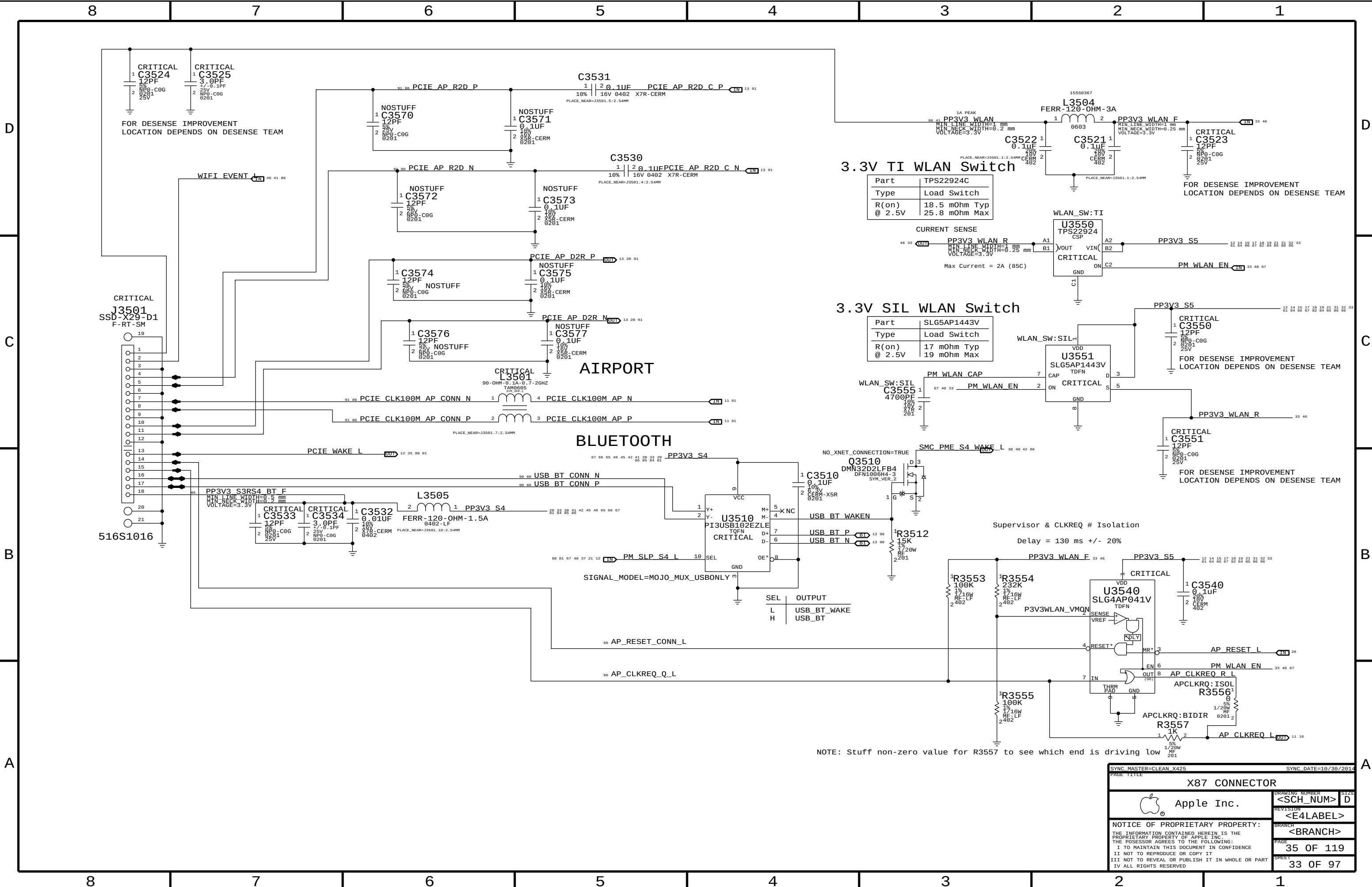
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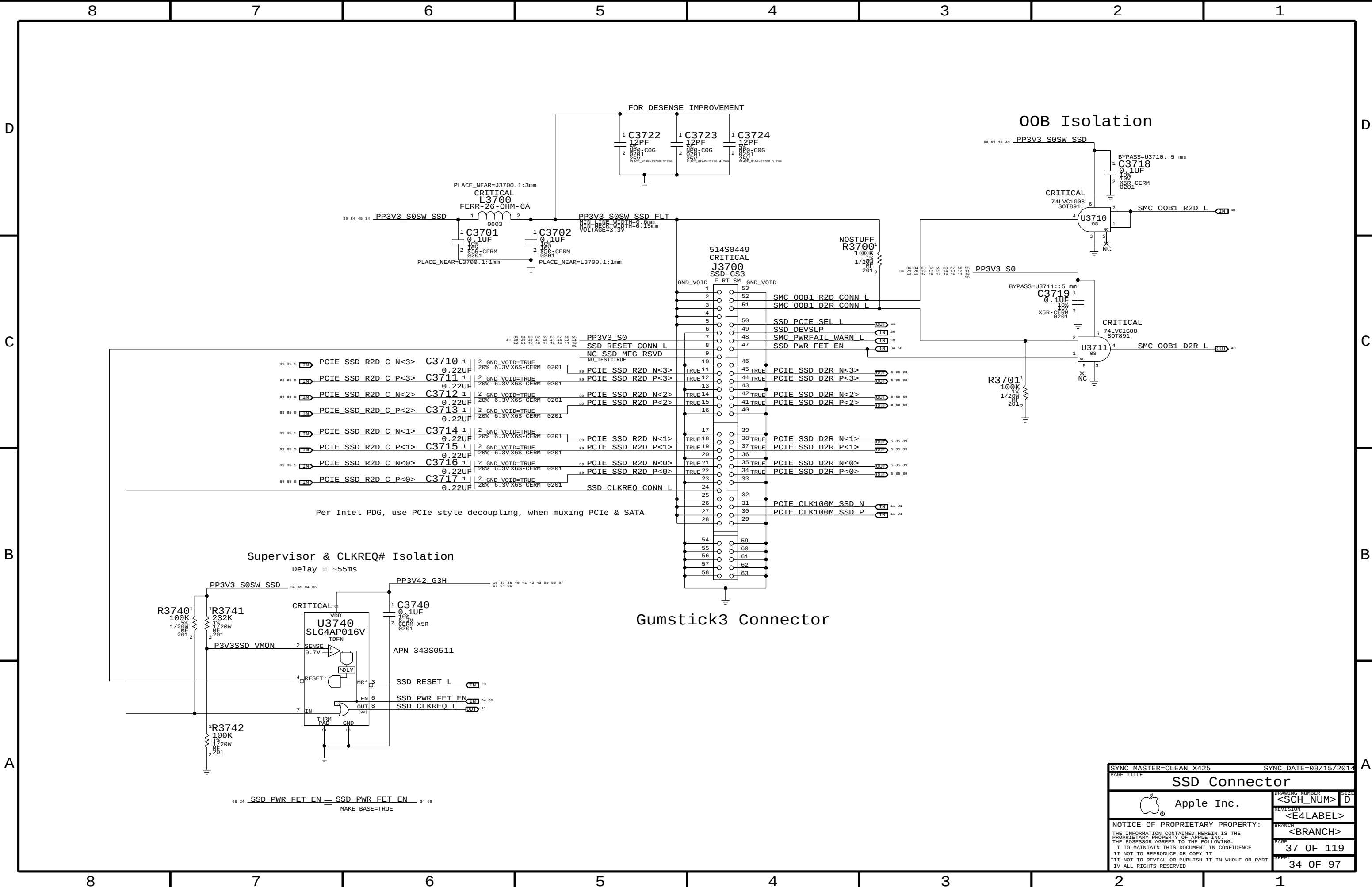
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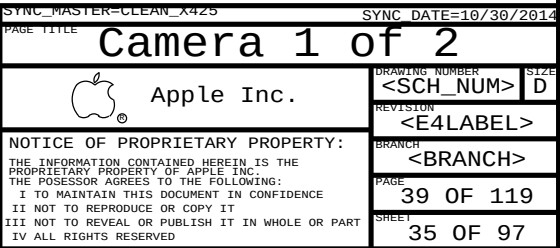


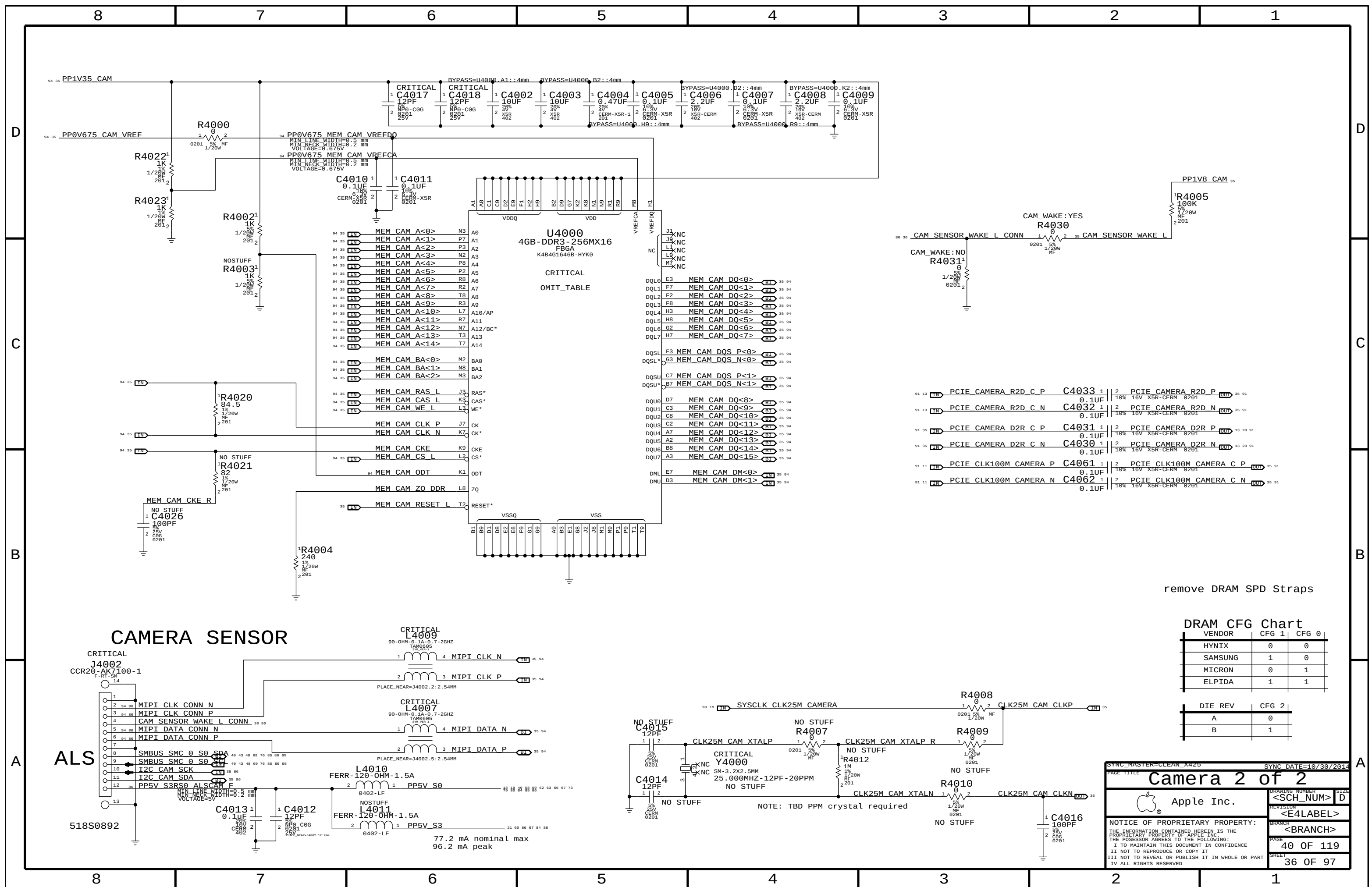


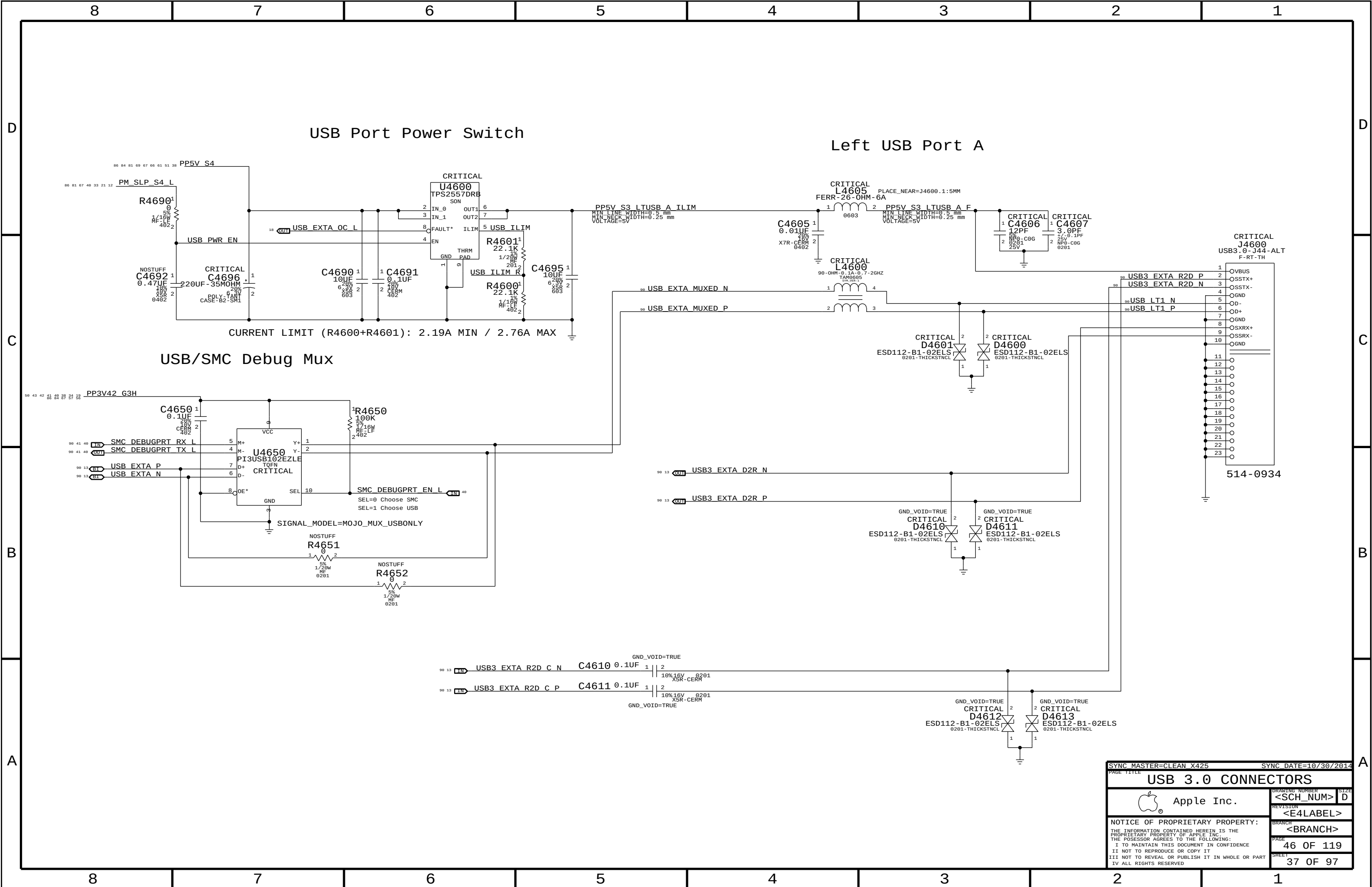
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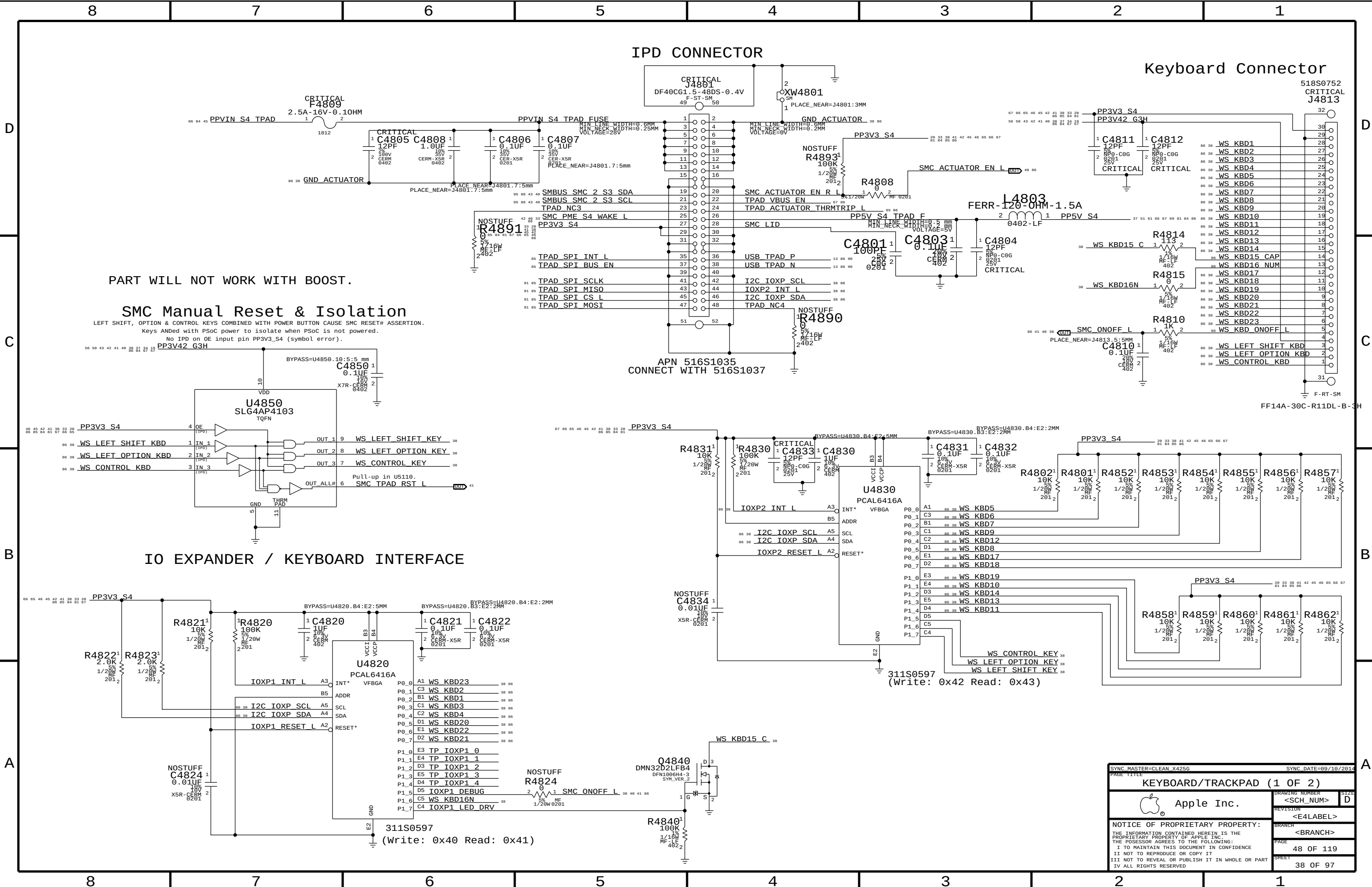


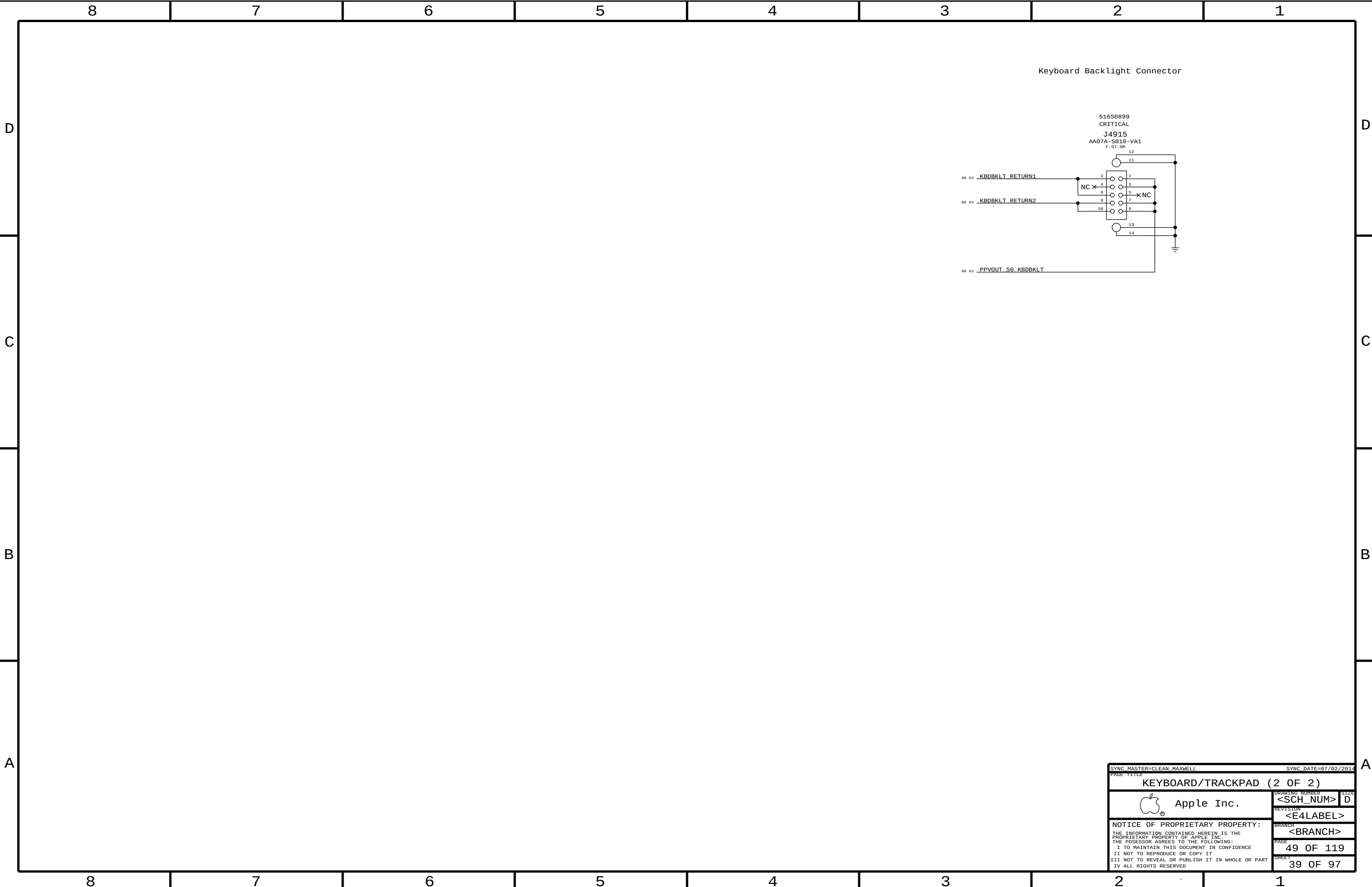
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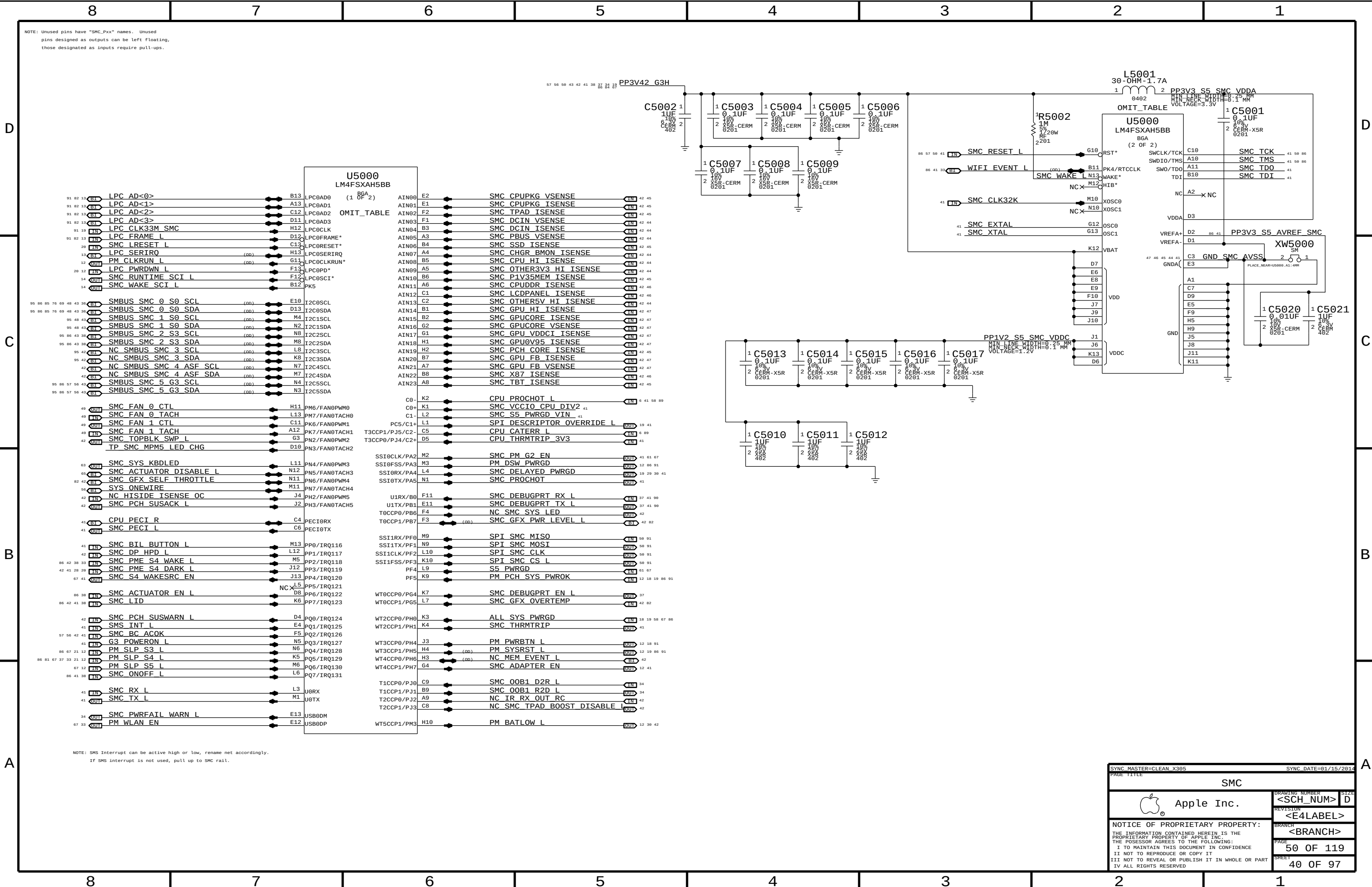


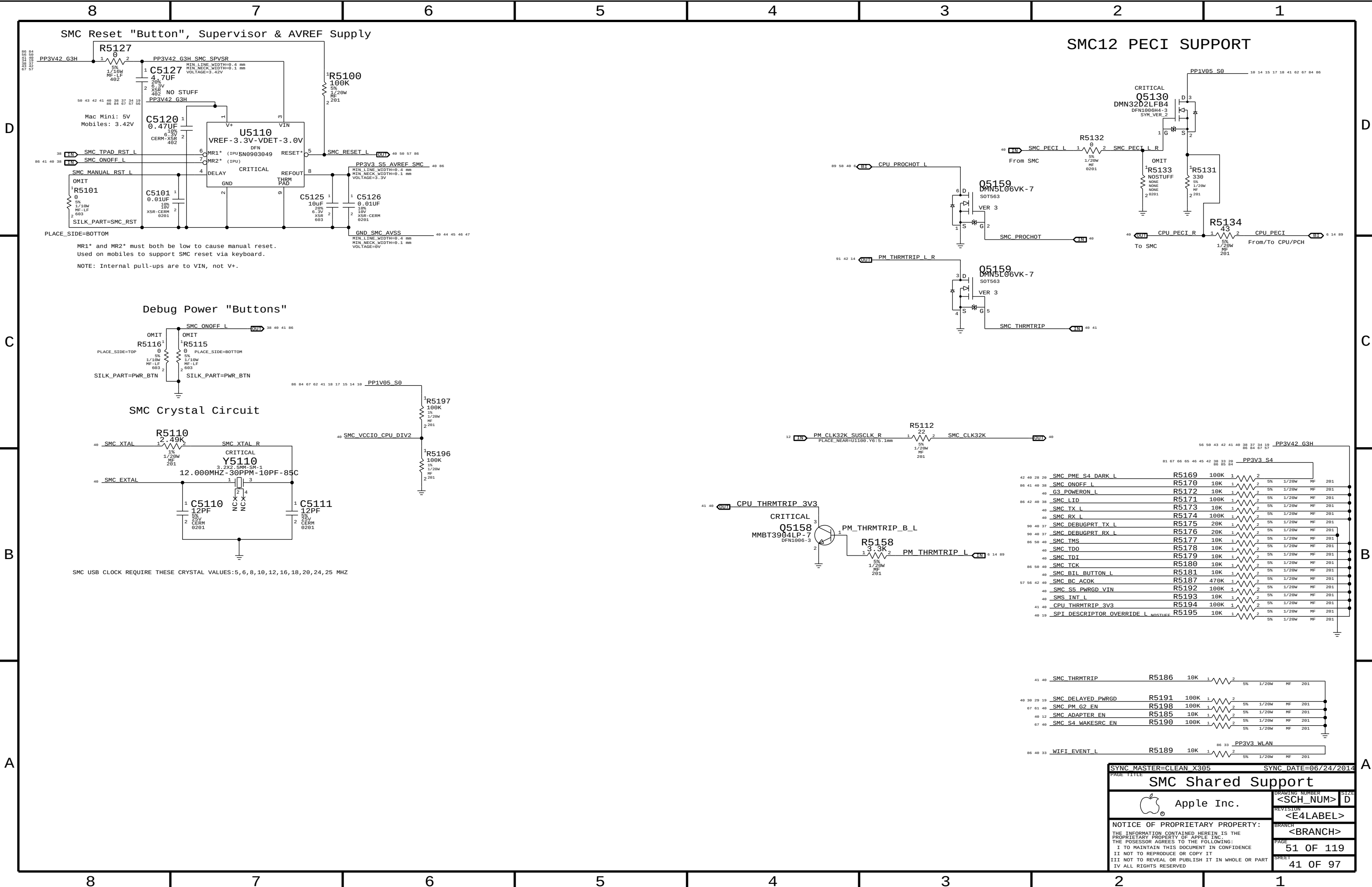












SMC Reset "Button", Supervisor & AVREF Supply

SMC12 PECE SUPPORT

Debug Power "Buttons"

SMC Crystal Circuit

SMC12 PECE SUPPORT

SMC USB CLOCK REQUIRE THESE CRYSTAL VALUES: 5, 6, 8, 10, 12, 16, 18, 20, 24, 25 MHZ

42 40 28 20	SMC PME S4 DARK L	R5169	100K	1	2	5%	1/20W	MF	201
86 41 40 38	SMC_ONOFF L	R5170	10K	1	2	5%	1/20W	MF	201
40	G3 POWERON L	R5172	10K	1	2	5%	1/20W	MF	201
86 42 40 38	SMC LID	R5171	100K	1	2	5%	1/20W	MF	201
40	SMC TX L	R5173	10K	1	2	5%	1/20W	MF	201
40	SMC RX L	R5174	100K	1	2	5%	1/20W	MF	201
90 40 37	SMC_DEBUGPT_TX L	R5175	20K	1	2	5%	1/20W	MF	201
90 40 37	SMC_DEBUGPT_RX L	R5176	20K	1	2	5%	1/20W	MF	201
86 50 40	SMC TMS	R5177	10K	1	2	5%	1/20W	MF	201
40	SMC TDO	R5178	10K	1	2	5%	1/20W	MF	201
40	SMC TDI	R5179	10K	1	2	5%	1/20W	MF	201
86 50 40	SMC TCK	R5180	10K	1	2	5%	1/20W	MF	201
40	SMC_BIL_BUTTON L	R5181	10K	1	2	5%	1/20W	MF	201
57 56 42 40	SMC_BC_ACOK	R5187	470K	1	2	5%	1/20W	MF	201
40	SMC_S5_PWRGD_VIN	R5192	100K	1	2	5%	1/20W	MF	201
40	SMC_INT L	R5193	10K	1	2	5%	1/20W	MF	201
41 40	CPU_THRMTRIP_3V3	R5194	100K	1	2	5%	1/20W	MF	201
40 19	SPI_DESCRIPTOR_OVERRIDE L	R5195	10K	1	2	5%	1/20W	MF	201

41 40	SMC_THRMTRIP	R5186	10K	1	2	5%	1/20W	MF	201
40 30 29 19	SMC_DELAYED_PWRGD	R5191	100K	1	2	5%	1/20W	MF	201
67 61 40	SMC_PM_G2_EN	R5198	100K	1	2	5%	1/20W	MF	201
40 12	SMC_ADAPTER_EN	R5185	10K	1	2	5%	1/20W	MF	201
67 40	SMC_S4_WAKESRC_EN	R5190	100K	1	2	5%	1/20W	MF	201

SYNC MASTER=CLEAN X305

SYNC DATE=06/24/2014

SMC Shared Support

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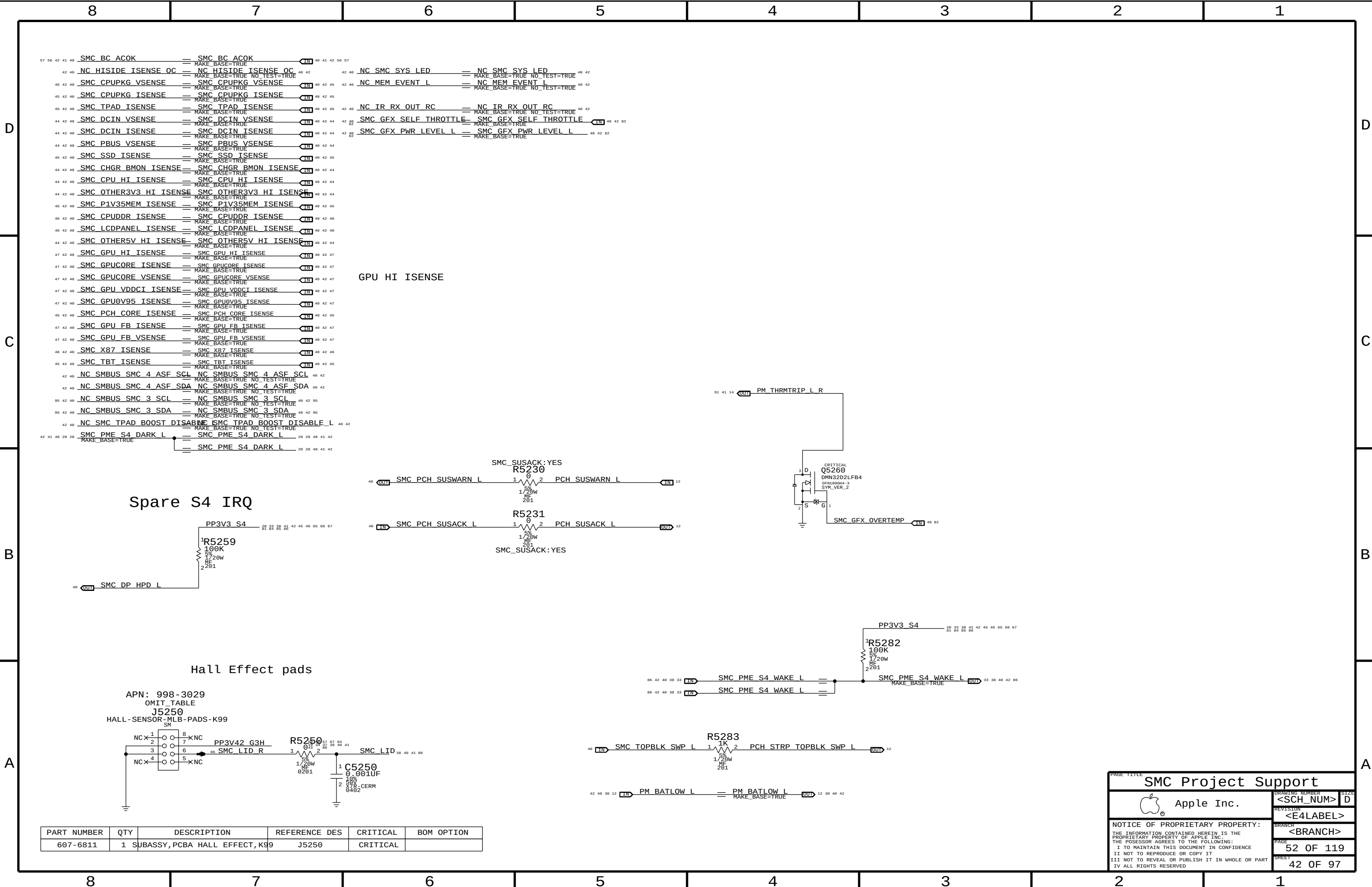
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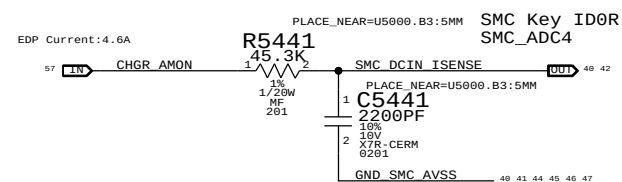
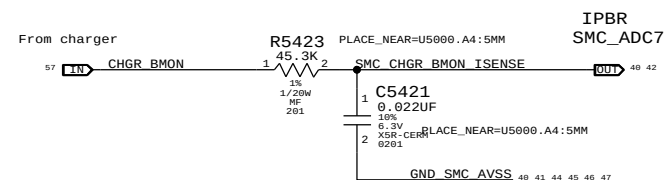
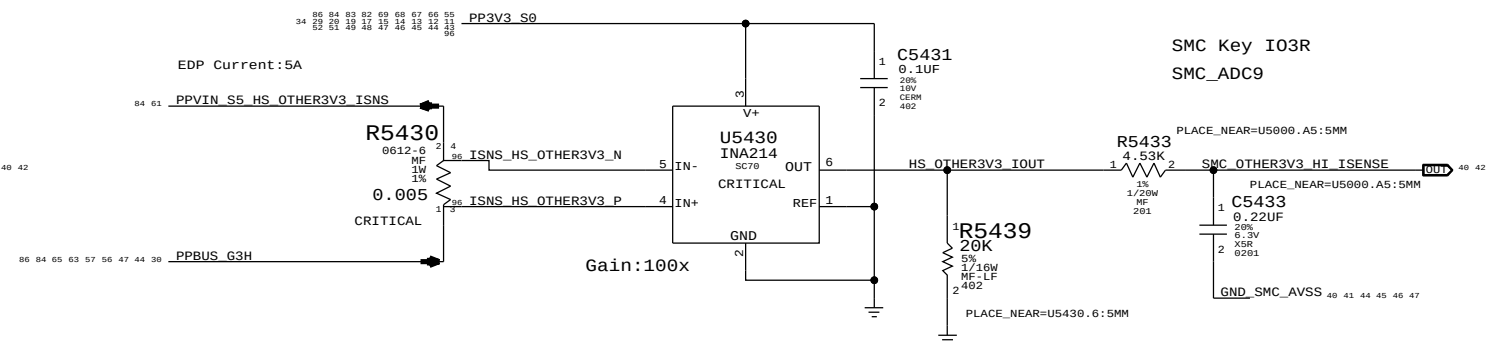
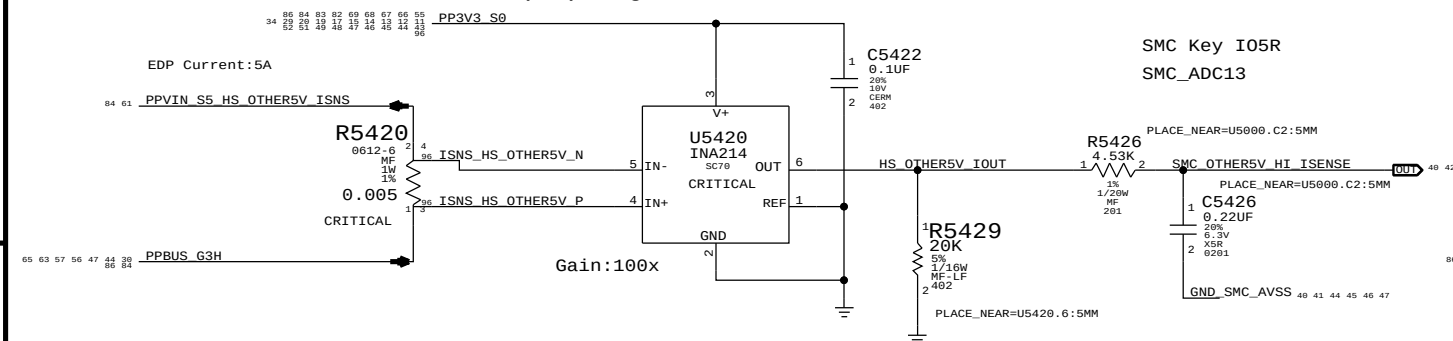
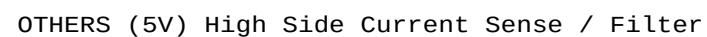
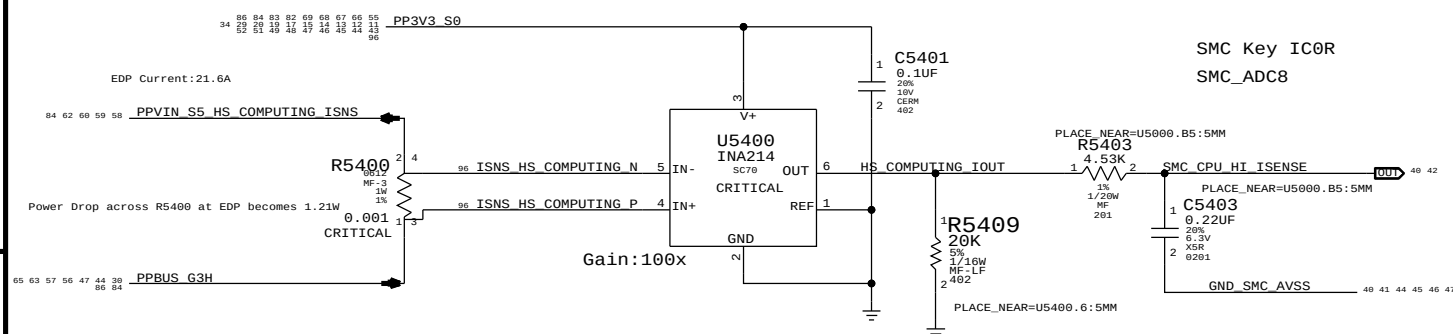
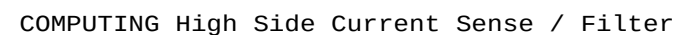
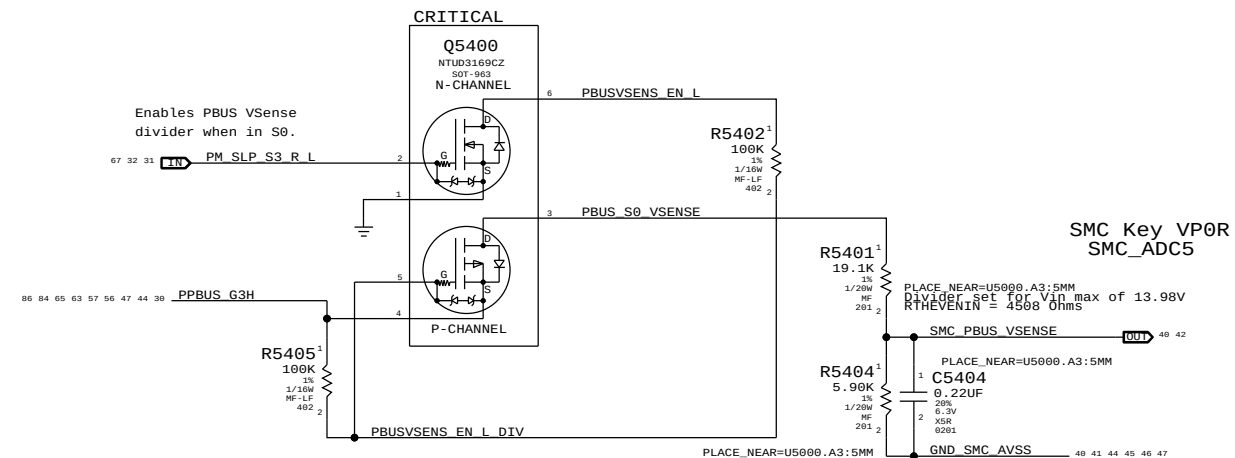
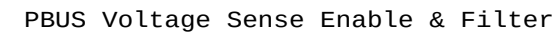
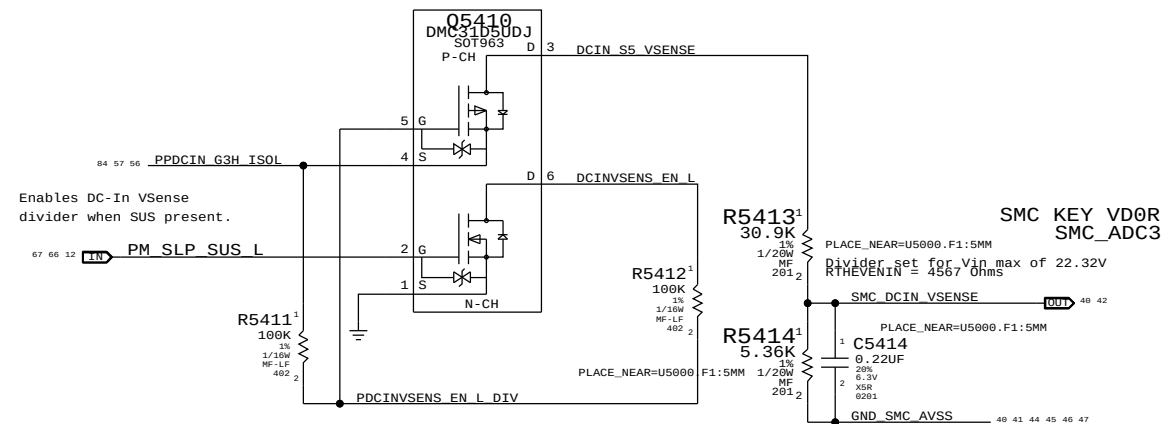
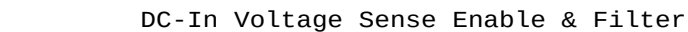
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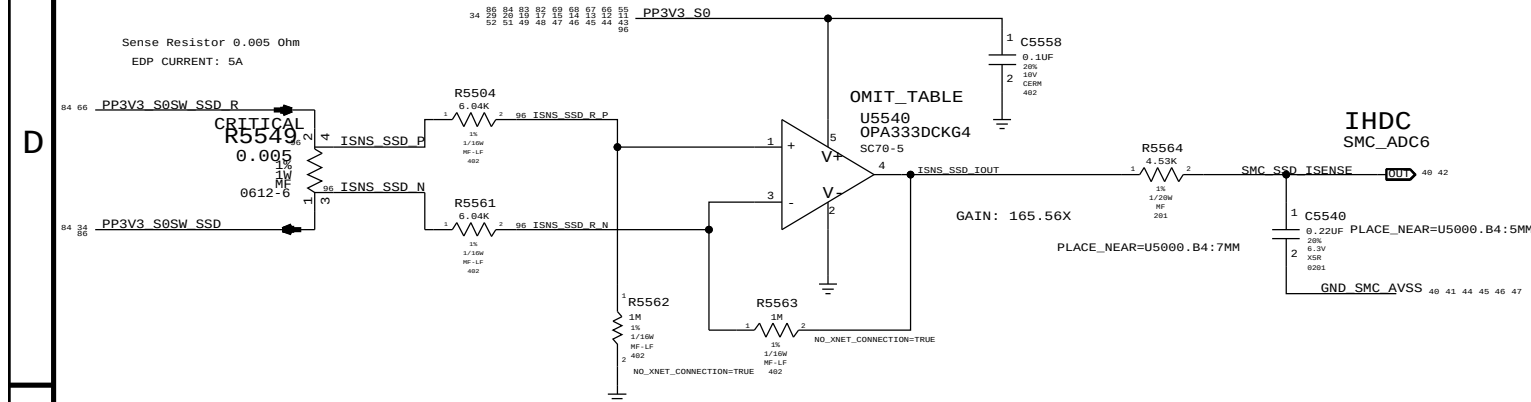
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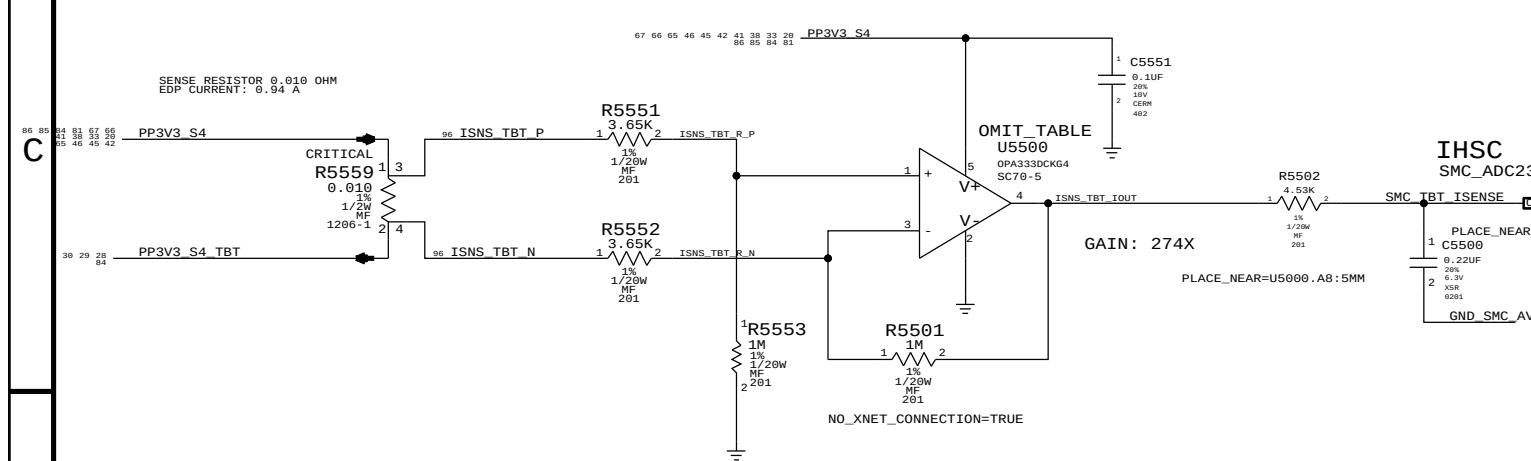


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PAGE TITLE			
High Side Voltage and Current Sensing			
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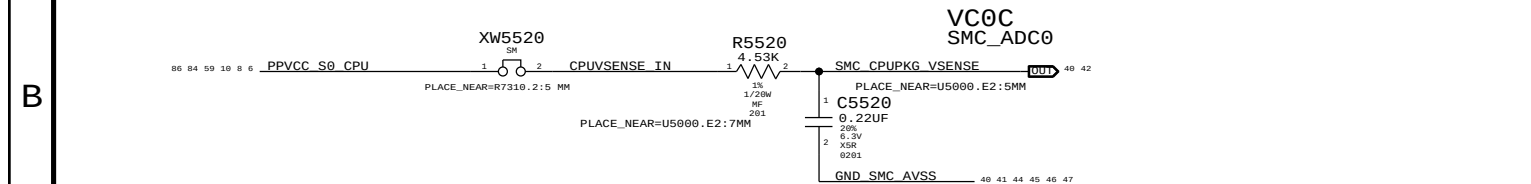
SSD CURRENT SENSE



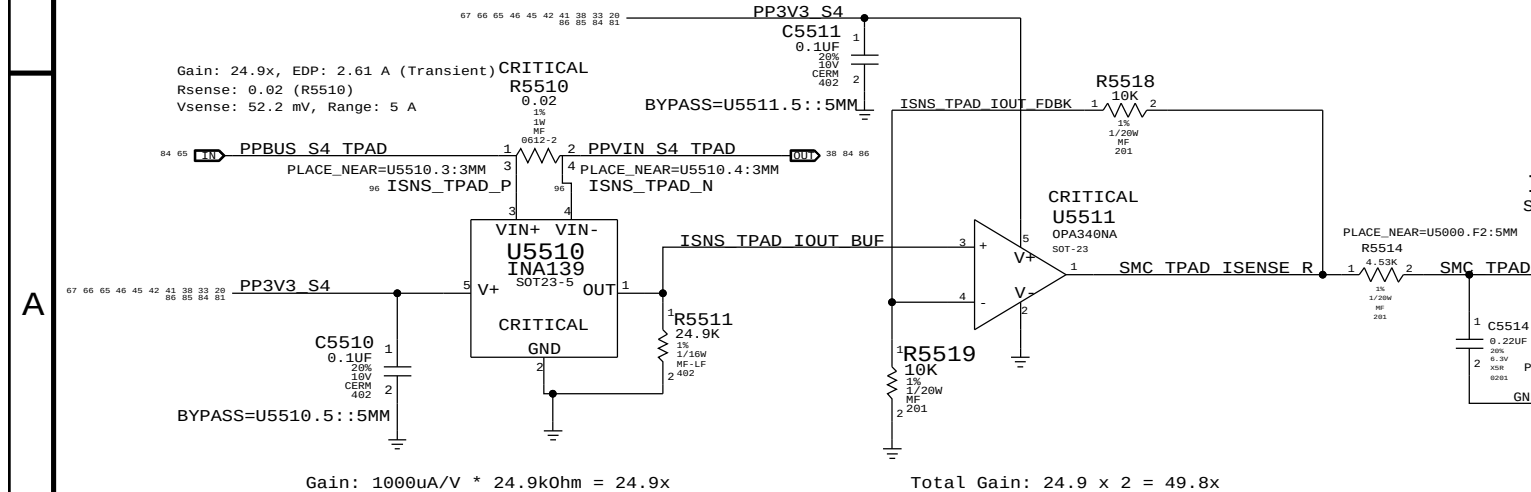
TBT Router CURRENT SENSE



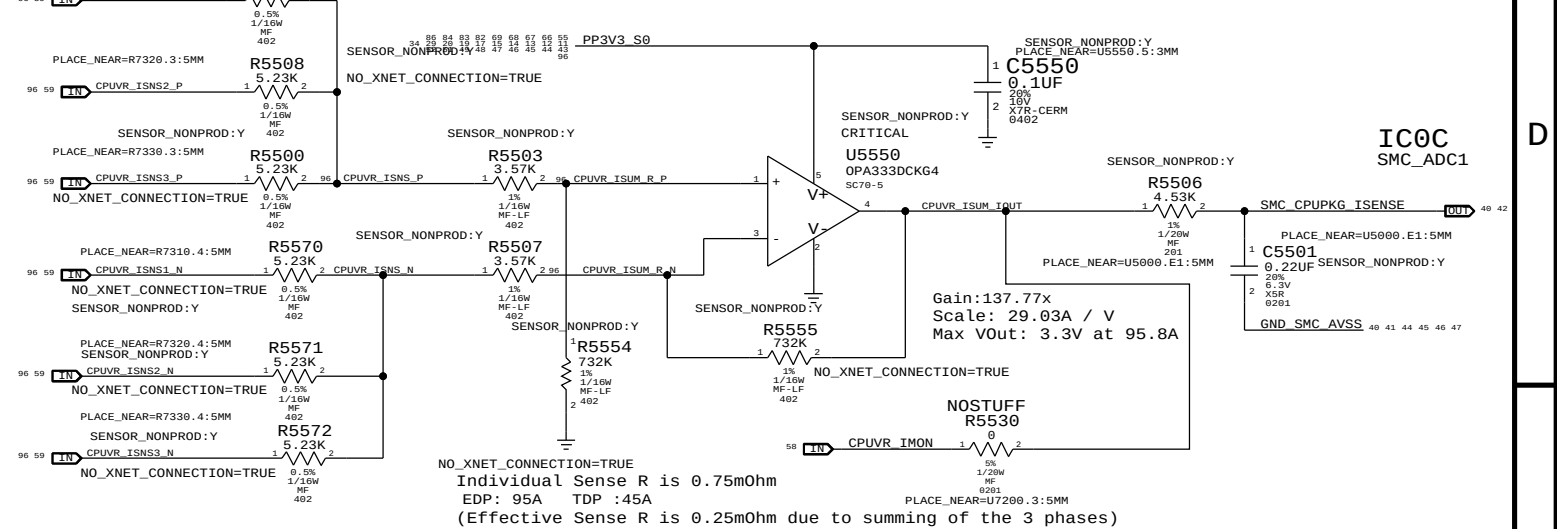
CPU Vcore Voltage Sense / Filter



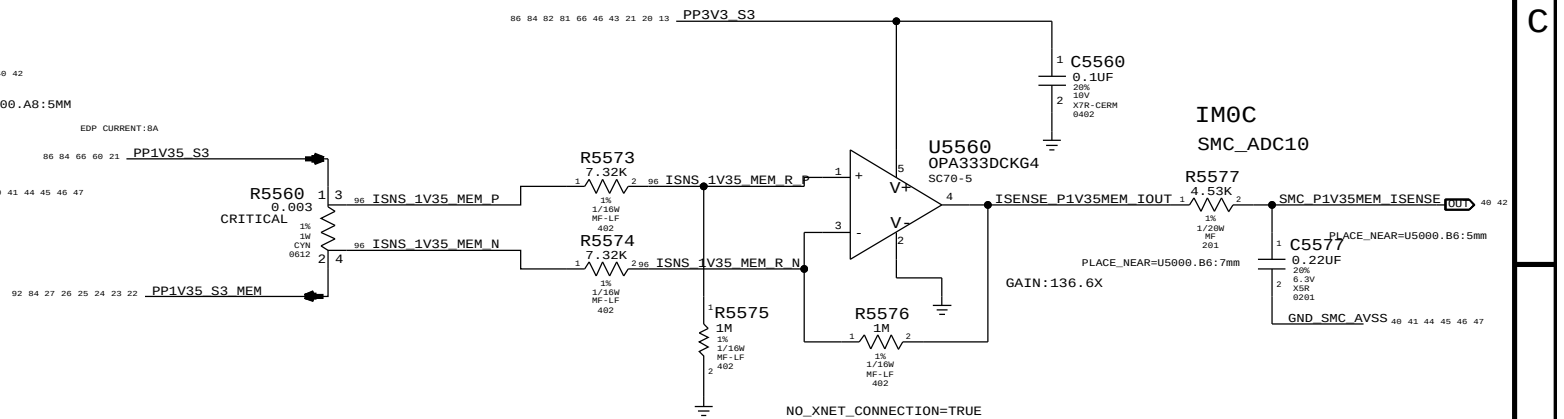
TRACKPAD CURRENT SENSE



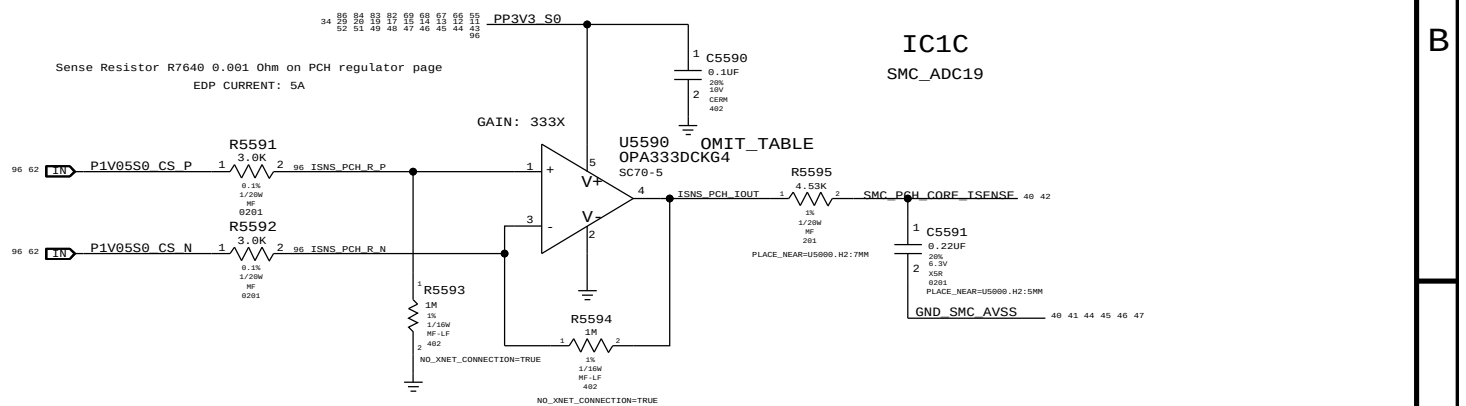
CPU PKG Load Side Current Sense / Filter



DDR3L 1.35V DRAM ONLY CURRENT SENSE / FILTER



PCH CORE CURRENT SENSE



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00107	3	IC, OPAMP, NCS333Q3, SC70-5	U5500, U5540, U5590	CRITICAL	

SYNC MASTER=CLEAN_X4256

SYNC DATE=09/10/2014

Load Side Voltage and Current Sensing

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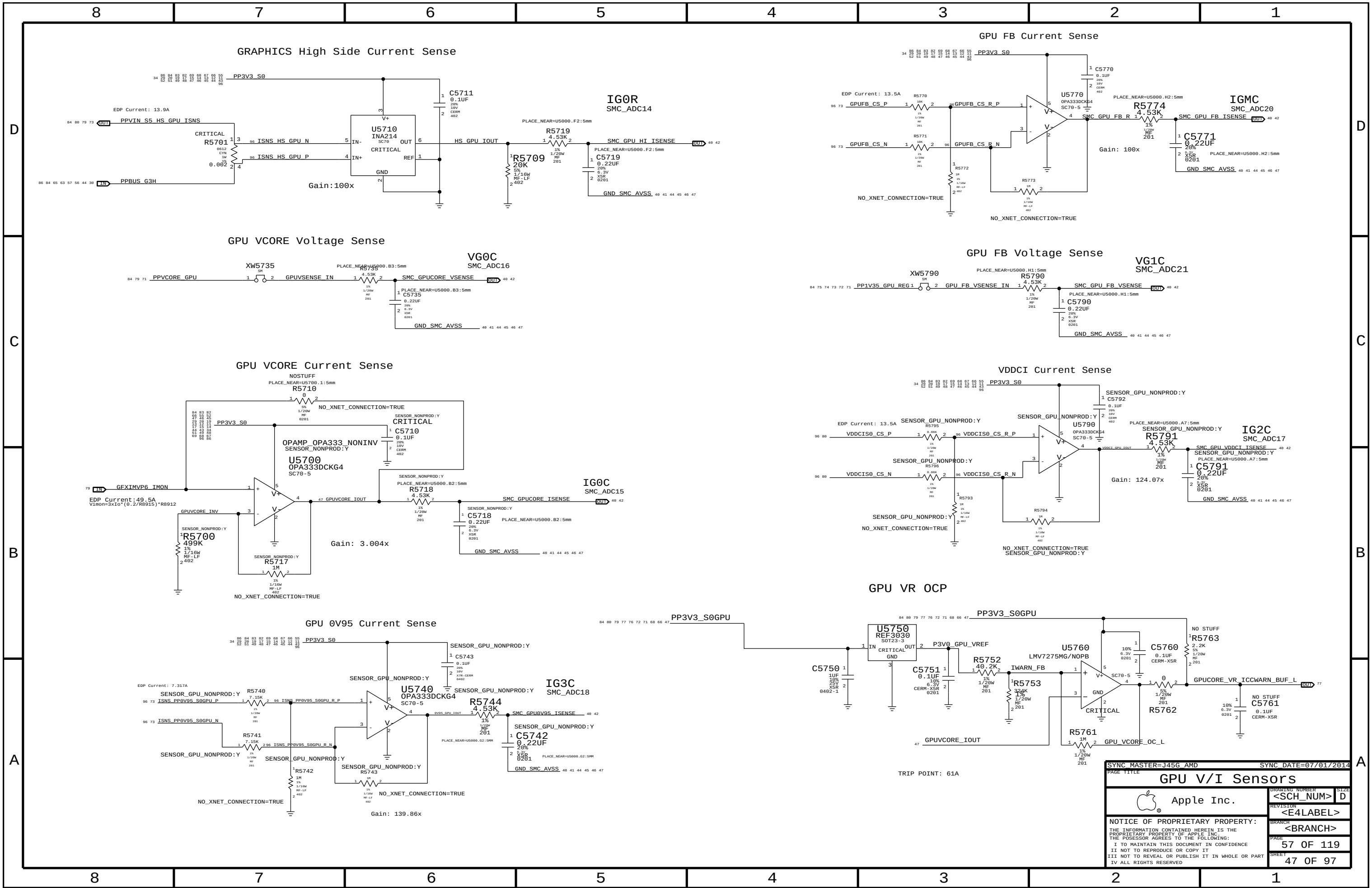
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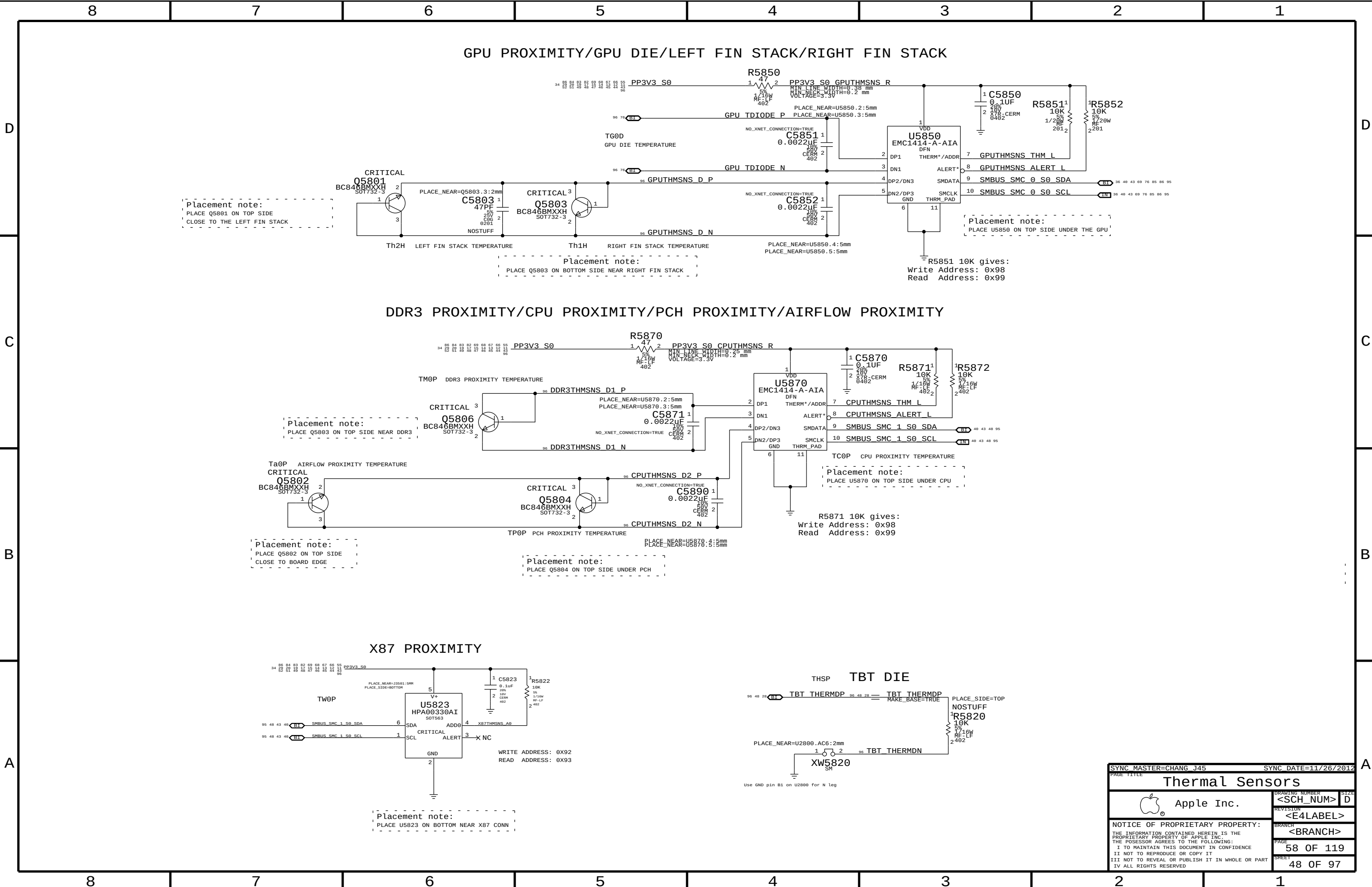
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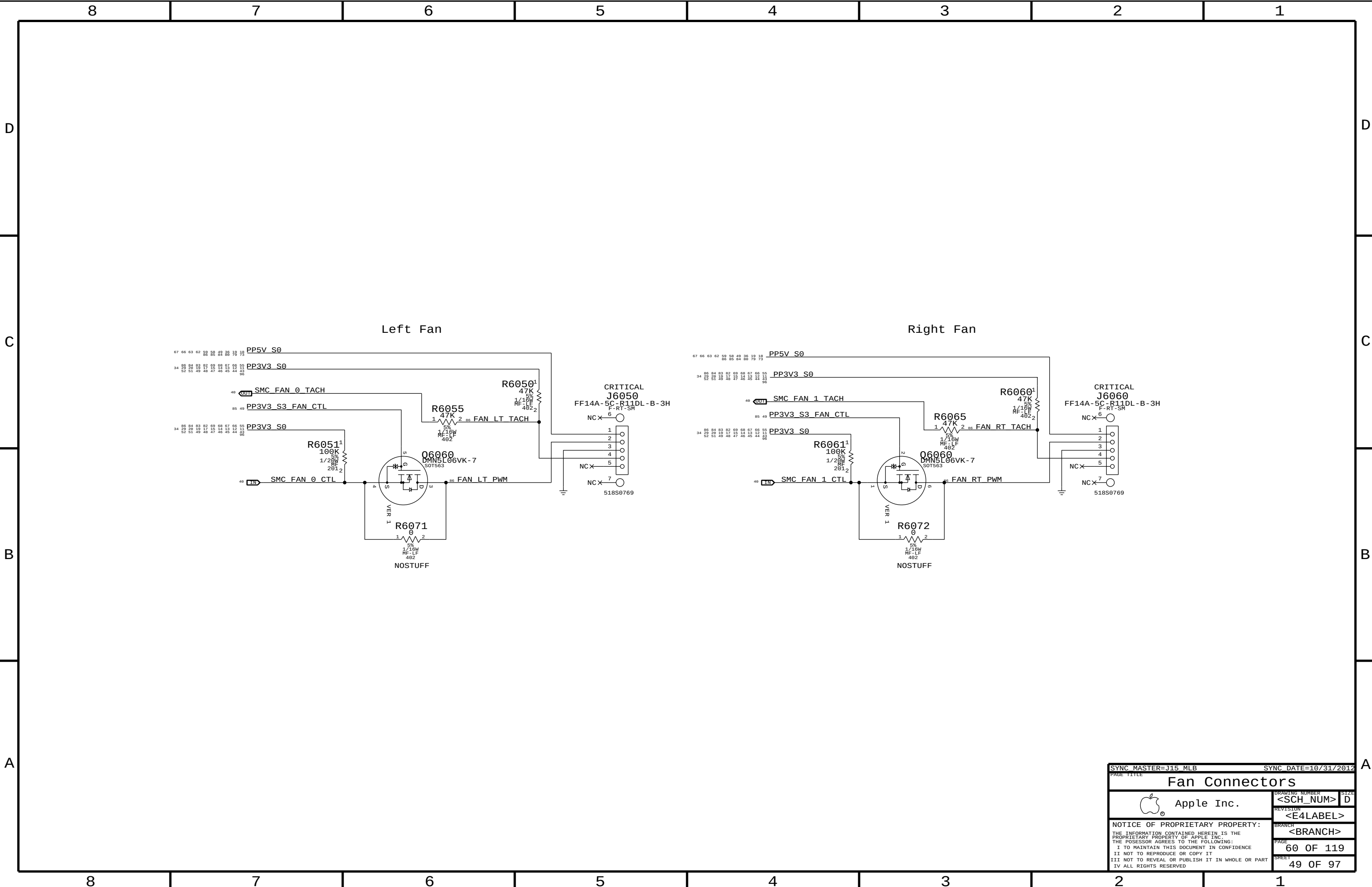
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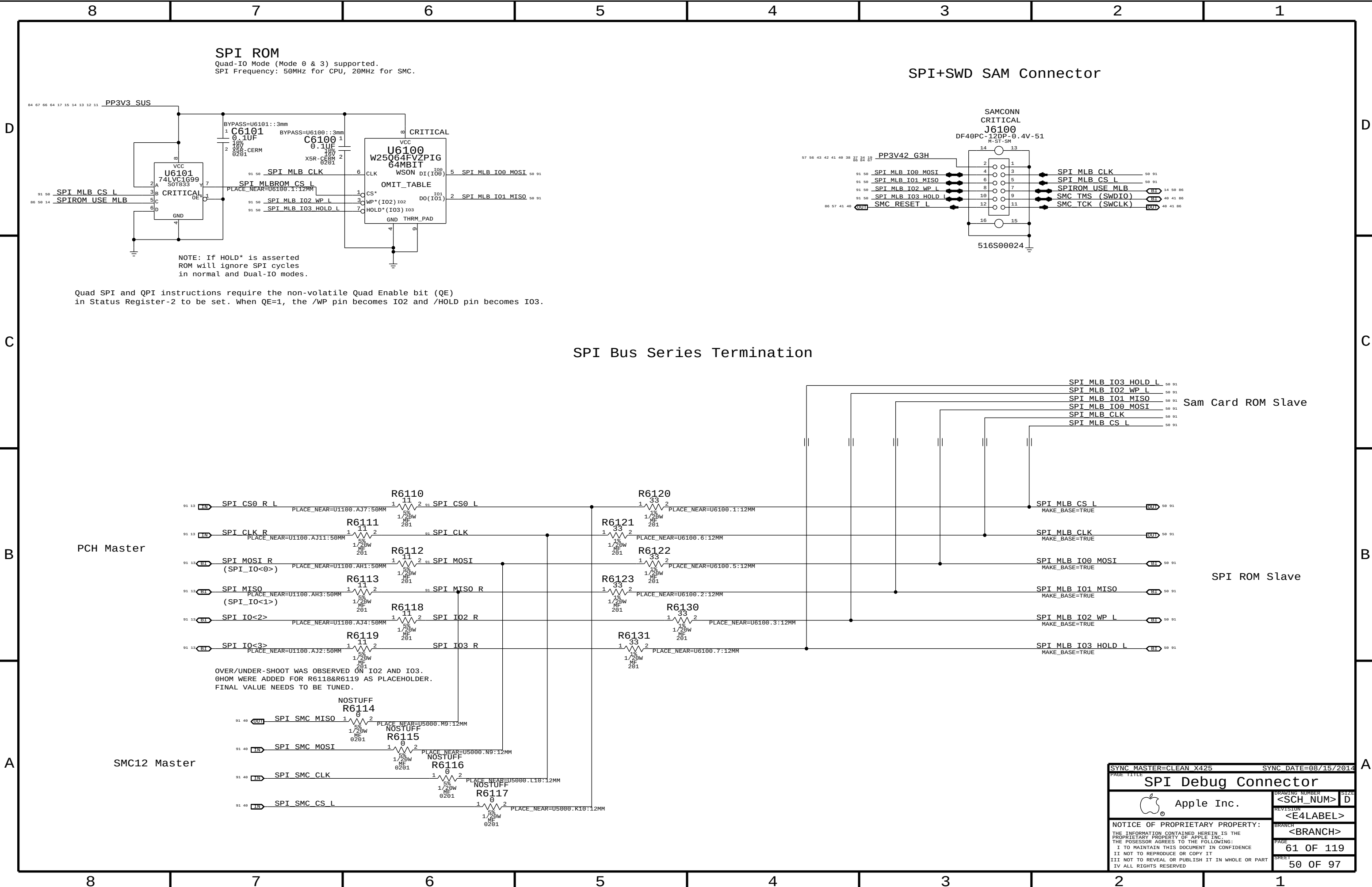
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SPI ROM
Quad-I/O Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.

SPI+SWD SAM Connector

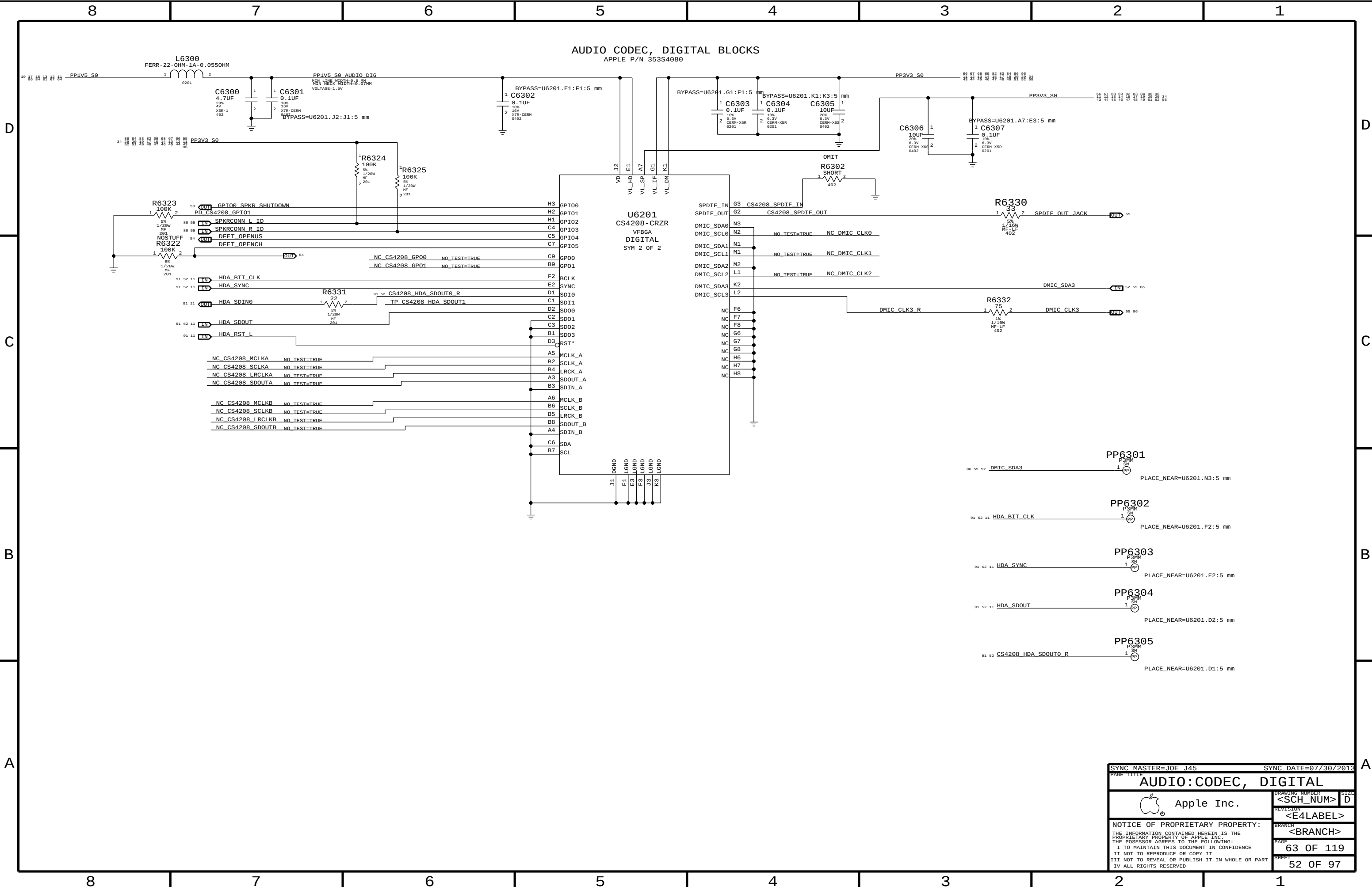
Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE) in Status Register-2 to be set. When QE=1, the /WP pin becomes IO2 and /HOLD pin becomes IO3.

SPI Bus Series Termination

Sam Card ROM Slave

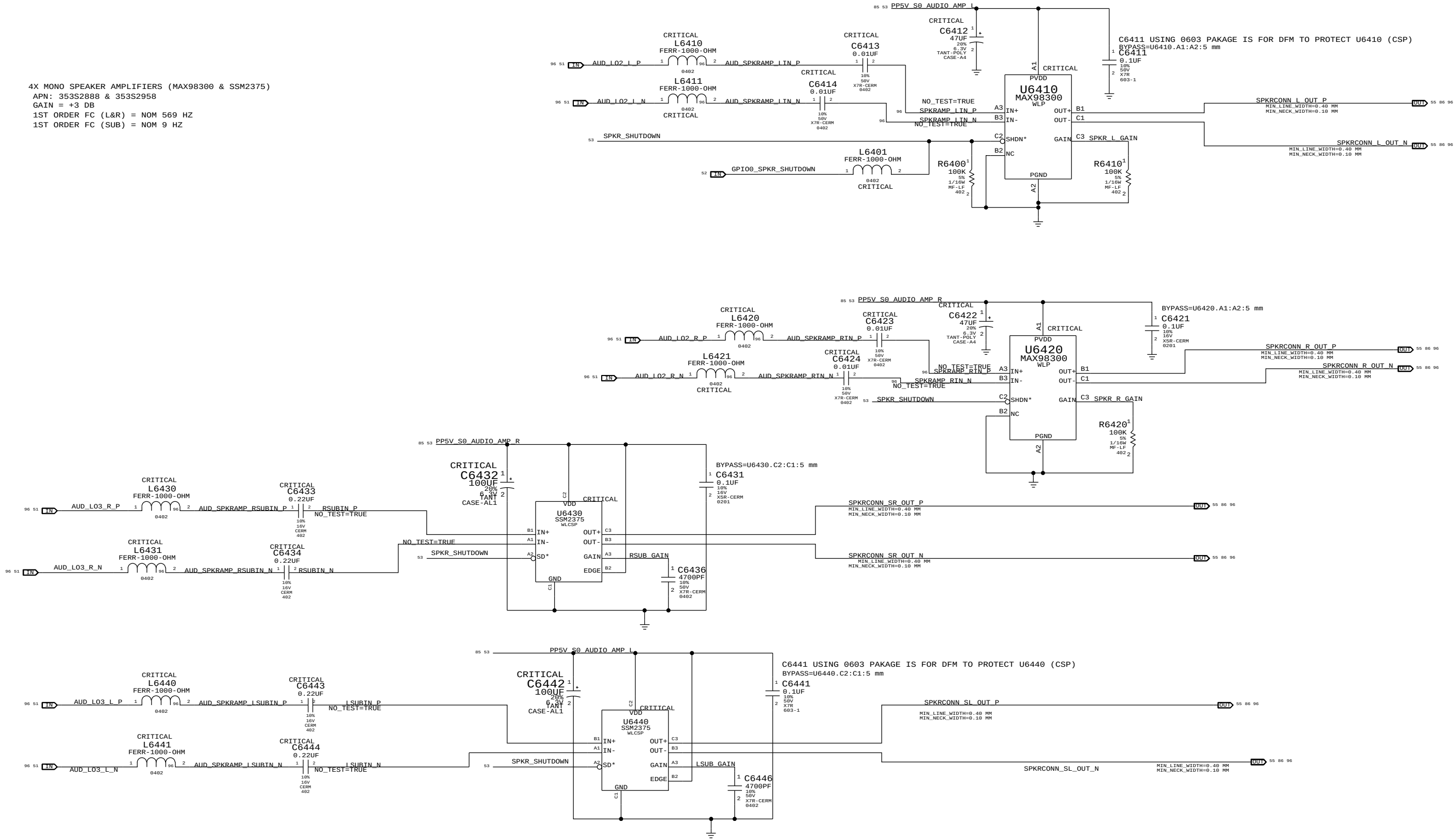
SPI ROM Slave

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		SHEET 50 OF 97	

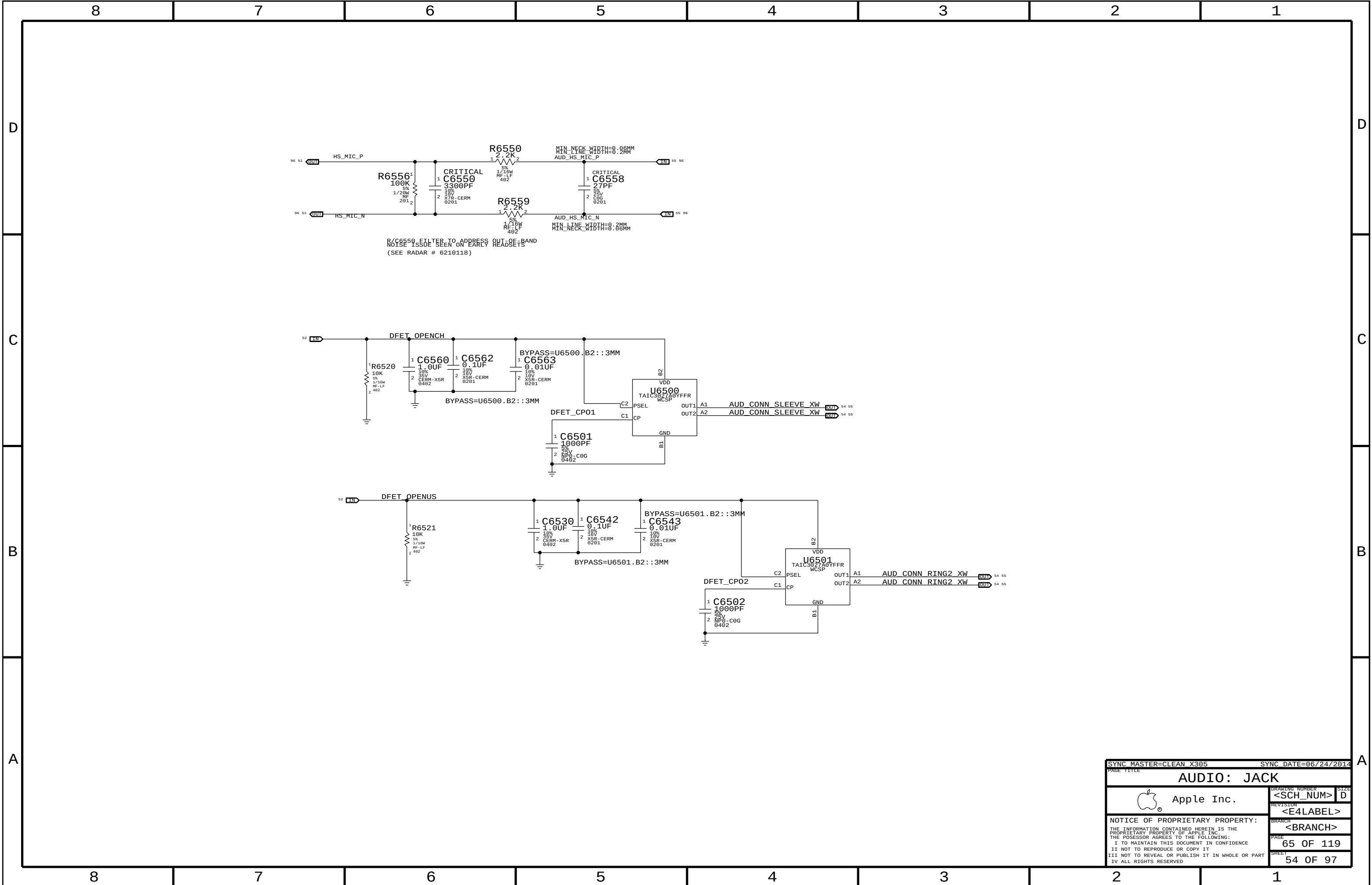


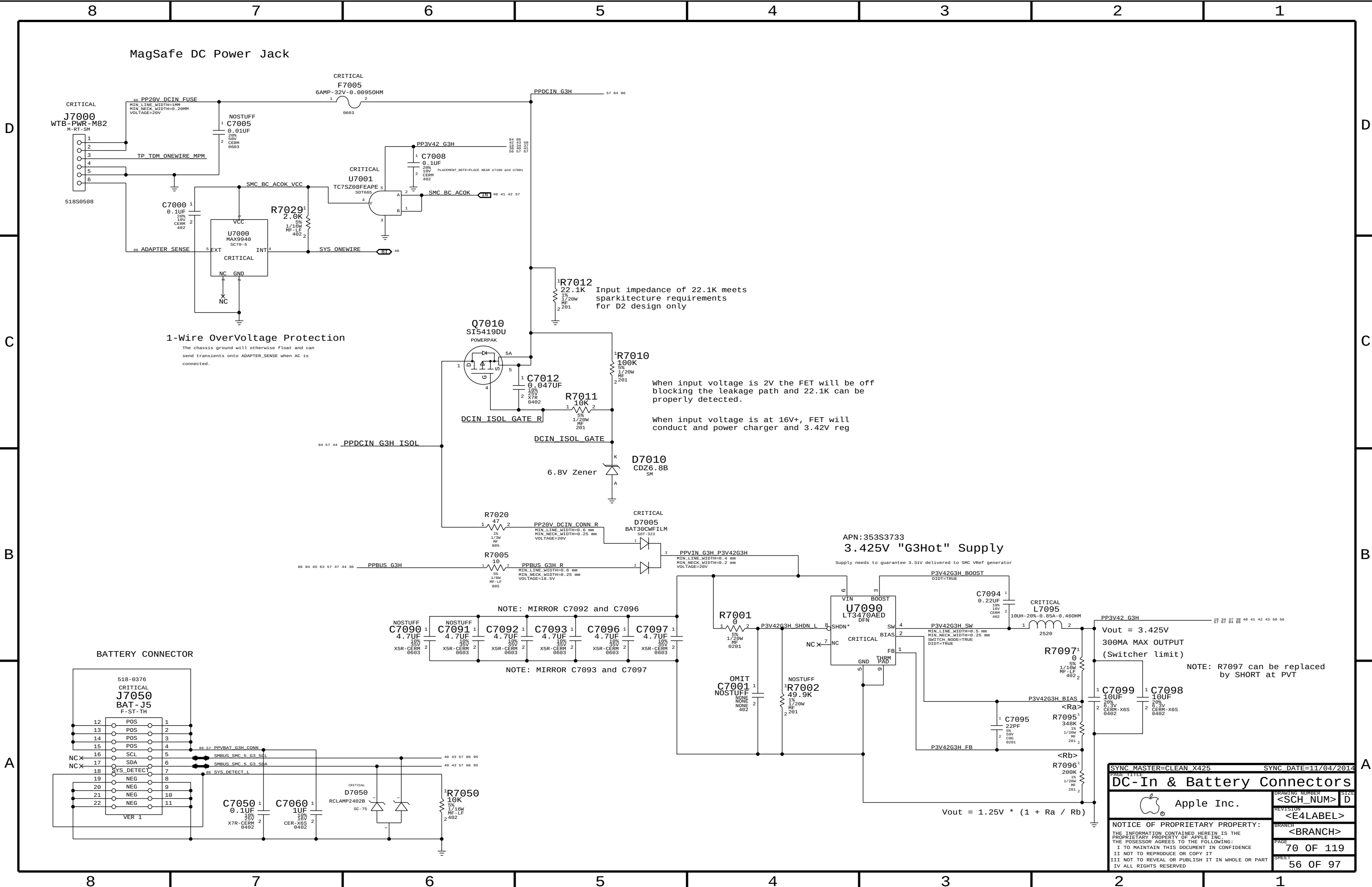
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		<BRANCH>	
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		SHEET	52 OF 97

4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



SYNC MASTER=JOE J45		SYNC DATE=07/30/2013	
PAGE TITLE		AUDIO: SPEAKER AMP	
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
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		PAGE	64 OF 119
		SHEET	53 OF 97





MagSafe DC Power Jack

1-Wire OverVoltage Protection

The chassis ground will otherwise float and can send transients onto ADAPTER_SENSE when AC is connected.

Input impedance of 22.1K meets sparkiterture requirements for D2 design only

When input voltage is 2V the FET will be off blocking the leakage path and 22.1K can be properly detected.

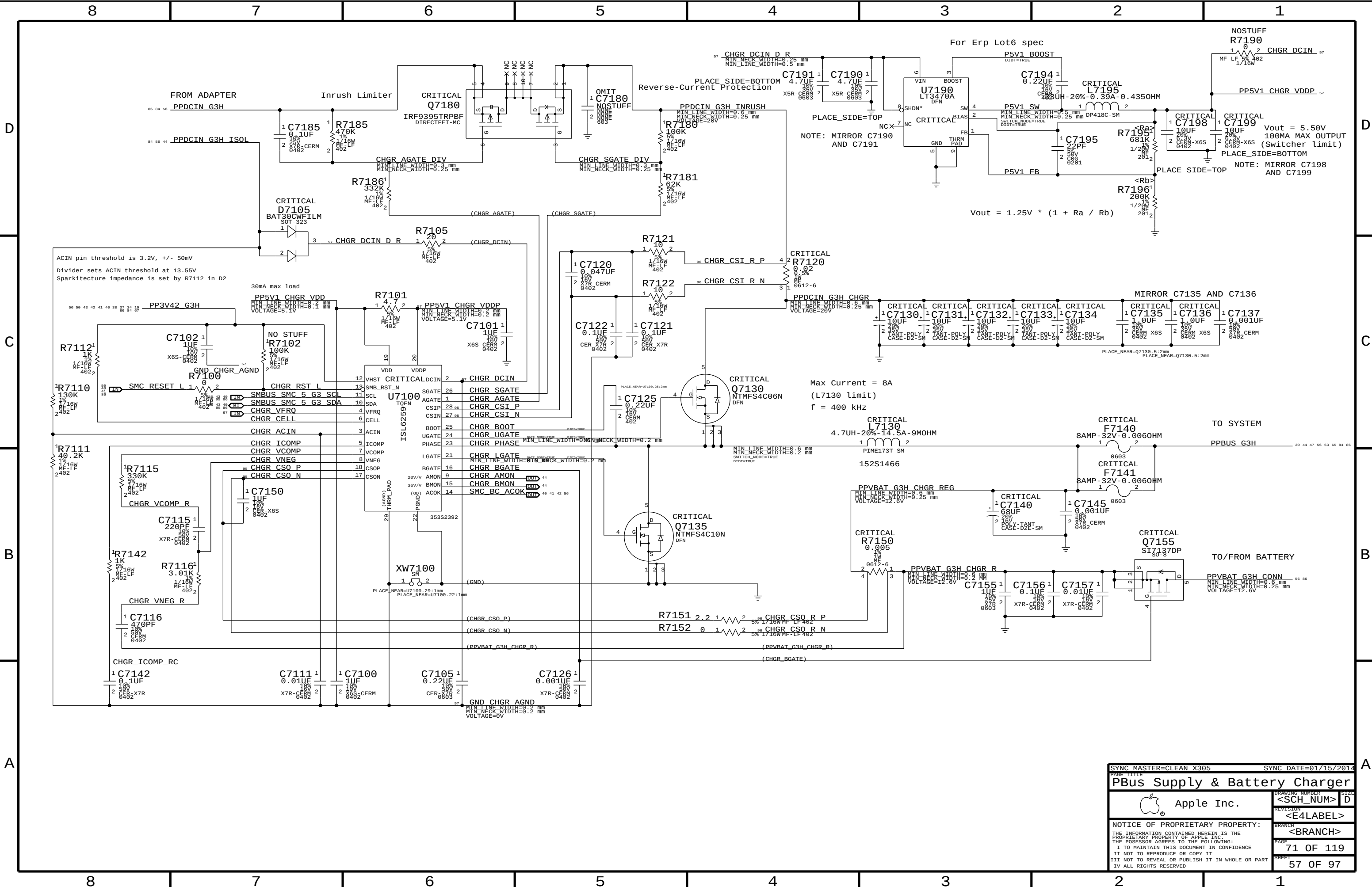
When input voltage is at 16V+, FET will conduct and power charger and 3.42V reg

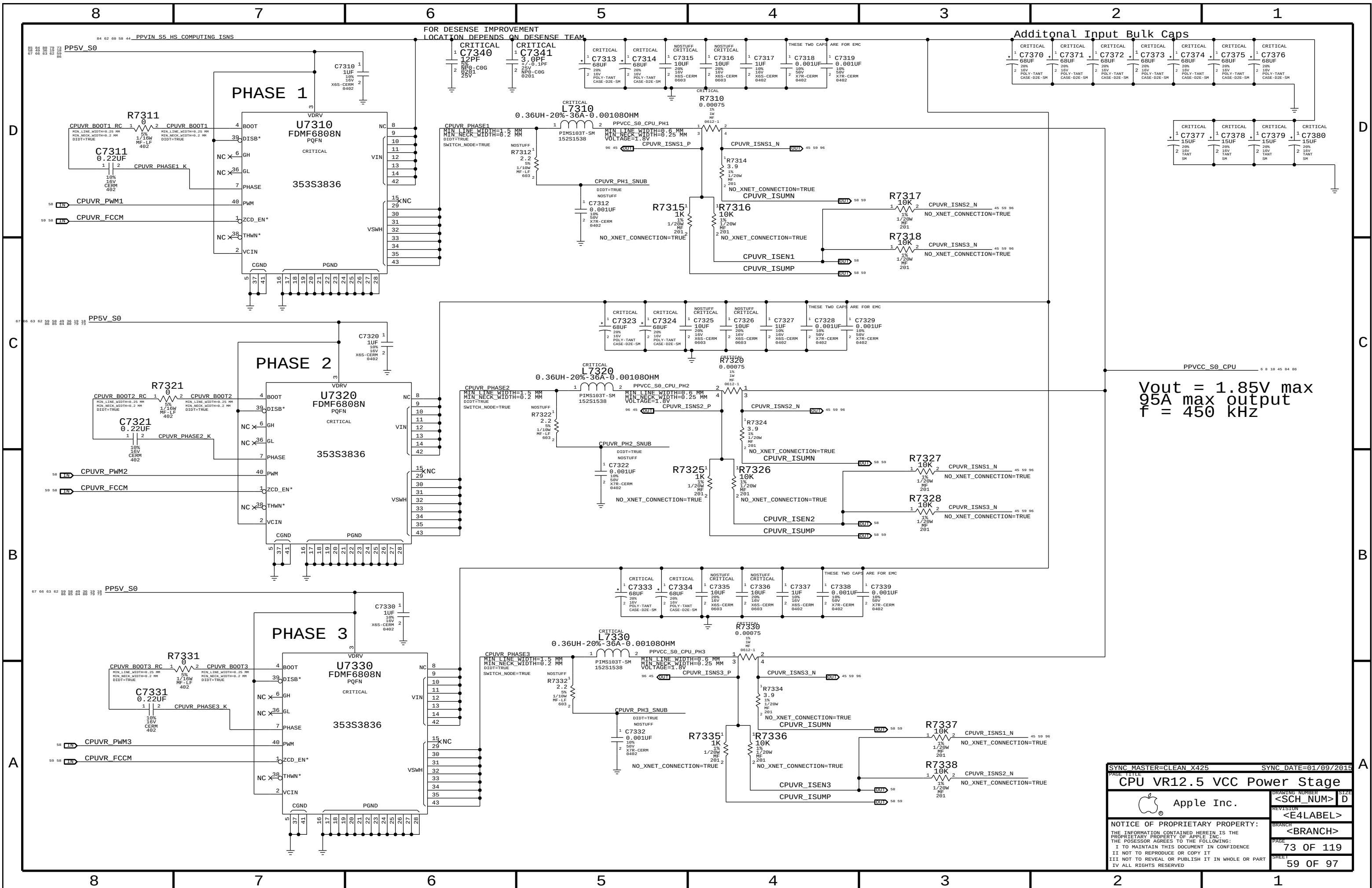
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator

NOTE: R7097 can be replaced by SHORT at PVT

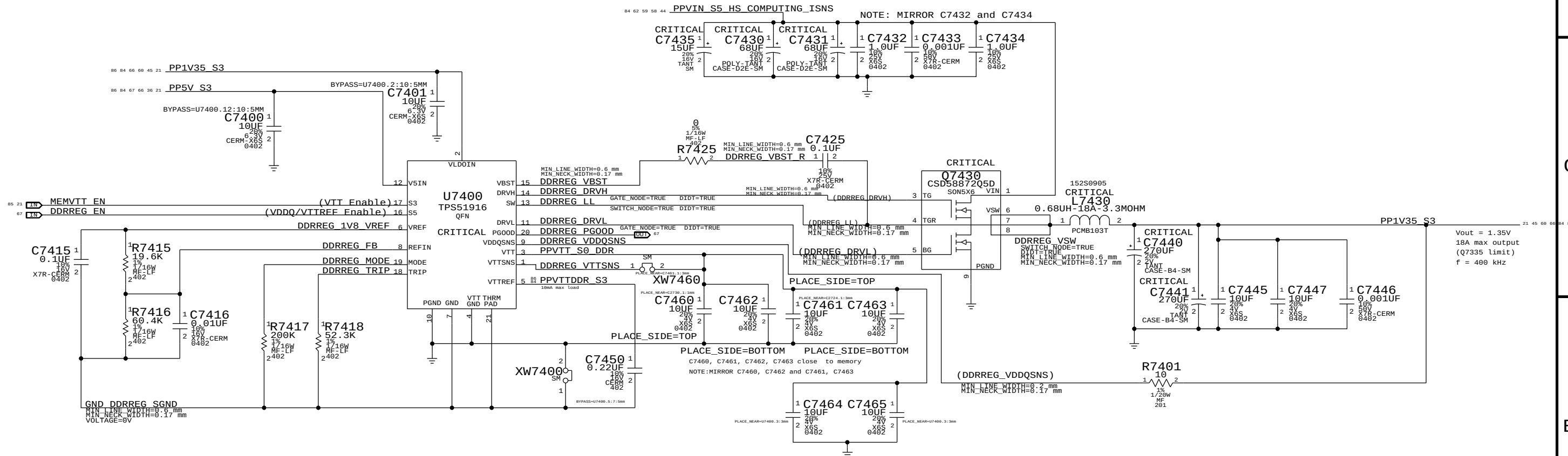
SYNC MASTER=CLEAN X425		SYNC DATE=11/04/2014	
PAGE TITLE			
DC-In & Battery Connectors			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	70 OF 119
		SHEET	56 OF 97



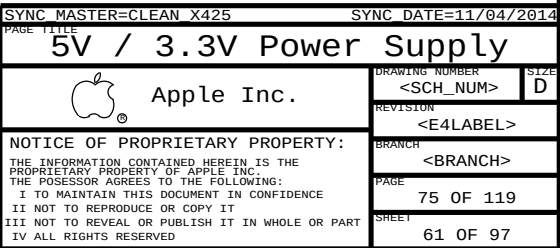


SYNC MASTER=CLEAN X425		SYNC DATE=01/09/2015	
PAGE TITLE			
CPU VR12.5 VCC Power Stage			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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IV ALL RIGHTS RESERVED		59 OF 97	

DDR3L (1V35 S3) REGULATOR



SYNC MASTER=CLEAN X305		SYNC DATE=01/15/2014	
PAGE TITLE			
1.35V DDR3L SUPPLY			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	SIZE D
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		BRANCH <BRANCH>	
		PAGE 74 OF 119	
		SHEET 60 OF 97	



Page Notes

Power aliases required by this page:
- =PPVIN_S0_LCDBKLT (9-12.6V LCD Backlight Input)
- =PPSV_S0_BKLTCTRL (5V Backlight Driver Input)
- =PPSV_S0_KBDLED (5V Keyboard Backlight Input)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7723,R7724		BKLT:PROD

D

C

B

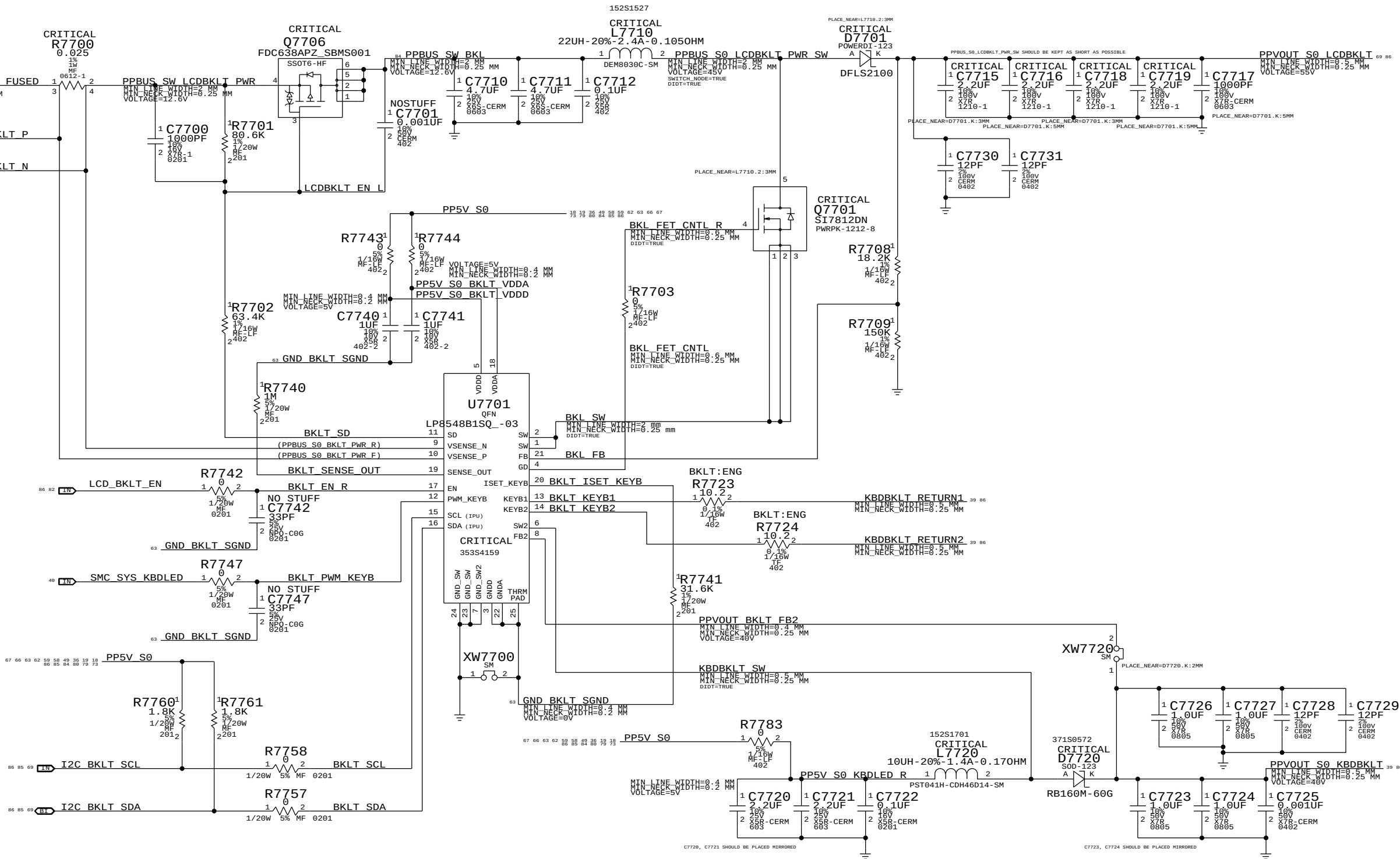
A

D

C

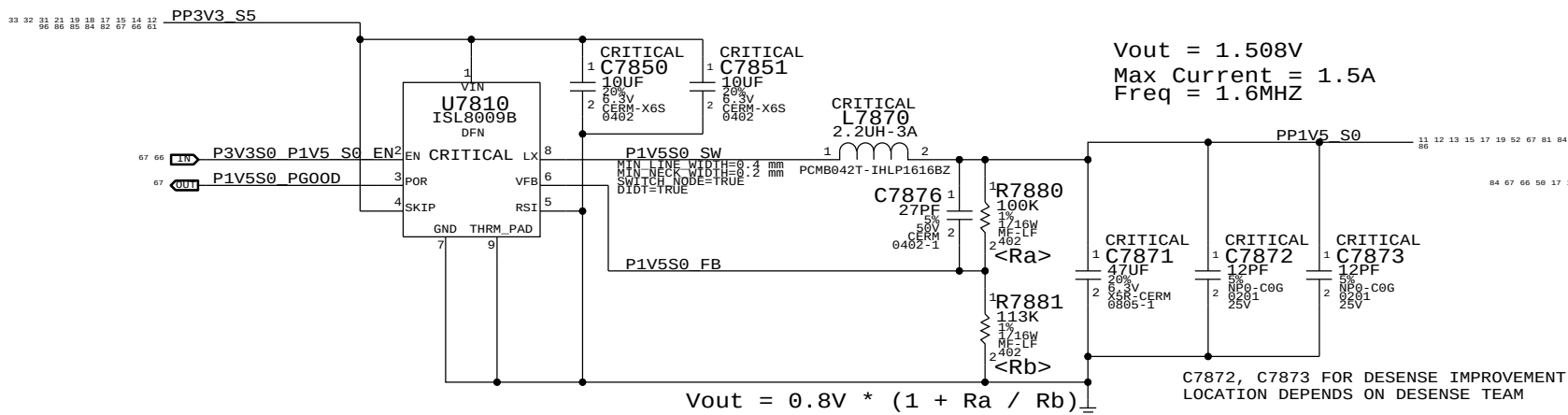
B

A



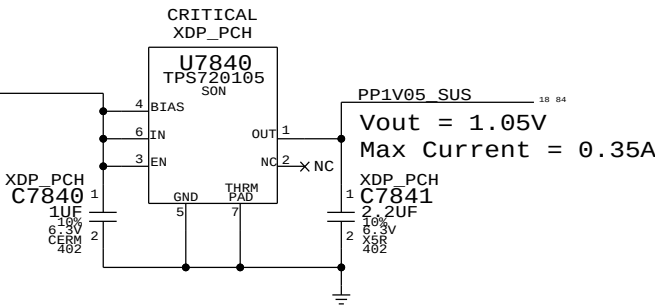
SYNC MASTER=CLEAN X425		SYNC DATE=10/30/2014	
PAGE TITLE			
LCD/KBD Backlight Driver			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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		PAGE	77 OF 119
		SHEET	63 OF 97

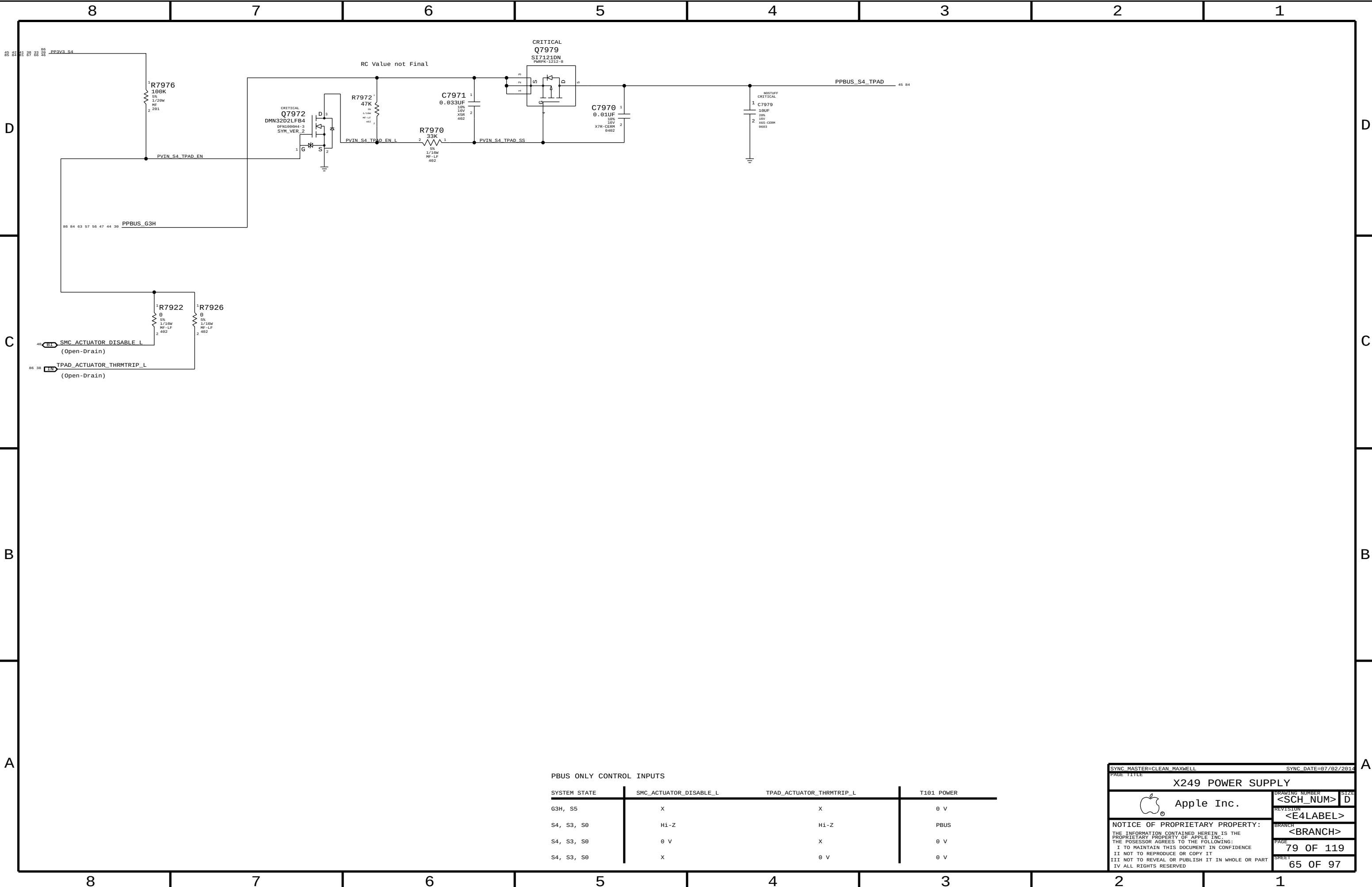
1.5V S0 Regulator



1.05V SUS LD0

Lynx Point-H requires JTAG pull-ups to be powered at 1.05V in SUS.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V SUS, which burns 100mW in all S-states.





PBUS ONLY CONTROL INPUTS

SYSTEM STATE	SMC_ACTUATOR_DISABLE_L	TPAD_ACTUATOR_THRMTRIP_L	T101 POWER
G3H, S5	X	X	0 V
S4, S3, S0	Hi-Z	Hi-Z	PBUS
S4, S3, S0	0 V	X	0 V
S4, S3, S0	X	0 V	0 V

SYNC MASTER=CLEAN MAXWELL

SYNC DATE=07/02/2014

PAGE TITLE

X249 POWER SUPPLY

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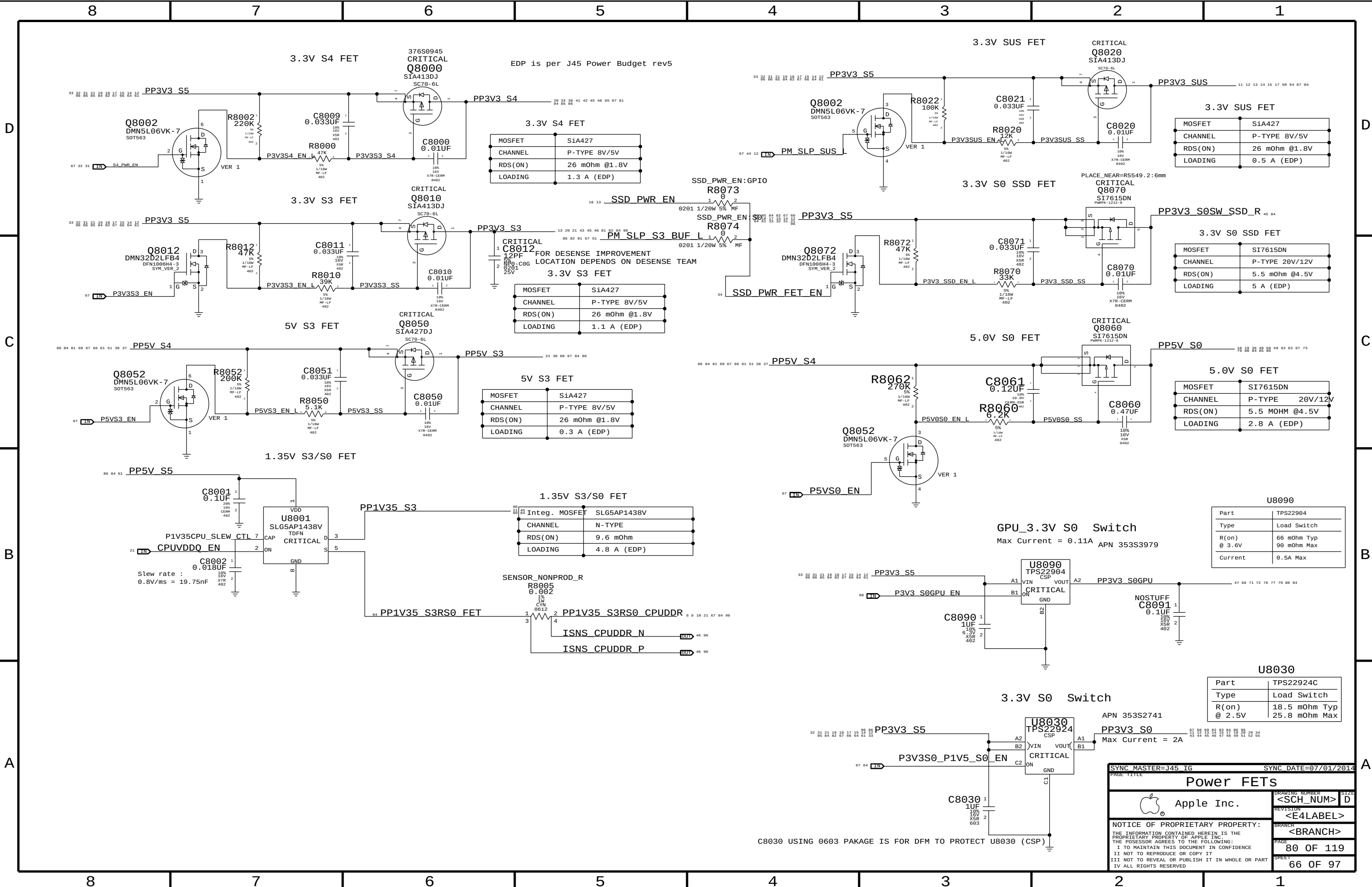
<BRANCH>

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EDP is per J45 Power Budget rev5

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.3 A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.1 A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.3 A (EDP)

MOSFET	SLG5AP1438V
CHANNEL	N-TYPE
RDS(ON)	9.6 mOhm
LOADING	4.8 A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.5 A (EDP)

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 mOhm @4.5V
LOADING	5 A (EDP)

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 MOHM @4.5V
LOADING	2.8 A (EDP)

Part	TPS22904
Type	Load Switch
R(on) @ 3.6V	66 mOhm Typ 90 mOhm Max
Current	0.5A Max

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SYNC MASTER=J45 IG

SYNC DATE=07/01/2014

Power FETs

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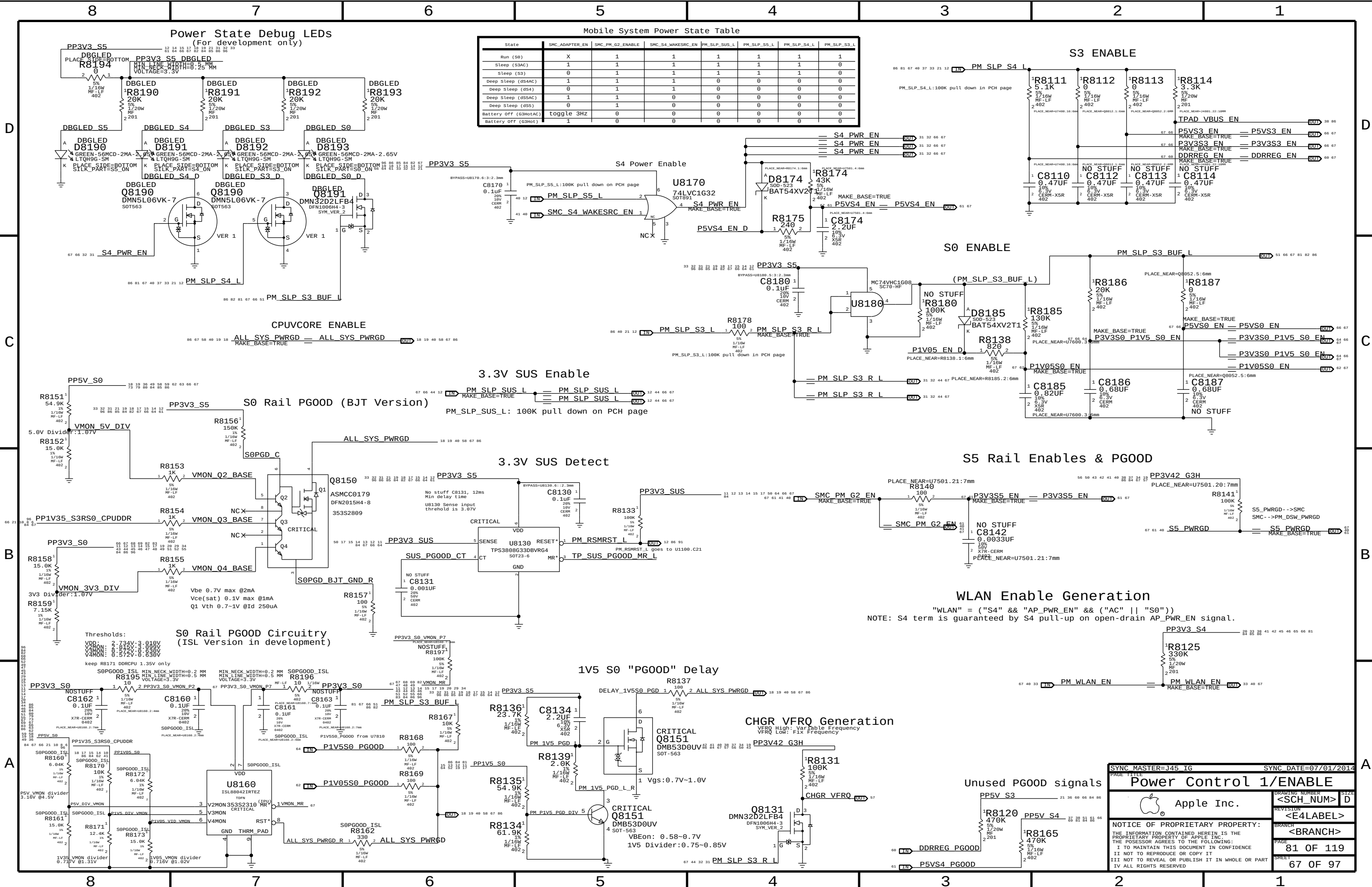
PAGE

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SHEET

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C8030 USING 0603 PACKAGE IS FOR DFM TO PROTECT U8030 (CSP)



Mobile System Power State Table							
State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SLP_SUS_L	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (dS4AC)	1	1	1	0	0	0	0
Deep Sleep (dS4)	0	1	1	0	0	0	0
Deep Sleep (dS5AC)	1	1	0	0	0	0	0
Deep Sleep (dS5)	1	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC MASTER=J45 IG

SYNC DATE=07/01/2014

Power Control 1/ENABLE

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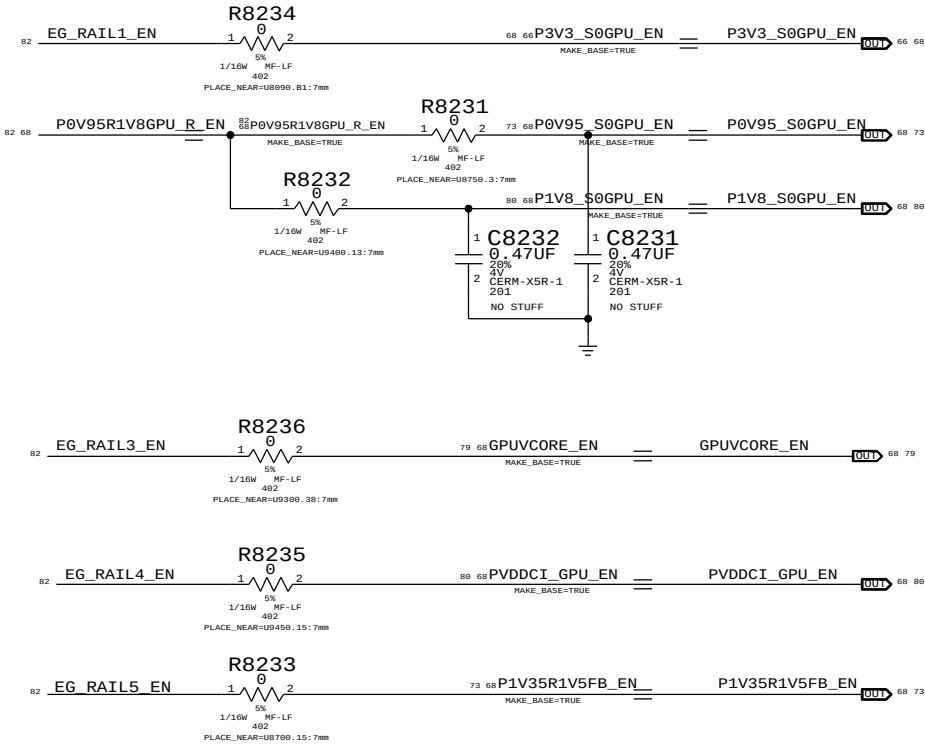
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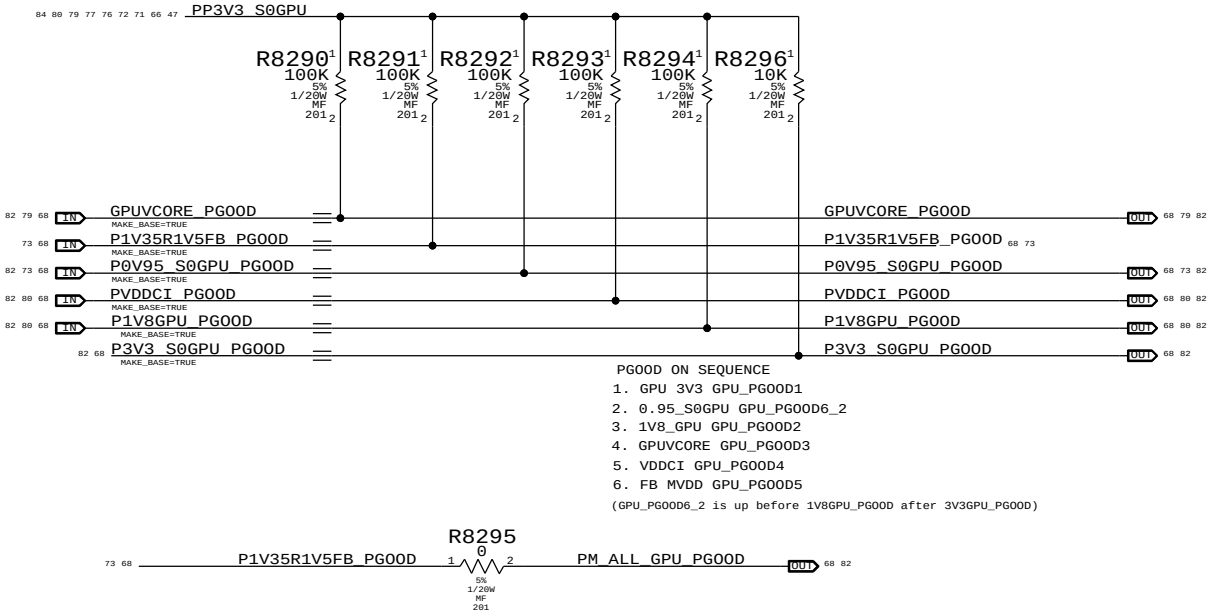
SHEET
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GPU Rails Power UP Sequencing

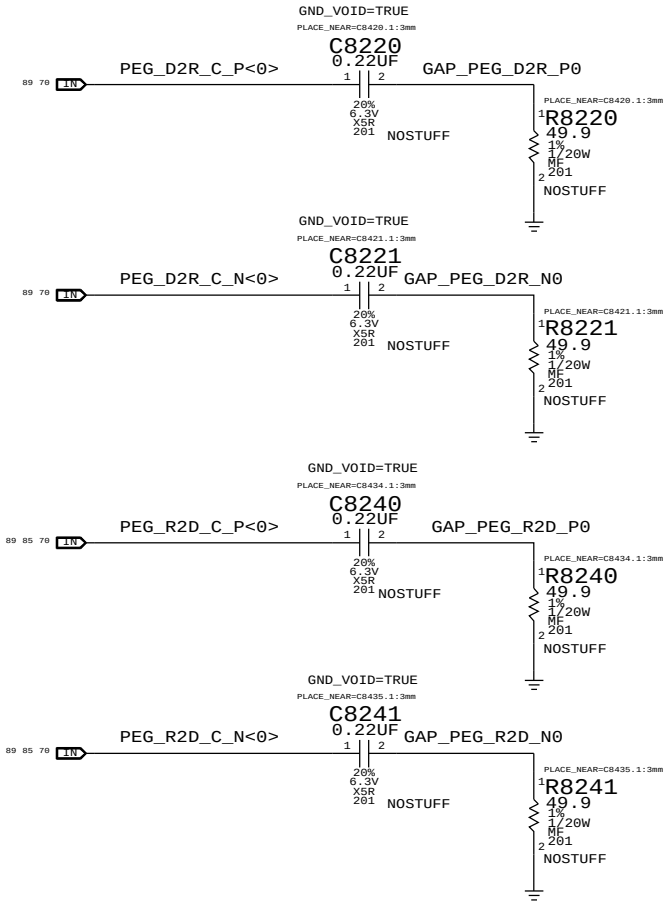
- Venus GPU requires rails to come up in the following order:
- 1) GPU_3.3V
 - 2) GPU_0V95 (BIF_VDDC) & GPU_1V8 (VDD_CT)
 - 3) GPUVCORE
 - 4) VDDCI
 - 5) FB VRAM MVDD



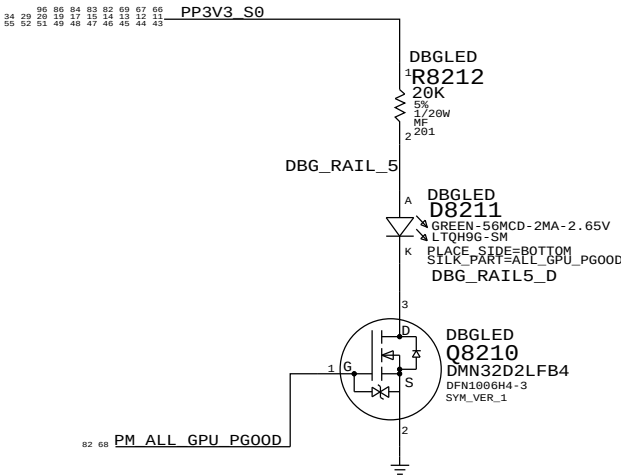
EXT GPU PWRGD Pullup



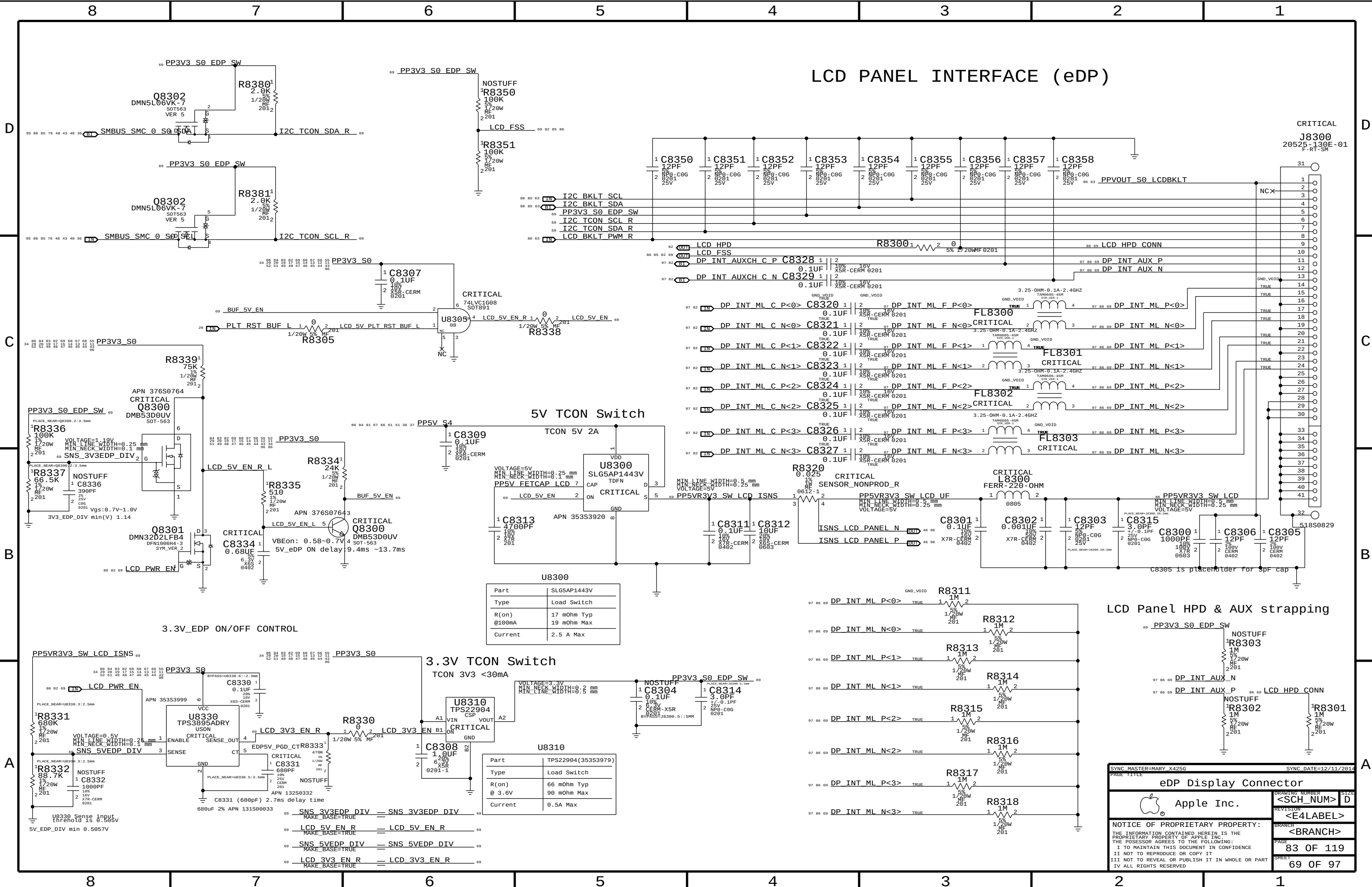
PCIE TEST STRUCTURES (FOR LAB USE)
Pending Layout. Can add more.



Power State Debug LEDs
(For development only)



SYNC MASTER=MARY_X425G		SYNC DATE=89/11/2814	
PAGE TITLE			
Power Sequencing EG/PG00D			
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		REVISION	
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LCD PANEL INTERFACE (eDP)

CRITICAL
J8300
20525-130E-01
F-RT-SM

5V TCON Switch

Part	SLG5AP1443V
Type	Load Switch
R(on) @100mA	17 mOhm Typ 19 mOhm Max
Current	2.5 A Max

3.3V TCON Switch

Part	TPS22904(353S3979)
Type	Load Switch
R(on) @ 3.6V	66 mOhm Typ 90 mOhm Max
Current	0.5A Max

LCD Panel HPD & AUX strapping

SYNC MASTER=MARY_X425G

SYNC DATE=12/11/2014

PAGE TITLE

eDP Display Connector

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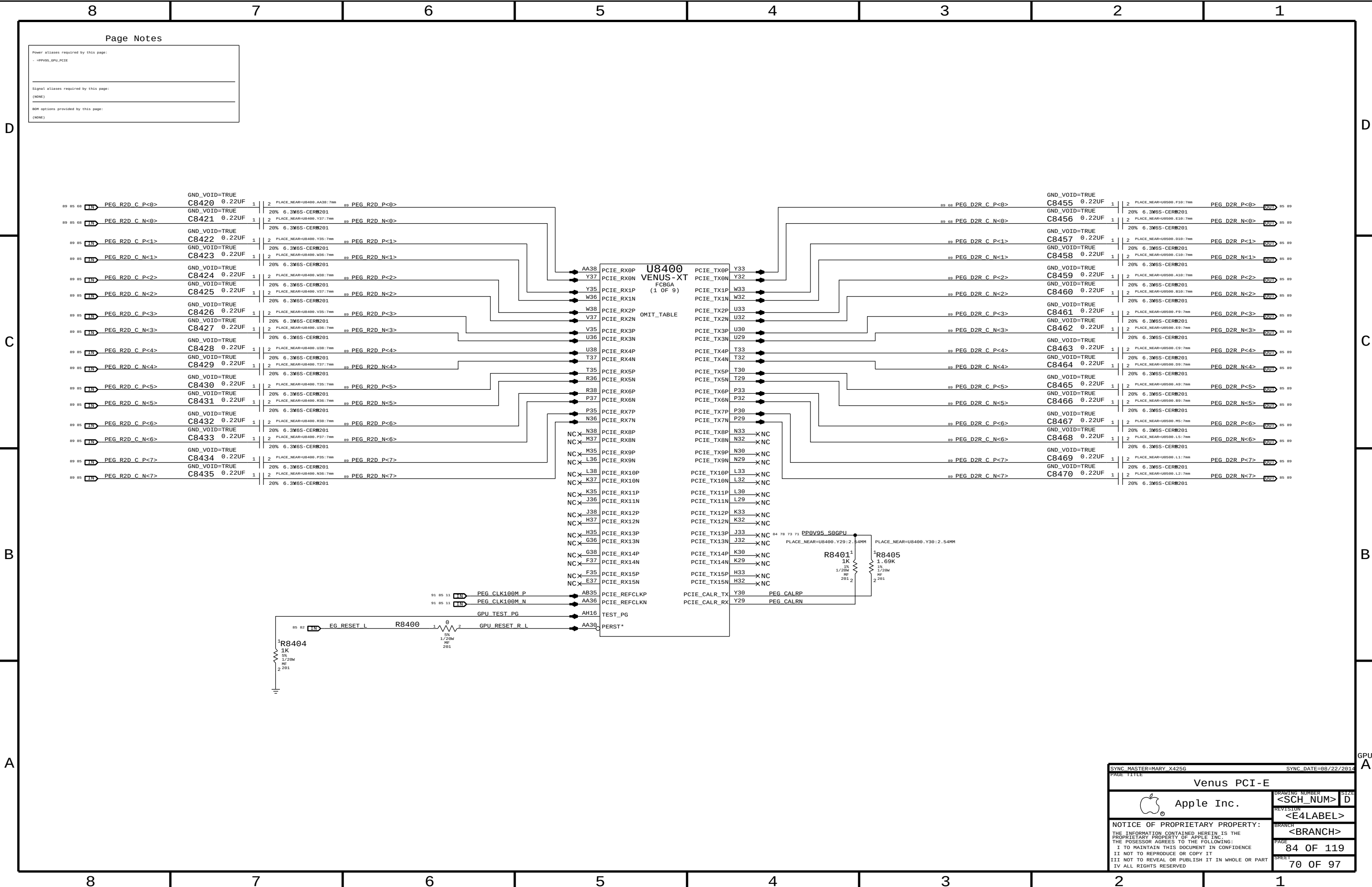
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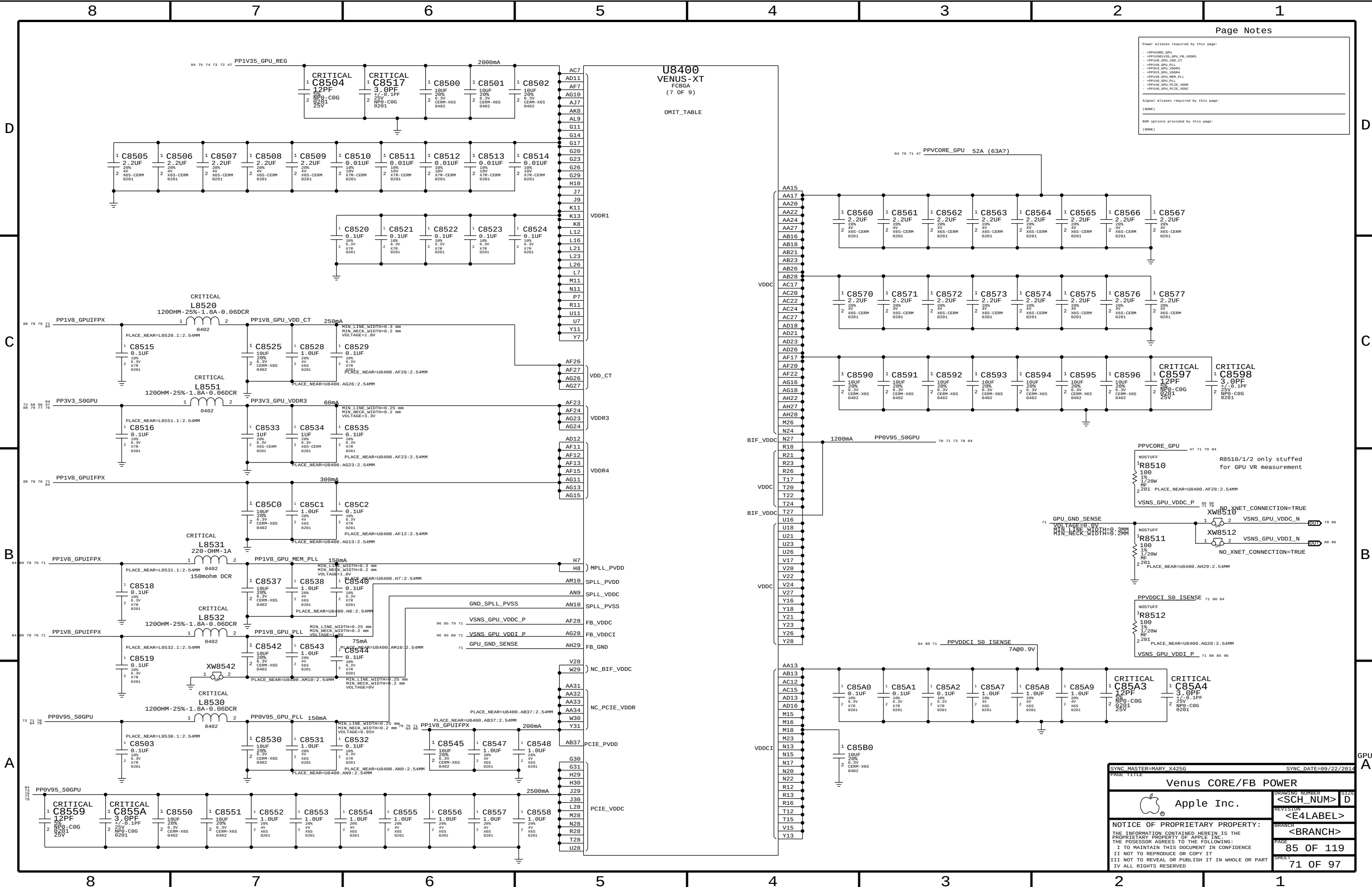


Page Notes

Power aliases required by this page:
- ->PP0V95_GPU_PCIE

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



Page Notes

Power aliases required by this page:

- PPVCORE_GPU
- PP1V35GPU_VDD_FB_VDD01
- PP1V8_GPU_VDD_CT
- PP3V3_GPU_PLL
- PP3V3_GPU_VDD03
- PP3V3_GPU_VDD04
- PP1V8_GPU_MEM_PLL
- PP1V8_GPU_PLL
- PP1V8_GPU_PVDD_P
- PP1V8_GPU_PVDD_P
- PP1V8_GPU_PVDD_P

Signal aliases required by this page:

(NONE)

BOX options provided by this page:

(NONE)

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SYNC MASTER=MARY_X4256

SYNC DATE=89/22/2814

Venus CORE/FB POWER

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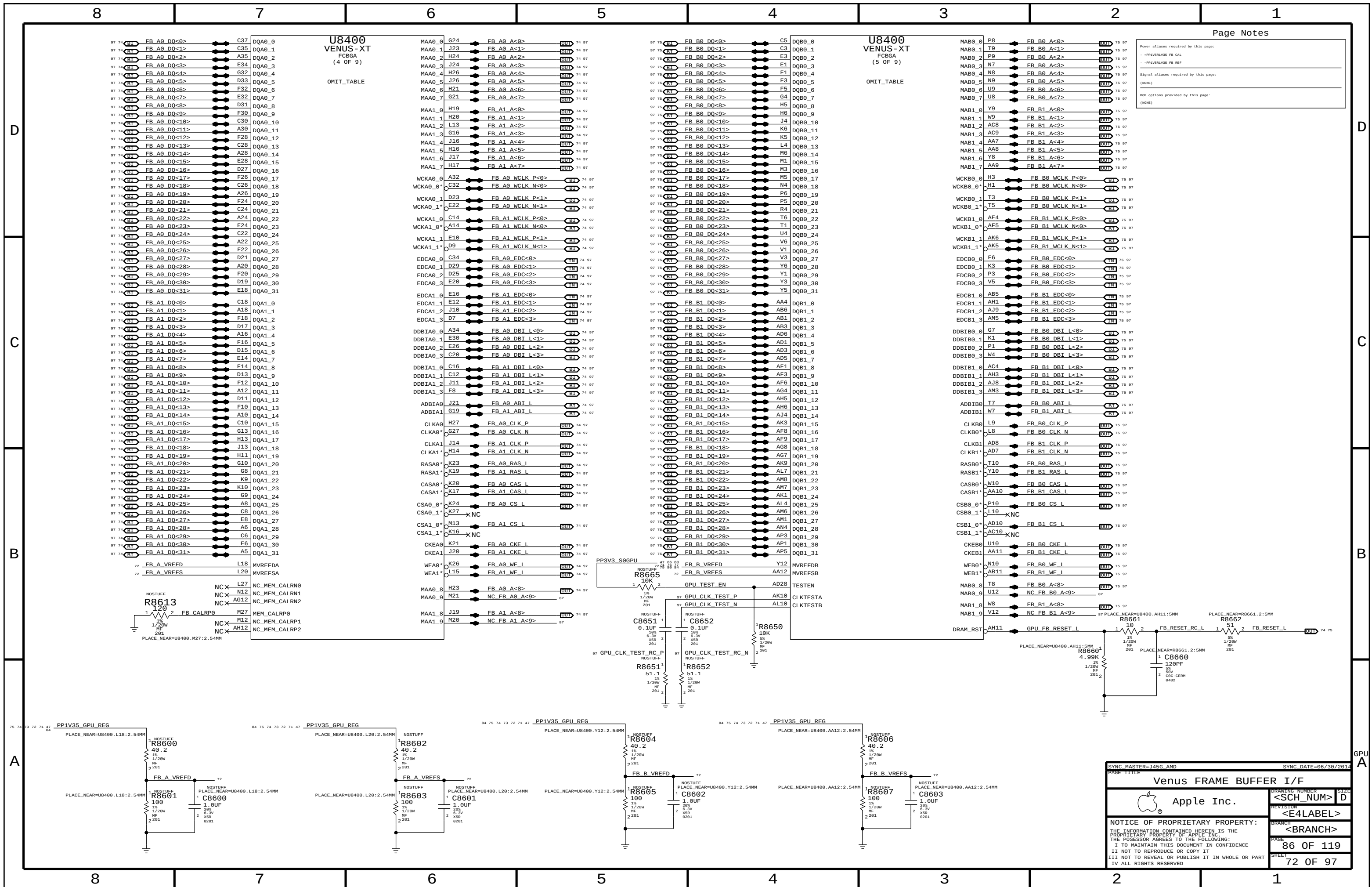
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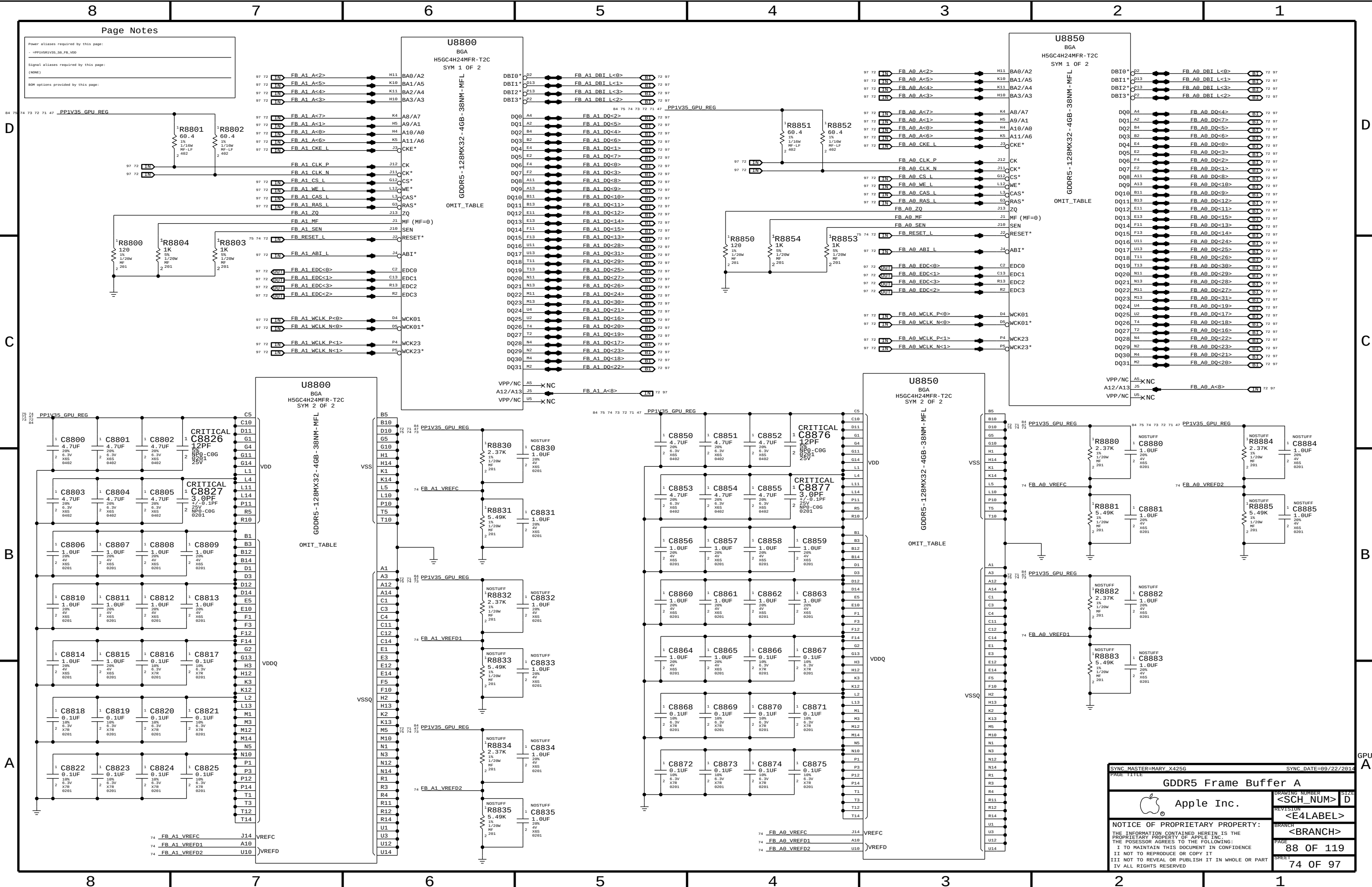
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Page Notes

Power aliases required by this page:

- PP1V35V35_S8_FB_VDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

U8800
BGA
H5GC4H24MFR-T2C
SYM 1 OF 2

U8850
BGA
H5GC4H24MFR-T2C
SYM 1 OF 2

OMIT_TABLE

OMIT_TABLE

U8800
BGA
H5GC4H24MFR-T2C
SYM 2 OF 2

U8850
BGA
H5GC4H24MFR-T2C
SYM 2 OF 2

OMIT_TABLE

OMIT_TABLE

SYNC MASTER=MARY_X425G

SYNC DATE=89/22/2014

PAGE TITLE

GDDR5 Frame Buffer A

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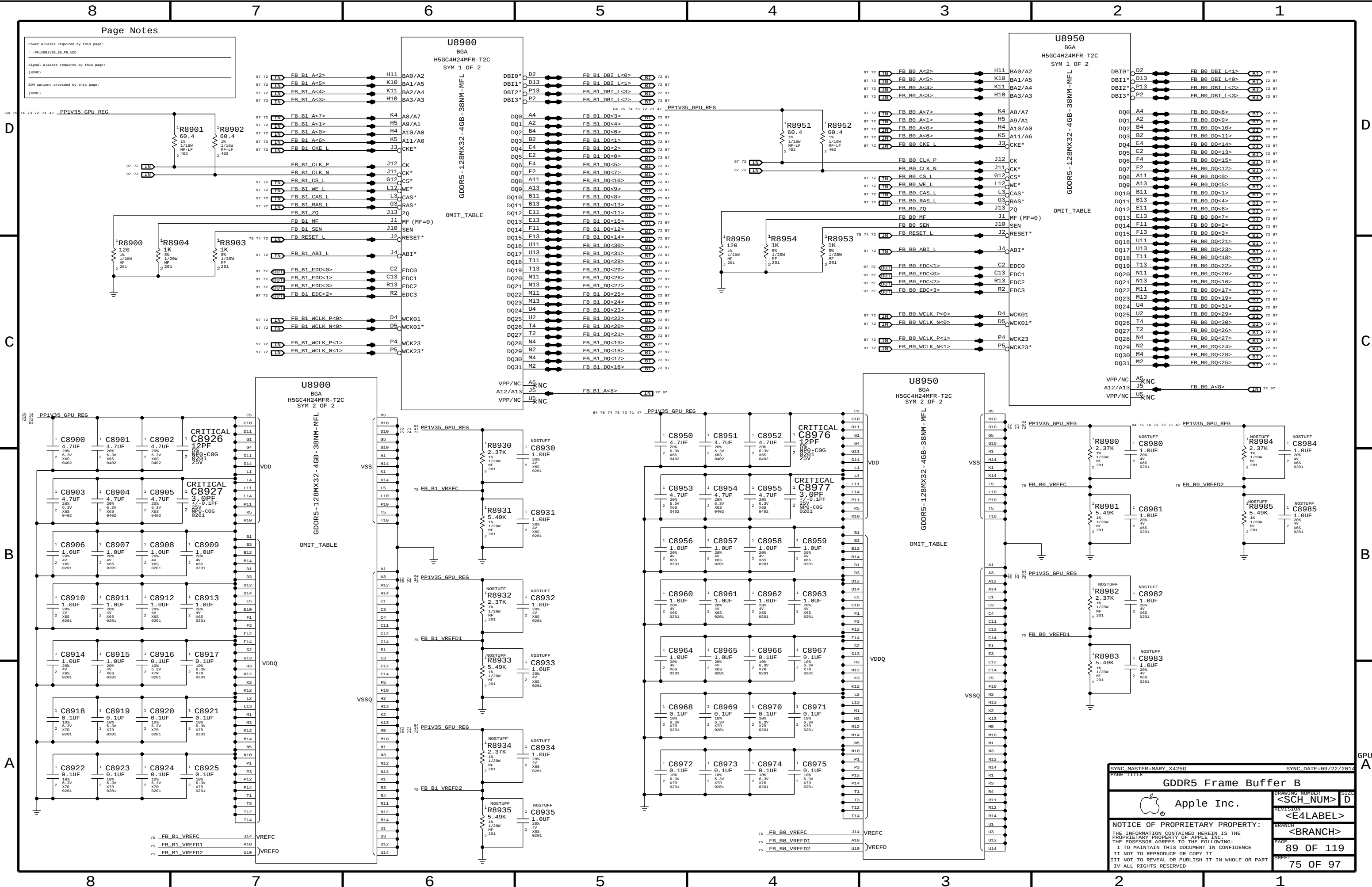
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Page Notes

Power aliases required by this page:

- PP1V35_VB_FB_VDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

SYNC MASTER=MARY_X425G

SYNC DATE=89/22/2014

PAGE TITLE

GDDR5 Frame Buffer B

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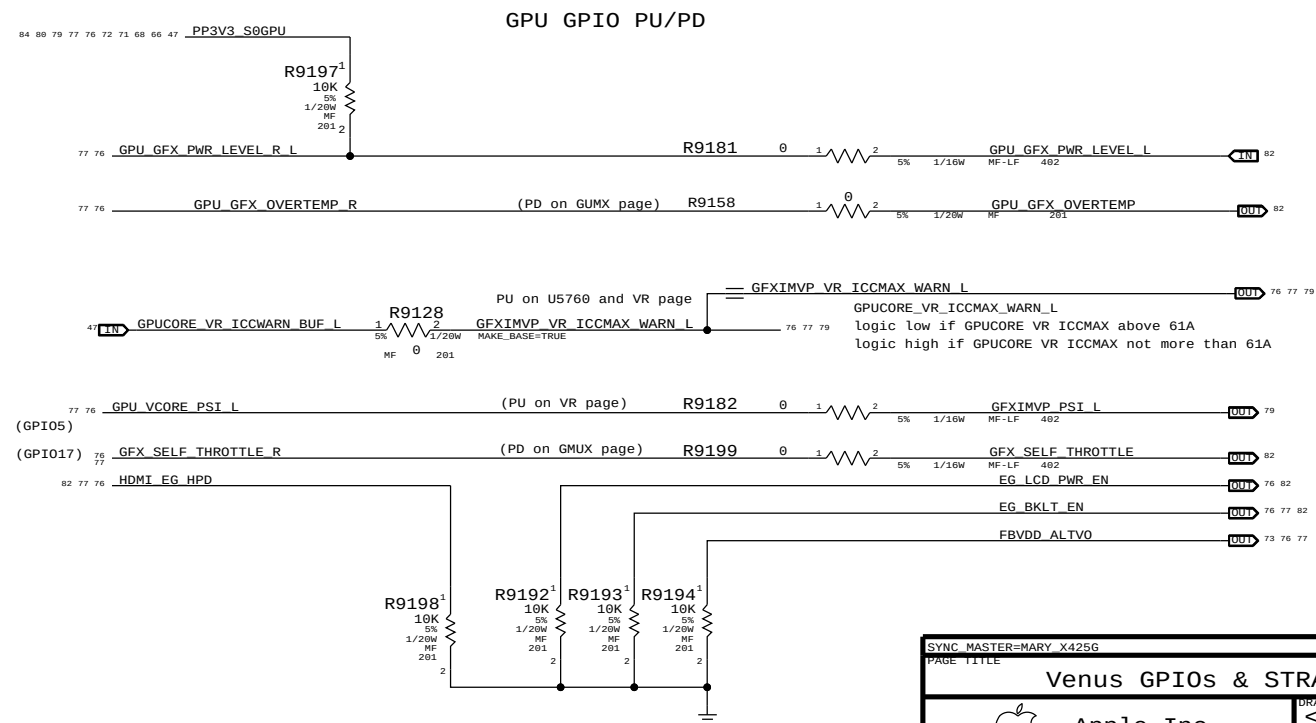
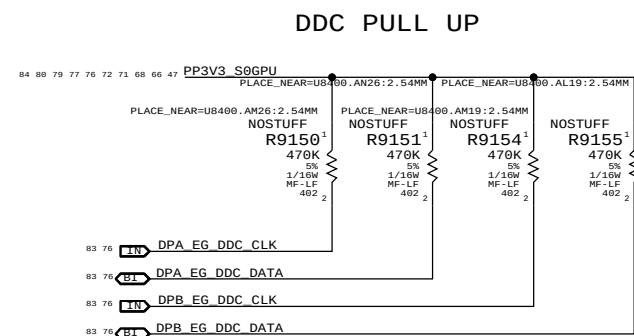
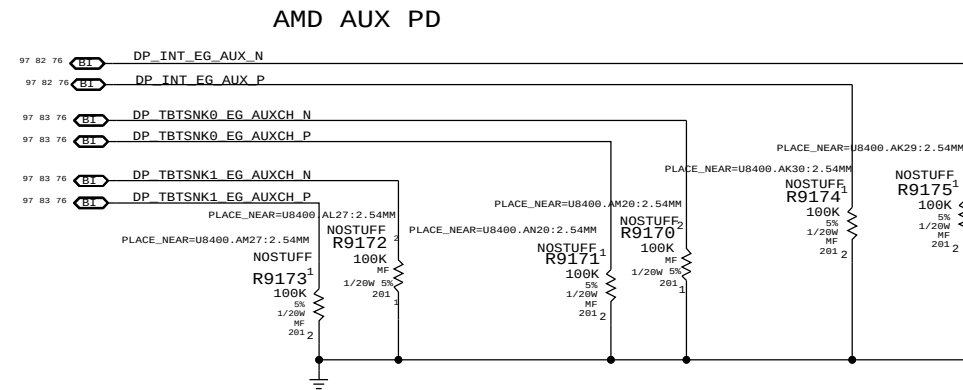
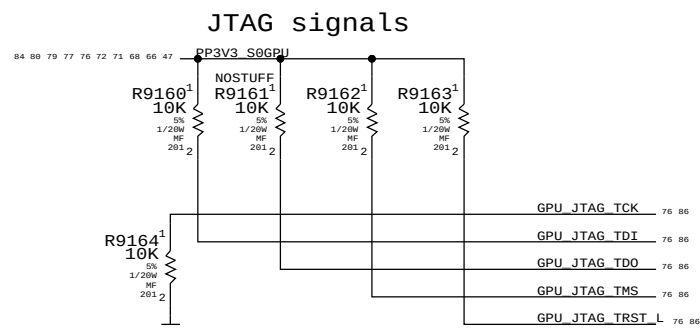
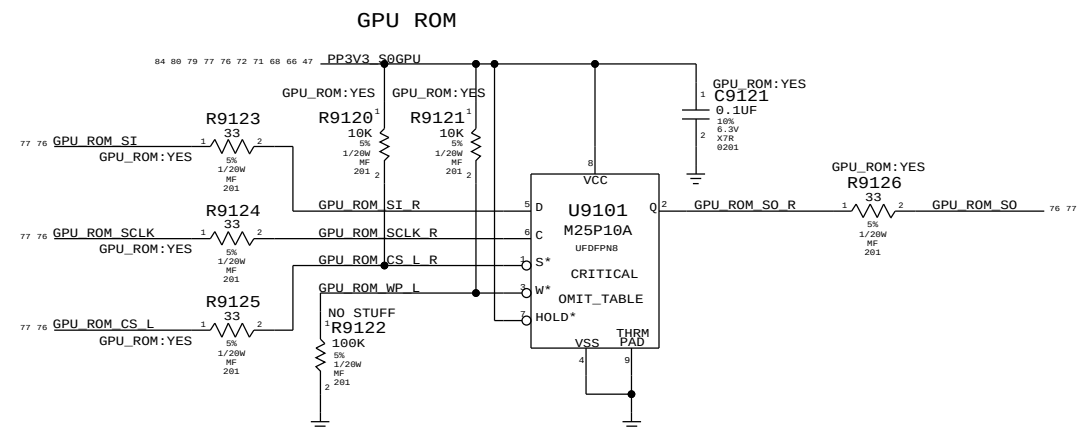
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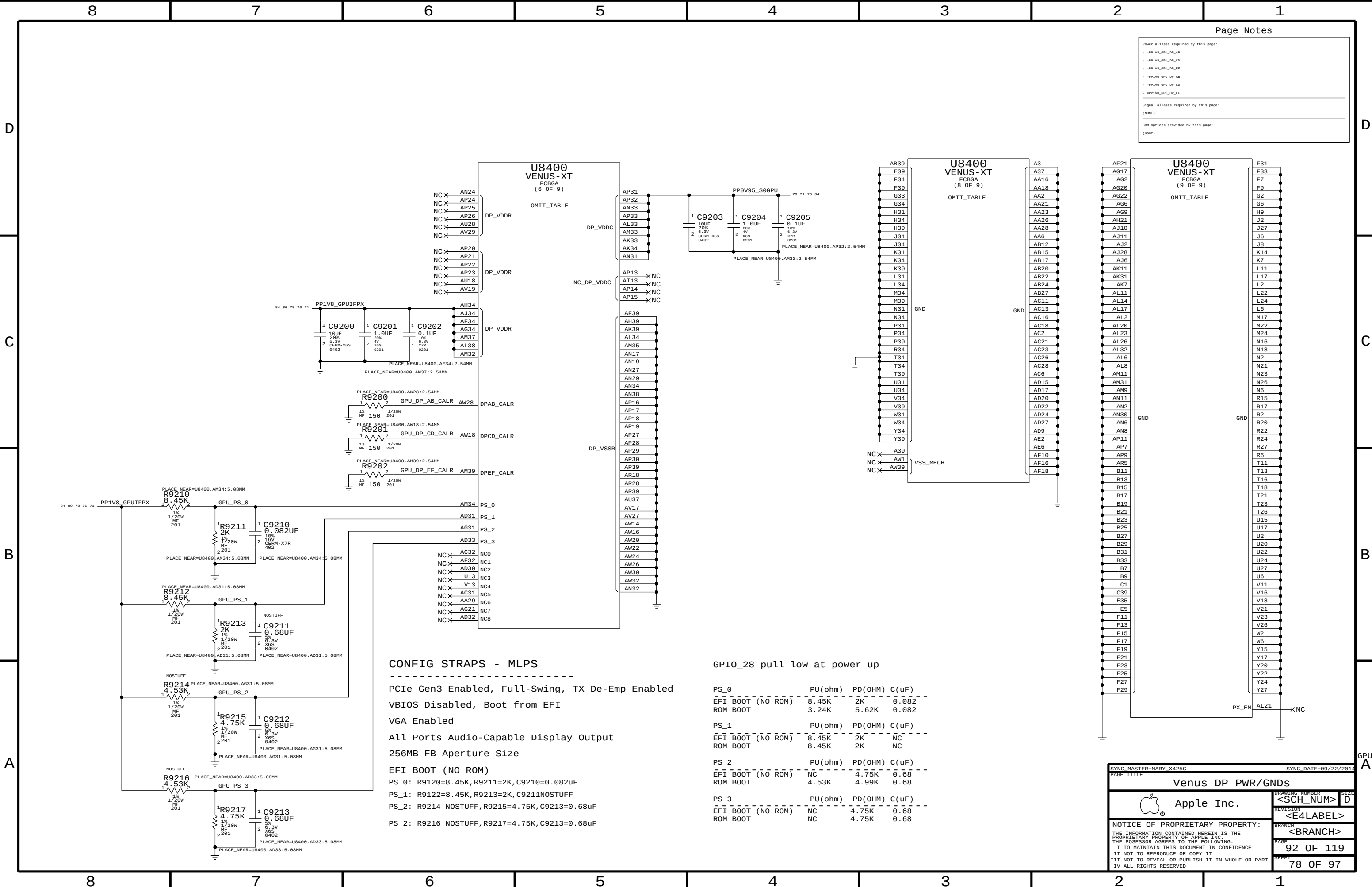
SHEET

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GPU GPIO TABLE									
		Native Func		GPIOs					
88	77	76	GFX_VDDCI_ALTV	GP	==	GFX_VDDCI_ALTV	76	77	88
						MAKE_BASE=TRUE			
77	76	75	NC_GPU_GPIO_1	GP	==	NC_GPU_GPIO_1	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	NC_GPU_GPIO_2	GP	==	NC_GPU_GPIO_2	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	GPU_GFX_PWR_LEVEL_R_L	GP	==	GPU_GFX_PWR_LEVEL_R_L	76	77	
						MAKE_BASE=TRUE			
79	77	76	GFXIMVP_VR_ICCMAx_WARN_L	GP	==	GFXIMVP_VR_ICCMAx_WARN_L	76	77	79
						MAKE_BASE=TRUE			
82	77	76	EG_BKLT_EN	GP	==	EG_BKLT_EN	76	77	82
						MAKE_BASE=TRUE			
77	76	75	GPU_ROM_S0	GP	==	GPU_ROM_S0	76	77	
						MAKE_BASE=TRUE			
77	76	75	GPU_ROM_SI	GP	==	GPU_ROM_SI	76	77	
						MAKE_BASE=TRUE			
77	76	75	GPU_ROM_SCLK	GP	==	GPU_ROM_SCLK	76	77	
						MAKE_BASE=TRUE			
79	77	76	GPU_VCORE_VID4	GP	==	GPU_VCORE_VID4	76	77	79
						MAKE_BASE=TRUE			
79	77	76	GPU_VCORE_VID5	GP	==	GPU_VCORE_VID5	76	77	79
						MAKE_BASE=TRUE			
79	77	76	GPU_VCORE_VID0	GP	==	GPU_VCORE_VID0	76	77	79
						MAKE_BASE=TRUE			
82	77	76	DP_TBTSNK1_HPD_EG	GP	==	DP_TBTSNK1_HPD_EG	76	77	82
						MAKE_BASE=TRUE			
79	77	76	GPU_VCORE_VID1	GP	==	GPU_VCORE_VID1	76	77	79
						MAKE_BASE=TRUE			
Native Func									
					GPIOs				
79	77	76	GPU_VCORE_VID2	GP	==	GPU_VCORE_VID2	76	77	79
						MAKE_BASE=TRUE			
77	76	75	GFX_SELF_THROTTLE_R	GP	==	GFX_SELF_THROTTLE_R	76	77	
						MAKE_BASE=TRUE			
82	77	76	HDMI_EG_HPD	GP	==	HDMI_EG_HPD	76	77	82
						MAKE_BASE=TRUE			
77	76	75	GPU_GFX_OVERTEMP_R	GP	==	GPU_GFX_OVERTEMP_R	76	77	
						MAKE_BASE=TRUE			
79	77	76	GPU_VCORE_VID3	GP	==	GPU_VCORE_VID3	76	77	79
						MAKE_BASE=TRUE			
77	76	75	FBVDD_ALTV0	GP	==	FBVDD_ALTV0	76	77	
						MAKE_BASE=TRUE			
77	76	75	GPU_ROM_CS_L	GP	==	GPU_ROM_CS_L	76	77	
						MAKE_BASE=TRUE			
77	76	75	TP_CLKREQ_L	GP	==	TP_CLKREQ_L	76	77	
						MAKE_BASE=TRUE			
77	76	75	NC_GPU_GENERICA	GP	==	NC_GPU_GENERICA	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	NC_GPU_GENERICB	GP	==	NC_GPU_GENERICB	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	GPU_VCORE_PSI_L	GP	==	GPU_VCORE_PSI_L	76	77	
						MAKE_BASE=TRUE			
79	77	76	GFXIMVP_DPSLP_EN_R	GP	==	GFXIMVP_DPSLP_EN_R	76	77	79
						MAKE_BASE=TRUE			
77	76	75	NC_DP_EXTB_CA_DET_EG	GP	==	NC_DP_EXTB_CA_DET_EG	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	NC_DP_EXT_A_CA_DET_EG	GP	==	NC_DP_EXT_A_CA_DET_EG	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
77	76	75	NC_GPU_GPIO_33	GP	==	NC_GPU_GPIO_33	76	77	
						MAKE_BASE=TRUE	NO_TEST=TRUE		
82	77	76	DP_INT_EG_HPD	GP	==	DP_INT_EG_HPD	76	77	82
						MAKE_BASE=TRUE			
77	76	75	NC_GPU_GPIO_35	GP	==	NC_GPU_GPIO_35	76	77	
						MAKE_BASE=TRUE			
82	77	76	DP_TBTSNK0_HPD_EG	GP	==	DP_TBTSNK0_HPD_EG	76	77	82
						MAKE_BASE=TRUE			



SYNC MASTER-MARY X425G		SYNC DATE=11/07/2014	
PAGE TITLE			
Venus GPIOs & STRAPS			
	Apple Inc.		DRAWING NUMBER <SCH_NUM>
			REVISION <E4LABEL>
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		SHEET	77 OF 97



Page Notes

Power aliases required by this page:

- RPS1V8_GPU_DP_AB
- RPS1V8_GPU_DP_CD
- RPS1V8_GPU_DP_EF
- RPS1V8_GPU_DP_AB
- RPS1V8_GPU_DP_CD
- RPS1V8_GPU_DP_EF

Signal aliases required by this page:

(NONE)

BOM options provided by this page:

(NONE)

CONFIG STRAPS - MLPS

PCIe Gen3 Enabled, Full-Swing, TX De-Emp Enabled

VBIOS Disabled, Boot from EFI

VGA Enabled

All Ports Audio-Capable Display Output

256MB FB Aperture Size

EFI BOOT (NO ROM)

PS_0: R9120=8.45K, R9211=2K, C9210=0.082uF

PS_1: R9122=8.45K, R9213=2K, C9211=NOSTUFF

PS_2: R9214 NOSTUFF, R9215=4.75K, C9213=0.68uF

PS_2: R9216 NOSTUFF, R9217=4.75K, C9213=0.68uF

GPIO_28 pull low at power up

PS_0	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	8.45K	2K	0.082
ROM BOOT	3.24K	5.62K	0.082

PS_1	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	8.45K	2K	NC
ROM BOOT	8.45K	2K	NC

PS_2	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	NC	4.75K	0.68
ROM BOOT	4.53K	4.99K	0.68

PS_3	PU(ohm)	PD(OHM)	C(uF)
EFI BOOT (NO ROM)	NC	4.75K	0.68
ROM BOOT	NC	4.75K	0.68

SYNC MASTER=MARY_X425G

SYNC DATE=09/22/2014

PAGE TITLE

Venus DP PWR/GNDs

 Apple Inc.

DRAWING NUMBER

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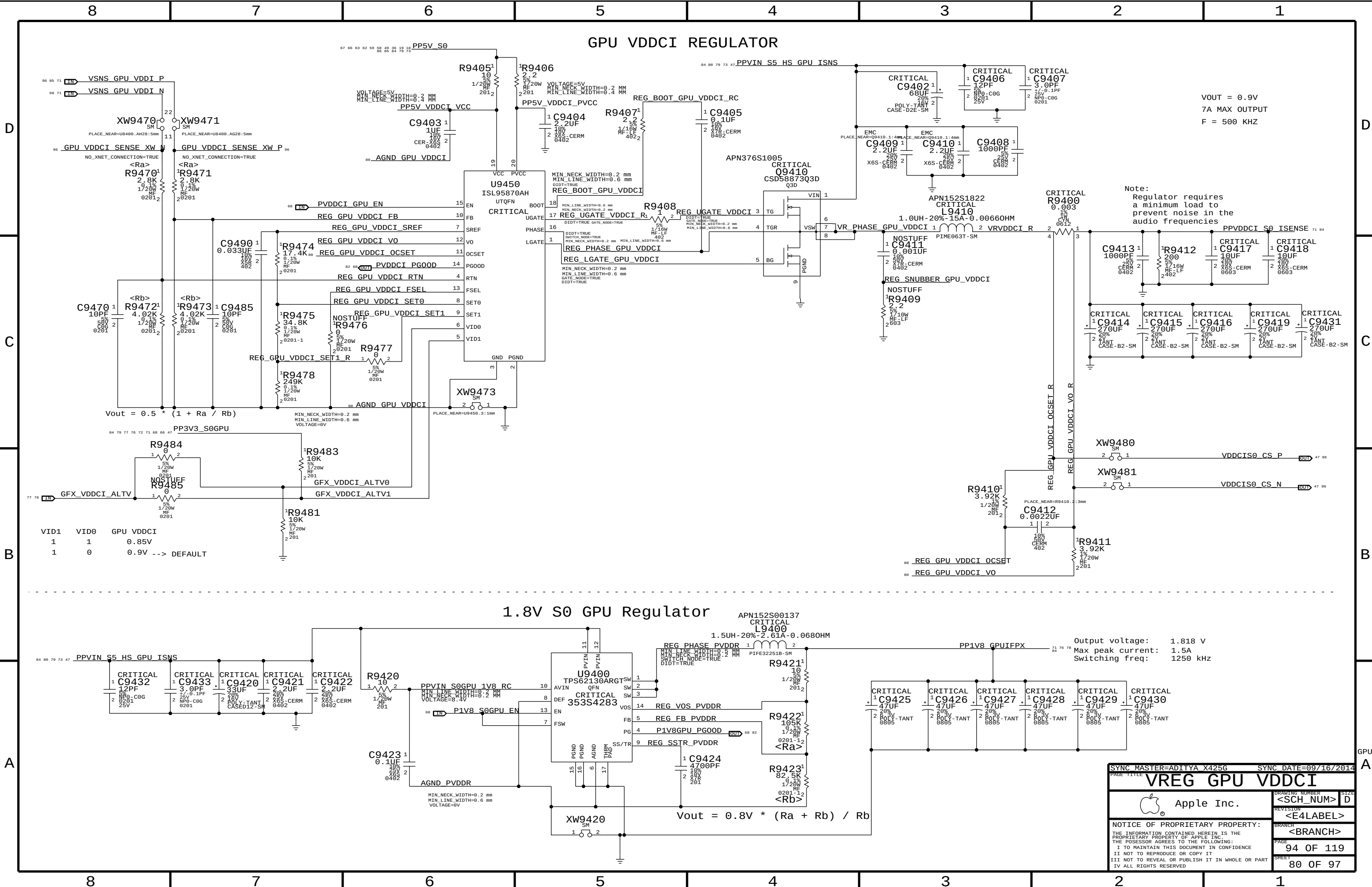
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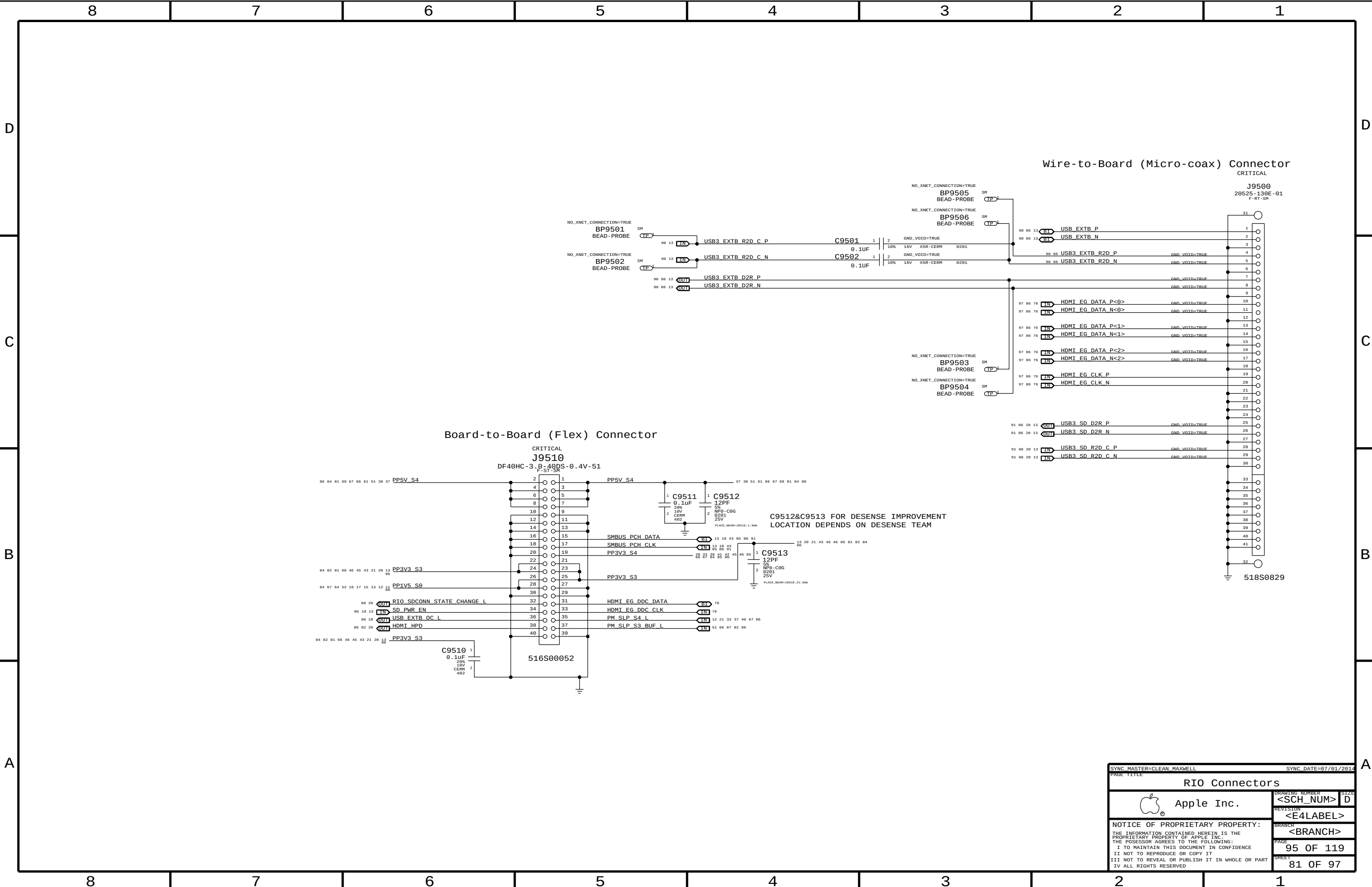
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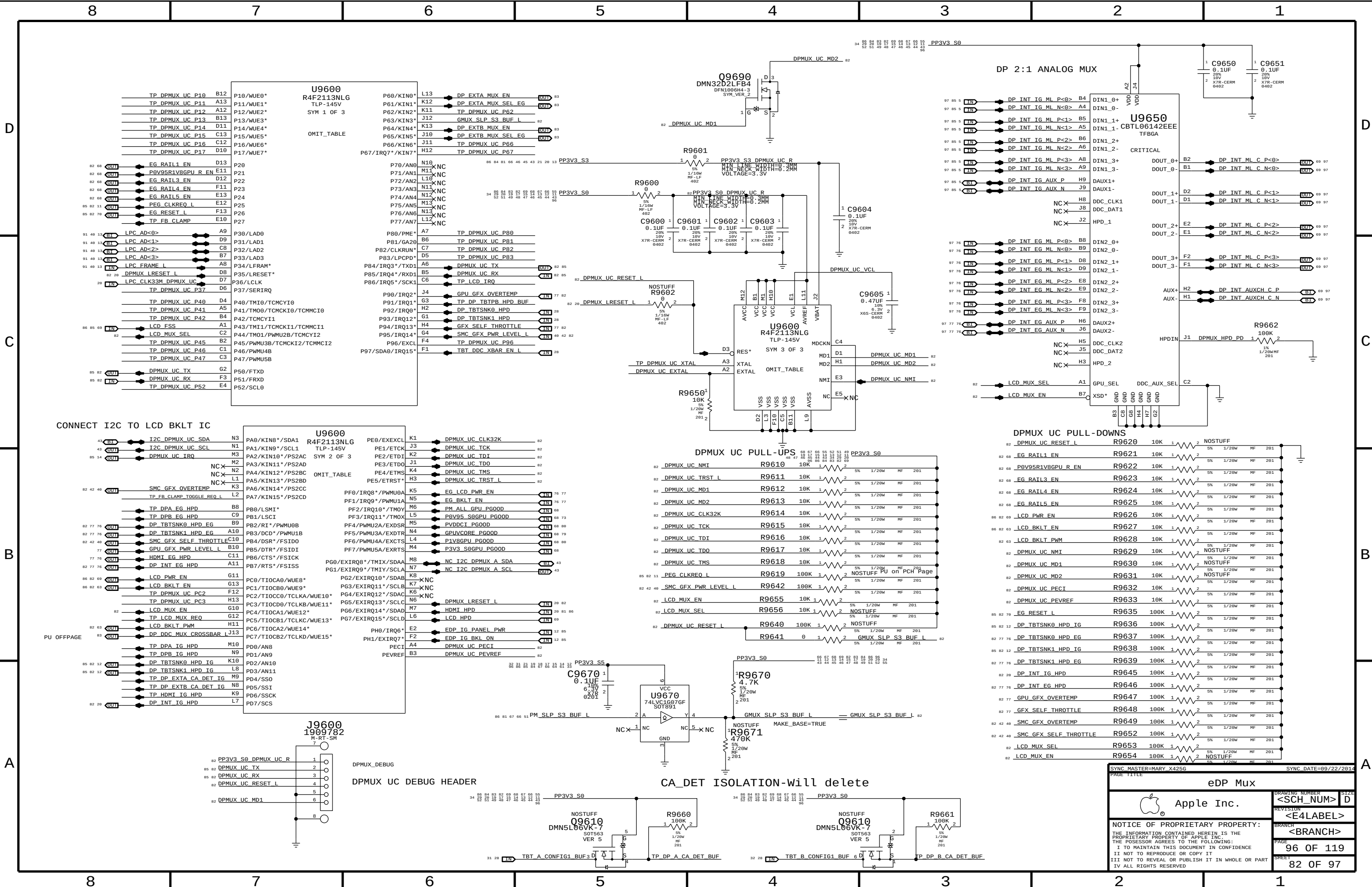


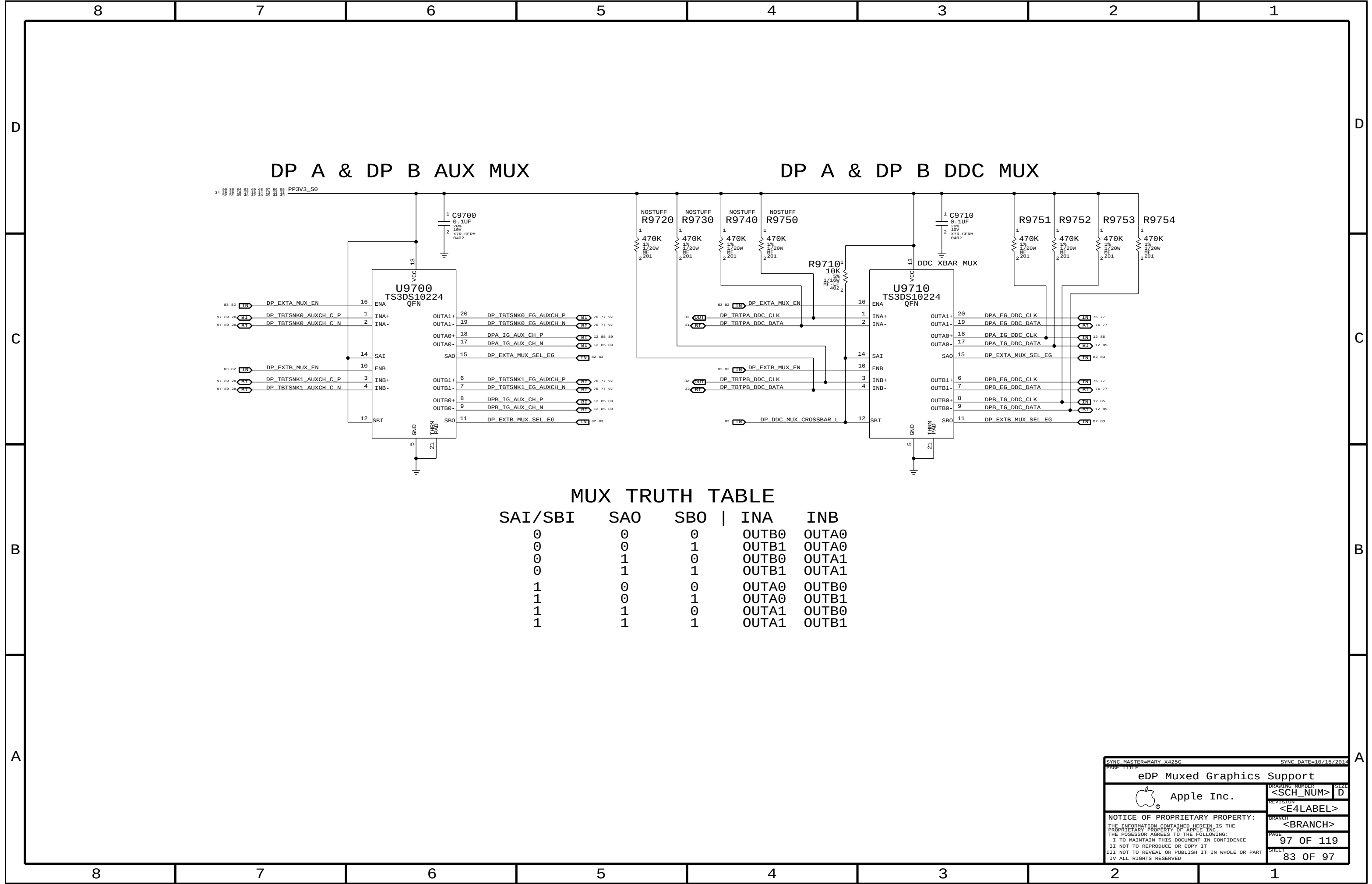
GPU VDDCI REGULATOR

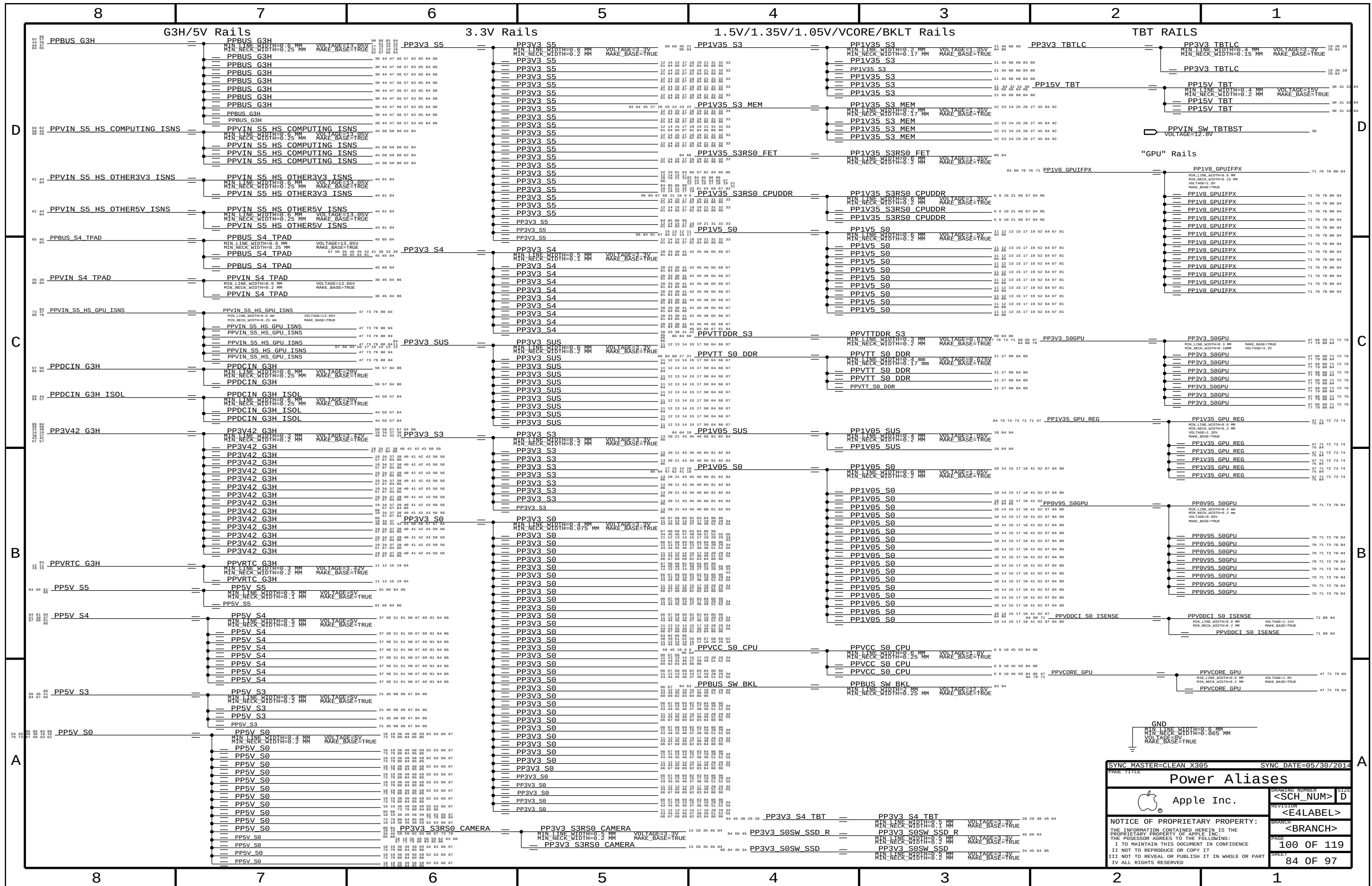
1.8V S0 GPU Regulator

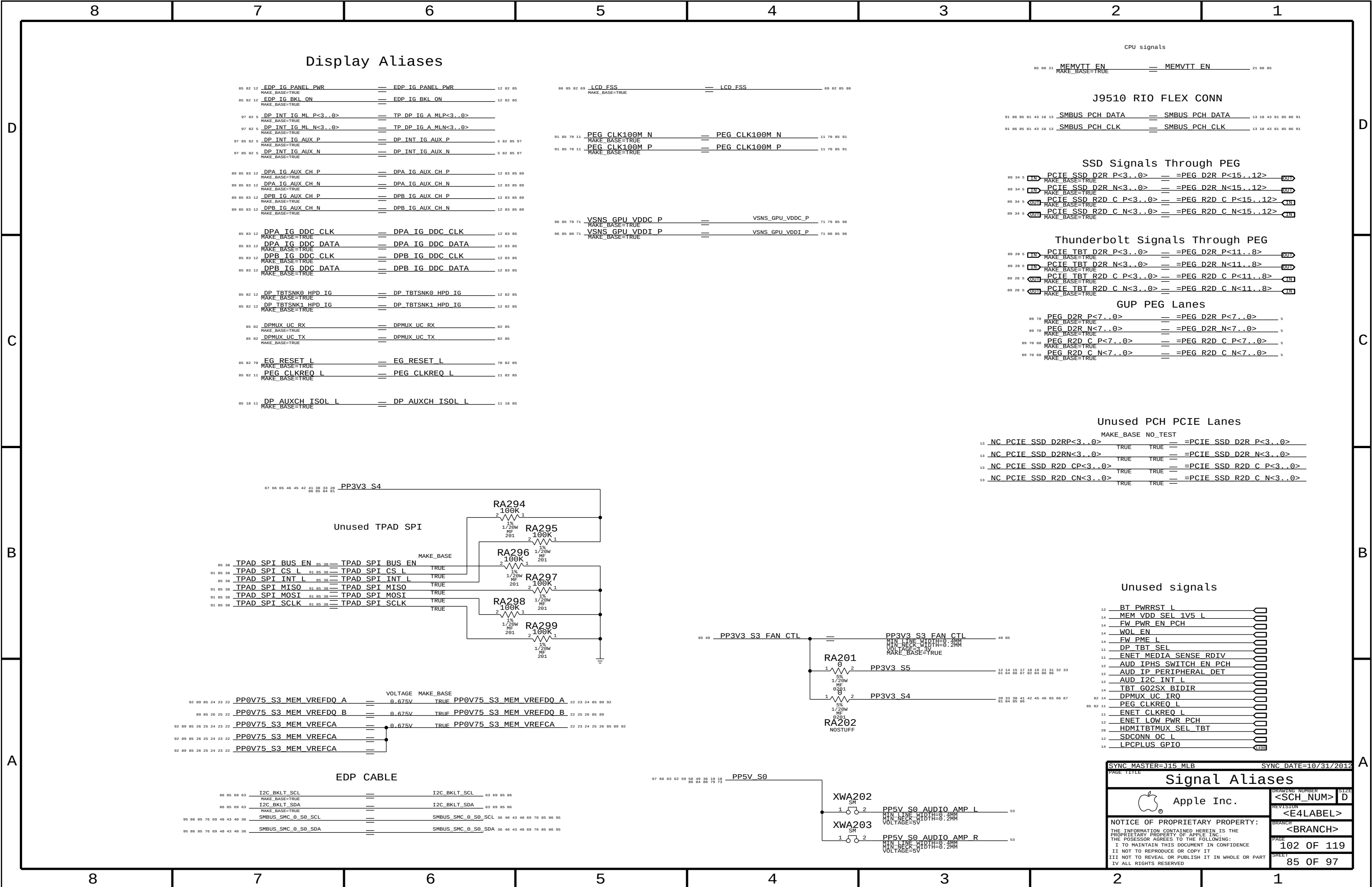
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PAGE TITLE		VREG GPU VDDCI	
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89 28 5

PCIE_TBT_D2R_P<3..0>

==

=PEG_D2R_P<11..8>

OUT

89 28 5

PCIE_TBT_D2R_N<3..0>

==

=PEG_D2R_N<11..8>

OUT

89 28 5

PCIE_TBT_R2D_C_P<3..0>

==

=PEG_R2D_C_P<11..8>

IN

89 28 5

PCIE_TBT_R2D_C_N<3..0>

==

=PEG_R2D_C_N<11..8>

IN

89 78

PEG_D2R_P<7..0>

==

=PEG_D2R_P<7..0>

5

89 78

PEG_D2R_N<7..0>

==

=PEG_D2R_N<7..0>

5

89 78 68

PEG_R2D_C_P<7..0>

==

=PEG_R2D_C_P<7..0>

5

89 78 68

PEG_R2D_C_N<7..0>

==

=PEG_R2D_C_N<7..0>

5

13

NC_PCIE_SSD_D2RP<3..0>

MAKE_BASE NO_TEST

TRUE TRUE

==

=PCIE_SSD_D2R_P<3..0>

13

NC_PCIE_SSD_D2RN<3..0>

TRUE TRUE

==

=PCIE_SSD_D2R_N<3..0>

13

NC_PCIE_SSD_R2D_CP<3..0>

TRUE TRUE

==

=PCIE_SSD_R2D_C_P<3..0>

13

NC_PCIE_SSD_R2D_CN<3..0>

TRUE TRUE

==

=PCIE_SSD_R2D_C_N<3..0>

67 66 65 46 45 42 41 38 33 28 26 86 85 84 81

PP3V3_S4

RA294

100K

2

1

RA295

100K

2

1

RA296

100K

2

1

RA297

100K

2

1

RA298

100K

2

1

RA299

100K

2

1

85 38

TPAD_SPI_BUS_EN

==

TPAD_SPI_BUS_EN

TRUE

91 85 38

TPAD_SPI_CS_L

==

TPAD_SPI_CS_L

TRUE

85 38

TPAD_SPI_INT_L

==

TPAD_SPI_INT_L

TRUE

91 85 38

TPAD_SPI_MISO

==

TPAD_SPI_MISO

TRUE

91 85 38

TPAD_SPI_MOSI

==

TPAD_SPI_MOSI

TRUE

91 85 38

TPAD_SPI_SCLK

==

TPAD_SPI_SCLK

TRUE

92 89 85 24 23 22

PP0V75_S3_MEM_VREFDQ_A

==

0.675V

TRUE

PP0V75_S3_MEM_VREFDQ_A

22 23 24 85 89 92

89 85 26 25 22

PP0V75_S3_MEM_VREFDQ_B

==

0.675V

TRUE

PP0V75_S3_MEM_VREFDQ_B

22 25 26 85 89

92 89 85 26 25 24 23 22

PP0V75_S3_MEM_VREFCA

==

0.675V

TRUE

PP0V75_S3_MEM_VREFCA

22 23 24 25 26 85 89 92

92 89 85 26 25 24 23 22

PP0V75_S3_MEM_VREFCA

==

92 89 85 26 25 24 23 22

PP0V75_S3_MEM_VREFCA

==

86 85 69 63

I2C_BKLT_SCL

==

I2C_BKLT_SCL

63 69 85 86

86 85 69 63

I2C_BKLT_SDA

==

I2C_BKLT_SDA

63 69 85 86

95 86 85 76 69 48 43 40 36

SMBUS_SMC_0_S0_SCL

==

SMBUS_SMC_0_S0_SCL

36 48 43 48 69 76 85 86 95

95 86 85 76 69 48 43 40 36

SMBUS_SMC_0_S0_SDA

==

SMBUS_SMC_0_S0_SDA

36 48 43 48 69 76 85 86 95

67 66 63 62 59 58 49 36 19 18 86 84 88 79 73

PP5V_S0

XWA202

SM

1

2

XWA203

SM

1

2

53

PP5V_S0_AUDIO_AMP_L

MIN_LANE_WIDTH=8.2MM

VOLTAGE=5V

53

PP5V_S0_AUDIO_AMP_R

MIN_LANE_WIDTH=8.2MM

VOLTAGE=5V

85 49

PP3V3_S3_FAN_CTL

==

PP3V3_S3_FAN_CTL

49 85

RA201

5%

1

2

RA202

5%

1

2

12

BT_PWRST_L

14

MEM_VDD_SEL_1V5_L

14

FW_PWR_EN_PCH

14

WOL_EN

14

FW_PME_L

11

DP_TBT_SEL

11

ENET_MEDIA_SENSE_RDIV

11

AUD_IPHS_SWITCH_EN_PCH

12

AUD_IP_PERIPHERAL_DET

12

AUD_I2C_INT_L

14

TBT_G02SX_BIDIR

82 14

DPMUX_UC_IRQ

85 82 11

PEG_CLKREQ_L

11

ENET_CLKREQ_L

12

ENET_LOW_PWR_PCH

28

HDMI_TBTMUX_SEL_TBT

12

SDCONN_OC_L

14

LPCPLUS_GPTIO

8

7

6

5

4

3

2

1

Signal Aliases

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Functional Test Points

J3501 - airport	
TRUE	AP CLKREQ 0 L
TRUE	AP RESET CONN L
TRUE	PCIE AP D2R PI N
TRUE	PCIE AP D2R PI P
TRUE	PCIE AP R2D N
TRUE	PCIE AP R2D P
TRUE	PCIE CLK100M AP CONN N
TRUE	PCIE CLK100M AP CONN P
TRUE	PCIE WAKE L
TRUE	PP3V3 S3RS4 BT F
TRUE	PP3V3 WLAN
TRUE	USB BT CONN N
TRUE	USB BT CONN P
TRUE	WIFI EVENT L
TRUE	GND

J4002 - Camera	
TRUE	MIPI CLK CONN N
TRUE	MIPI CLK CONN P
TRUE	CAM SENSOR WAKE L CONN
TRUE	MIPI DATA CONN N
TRUE	MIPI DATA CONN P
TRUE	SMBUS SMC 0 S0 SDA
TRUE	SMBUS SMC 0 S0 SCL
TRUE	I2C CAM SCK
TRUE	I2C CAM SDA
TRUE	PP5V S3RS0 ALSCAM F
TRUE	GND

J9500 - rio coax	
TRUE	HDMI EG CLK N
TRUE	HDMI EG CLK P
TRUE	HDMI EG DATA N<0>
TRUE	HDMI EG DATA N<1>
TRUE	HDMI EG DATA N<2>
TRUE	HDMI EG DATA P<0>
TRUE	HDMI EG DATA P<1>
TRUE	HDMI EG DATA P<2>

TRUE	USB3 SD D2R N
TRUE	USB3 SD D2R P
TRUE	USB3 SD R2D C N
TRUE	USB3 SD R2D C P
TRUE	USB3 EXTB D2R N
TRUE	USB3 EXTB D2R P
TRUE	USB3 EXTB R2D N
TRUE	USB3 EXTB R2D P
TRUE	USB EXTB N
TRUE	USB EXTB P
TRUE	GND

J9510 - rio flex	
TRUE	SD PWR EN
TRUE	HDMI DDC CLK
TRUE	HDMI DDC DATA
TRUE	HDMI HPD
TRUE	SMBUS PCH CLK
TRUE	SMBUS PCH DATA
TRUE	PM SLP S3 BUF L
TRUE	PM SLP S4 L
TRUE	PP3V3 S3
TRUE	PP3V3 S4
TRUE	PP5V S4
TRUE	RIO_SDCONN_STATE_CHANGE L
TRUE	USB EXTB OC L
TRUE	GND

J5150 - hall effect	
TRUE	PP3V42 G3H
TRUE	SMC LID R
TRUE	GND

J6050 - left fan	
TRUE	FAN LT PWM
TRUE	FAN LT TACH
TRUE	PP5V S0
TRUE	GND

J6060 - right fan	
TRUE	FAN RT PWM
TRUE	FAN RT TACH
TRUE	PP5V S0
TRUE	GND

J6100 - spi	
TRUE	PP3V42 G3H
TRUE	SMC RESET L
TRUE	SMC TCK
TRUE	SMC TMS
TRUE	SPIROM_USE_MLB
TRUE	GND

J4801 - ipd flex	
TRUE	USB TPAD N
TRUE	USB TPAD P
TRUE	IOXP2 INT L
TRUE	I2C IOXP SCL
TRUE	I2C IOXP SDA
TRUE	SMC PME S4 WAKE L
TRUE	TPAD_ACTUATOR_THRMTRIP L
TRUE	TPAD_VBUS_EN
TRUE	SMBUS_SMC_2_S3_SCL
TRUE	SMBUS_SMC_2_S3_SDA
TRUE	SMC_LID
TRUE	SMC_ACTUATOR_EN L
TRUE	PPVIN_S4_TPAD
TRUE	GND_ACTUATOR
TRUE	PP3V3_S4
TRUE	PP5V_S4
TRUE	GND

J4813 - keyboard	
TRUE	PP3V3 S4
TRUE	PP3V42 G3H
TRUE	WS_CONTROL_KBD
TRUE	WS_KBD1
TRUE	WS_KBD10
TRUE	WS_KBD11
TRUE	WS_KBD12
TRUE	WS_KBD13
TRUE	WS_KBD14
TRUE	WS_KBD15_CAP
TRUE	WS_KBD16_NUM
TRUE	WS_KBD17
TRUE	WS_KBD18
TRUE	WS_KBD19
TRUE	WS_KBD2
TRUE	WS_KBD20
TRUE	WS_KBD21
TRUE	WS_KBD22
TRUE	WS_KBD23
TRUE	WS_KBD3
TRUE	WS_KBD4
TRUE	WS_KBD5
TRUE	WS_KBD6
TRUE	WS_KBD7
TRUE	WS_KBD8
TRUE	WS_KBD9
TRUE	WS_KBD_ONOFF_L
TRUE	WS_LEFT_OPTION_KBD
TRUE	WS_LEFT_SHIFT_KBD
TRUE	GND

J4915 - kbd bklt	
TRUE	KBDBKLT_RETURN1
TRUE	KBDBKLT_RETURN2
TRUE	PPVOUT_S0_KBDBKLT
TRUE	GND

J6601 - mic	
TRUE	DMIC_CLK3
TRUE	PP3V3_S0
TRUE	DMIC_SDA2
TRUE	DMIC_SDA3
TRUE	GND

J6602 - L speaker	
TRUE	SPKRCONN_L_ID
TRUE	SPKRCONN_L_OUT_N
TRUE	SPKRCONN_L_OUT_P
TRUE	SPKRCONN_SL_OUT_N
TRUE	SPKRCONN_SL_OUT_P
TRUE	GND

J6603 - R speaker	
TRUE	SPKRCONN_R_ID
TRUE	SPKRCONN_R_OUT_N
TRUE	SPKRCONN_R_OUT_P
TRUE	SPKRCONN_SR_OUT_N
TRUE	SPKRCONN_SR_OUT_P
TRUE	GND

J7000 - DC PWR	
TRUE	ADAPTER_SENSE
TRUE	PP20V_DCIN_FUSE
TRUE	GND

J7050 - battery	
TRUE	PPVBAT_G3H_CONN
TRUE	SMBUS_SMC_5_G3_SCL
TRUE	SMBUS_SMC_5_G3_SDA
TRUE	SYS_DETECT_L
TRUE	GND

J8300 - edp	
TRUE	DP_INT_AUX_N
TRUE	DP_INT_AUX_P
TRUE	DP_INT_ML_N<0>
TRUE	DP_INT_ML_N<1>
TRUE	DP_INT_ML_N<2>
TRUE	DP_INT_ML_N<3>
TRUE	DP_INT_ML_P<0>
TRUE	DP_INT_ML_P<1>
TRUE	DP_INT_ML_P<2>
TRUE	DP_INT_ML_P<3>
TRUE	LCD_FSS
TRUE	LCD_HPD_CONN
TRUE	LCD_BKLT_PWM_R
TRUE	SMBUS_SMC_0_S0_SDA
TRUE	SMBUS_SMC_0_S0_SCL
TRUE	I2C_BKLT_SDA
TRUE	I2C_BKLT_SCL
TRUE	PP5VR3V3_SW_LCD
TRUE	PPVOUT_S0_LCDBKLT
TRUE	GND

Power Rails	
TRUE	PM_SLP_S3_L
TRUE	PPVTT_S0_DDR
TRUE	PP3V3_S0
TRUE	PP3V3_S3
TRUE	PP3V3_S5
TRUE	PP3V3_S5_AVREF_SMC
TRUE	PP3V42_G3H
TRUE	PP5V_S0
TRUE	PP5V_S3
TRUE	PP5V_S5
TRUE	PPBUS_G3H
TRUE	PPDCIN_G3H
TRUE	PPVCC_S0_CPU
TRUE	PPVTIDDR_S3
TRUE	PP3V3_S0SW_SSD
TRUE	PP1V5_S0
TRUE	PP1V35_S3

XDP	
TRUE	XDP_CPU_TCK
TRUE	XDP_PCH_TCK
TRUE	XDP_CPU_TDI
TRUE	XDP_CPU_TDO
TRUE	XDP_CPUPCH_TRST_L
TRUE	XDP_CPU_TMS
TRUE	XDP_PCH_TMS
TRUE	XDP_PCH_TDI
TRUE	XDP_PCH_TDO
TRUE	XDP_CPU_PREQ_L
TRUE	XDP_CPU_PRDY_L
TRUE	PM_RSMRST_L
TRUE	PM_PCH_PWROK
TRUE	PM_SYSRST_L
TRUE	CPU_CFG<3>
TRUE	PP1V05_S0
TRUE	GND

Power Sequence	
TRUE	SMC_ONOFF_L
TRUE	PM_DSW_PWRGD
TRUE	ALL_SYS_PWRGD
TRUE	PM_PCH_SYS_PWROK
TRUE	PLT_RESET_L
TRUE	LCD_PWR_EN
TRUE	LCD_BKLT_EN

GPU_VENUS_JTAG	
TRUE	GPU_JTAG_TCK
TRUE	GPU_JTAG_TDI
TRUE	GPU_JTAG_TDO
TRUE	GPU_JTAG_TMS
TRUE	GPU_JTAG_TRST_L
TRUE	GPU_PWRGOOD

SYNC MASTER=J15_MLB

SYNC DATE=10/31/2012

Functional Test Points

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8	7	6	5	4	3	2	1
NC NO_TESTS							
PCH				Thunderbolt		PLACEABLE BEAD-PROBES FOR TBT	
NO_TEST MAKE_BASE				NO_TEST MAKE_BASE			
87 13	NC USB3 SPARE D2RN	==	TRUE TRUE	NC USB3 SPARE D2RN	13 87	87 28	NC TBT XTAL250UT
87 13	NC USB3 SPARE D2RP	==	TRUE TRUE	NC USB3 SPARE D2RP	13 87		
87 13	NC USB3 SPARE R2D CN	==	TRUE TRUE	NC USB3 SPARE R2D CN	13 87		
87 13	NC USB3 SPARE R2D CP	==	TRUE TRUE	NC USB3 SPARE R2D CP	13 87		
90 87 13	NC USB3 EXTC D2RN	==	TRUE TRUE	NC USB3 EXTC D2RN	13 87 90		
90 87 13	NC USB3 EXTC D2RP	==	TRUE TRUE	NC USB3 EXTC D2RP	13 87 90		
90 87 13	NC USB3 EXTC R2D CN	==	TRUE TRUE	NC USB3 EXTC R2D CN	13 87 90		
90 87 13	NC USB3 EXTC R2D CP	==	TRUE TRUE	NC USB3 EXTC R2D CP	13 87 90		
90 87 13	NC USB3 EXTD D2RN	==	TRUE TRUE	NC USB3 EXTD D2RN	13 87 90		
90 87 13	NC USB3 EXTD D2RP	==	TRUE TRUE	NC USB3 EXTD D2RP	13 87 90		
90 87 13	NC USB3 EXTD R2D CN	==	TRUE TRUE	NC USB3 EXTD R2D CN	13 87 90		
90 87 13	NC USB3 EXTD R2D CP	==	TRUE TRUE	NC USB3 EXTD R2D CP	13 87 90		
				TP DP TBTSRC ML CP<3..0>		NC DP TBTSRC ML CP<3..0>	
				TP DP TBTSRC ML CN<3..0>		NC DP TBTSRC ML CN<3..0>	
				NC DP TBTSRC AUXCH CP		NC DP TBTSRC AUXCH CP	
				NC DP TBTSRC AUXCH CN		NC DP TBTSRC AUXCH CN	

X425G BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALL IN ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2, ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2, ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	0.1 MM	?
BGA_P2MM	*	0.2 MM	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP,BOTTOM	0.058 MM	?
1X_DIELECTRIC	ISL3, ISL4, ISL9, ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2, ISL5, ISL6, ISL7, ISL8, ISL11	0.101 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL
CPU_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP,BOTTOM	=2X_DIELECTRIC	?
CPU_8MIL	*	8 MIL	?	CPU_VID	*	0.457 MM	?
CPU_COMP	*	28 MIL	?	CPU_VREF	*	12 MIL	?
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: IVB PLATFORM DG , Tables 205-207

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_2SAME	*	=3X_DIELECTRIC	?	DMI_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
DMI_TXRX	*	=6X_DIELECTRIC	?	DMI_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
DMICK2N2S	*	=6X_DIELECTRIC	?	DMICK2N2S	TOP,BOTTOM	=10X_DIELECTRIC	?
DMICK2S2N	*	=3X_DIELECTRIC	?	DMICK2S2N	TOP,BOTTOM	=6X_DIELECTRIC	?
DMICK20THER	*	=4X_DIELECTRIC	?	DMICK20THER	TOP,BOTTOM	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_*	=SAME	*	DMI_2SAME
DMI_N2S	DMI_S2N	*	DMI_TXRX
DMI_S2N	DMI_N2S	*	DMI_TXRX
CLK_DMI	DMI_N2S	*	DMICK2N2S
CLK_DMI	DMI_S2N	*	DMICK2S2N
CLK_DMI		*	DMICK20THER

PEG - SSD & TBT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PEG_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_2SAME	*	=3X_DIELECTRIC	?	PEG_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PEG_TXRX	*	=6X_DIELECTRIC	?	PEG_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PEG_20THER	*	=4X_DIELECTRIC	?	PEG_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
PEG_2CLK	*	=7X_DIELECTRIC	?	PEG_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG3_2SAME	*	=4X_DIELECTRIC	?	PEG3_2SAME	TOP,BOTTOM	=6X_DIELECTRIC	?
PEG3_TXRX	*	=8X_DIELECTRIC	?	PEG3_TXRX	TOP,BOTTOM	=12X_DIELECTRIC	?
PEG3_20THER	*	=5X_DIELECTRIC	?	PEG3_20THER	TOP,BOTTOM	=8X_DIELECTRIC	?
PEG3_2CLK	*	=8X_DIELECTRIC	?	PEG3_2CLK	TOP,BOTTOM	=12X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PEG_*	=SAME	*	PEG_2SAME	PEG3_*	=SAME	*	PEG3_2SAME
PEG_R2D	PEG_D2R	*	PEG_TXRX	PEG3_R2D	PEG3_D2R	*	PEG3_TXRX
PEG_*	*	*	PEG_20THER	PEG3_*	*	*	PEG3_20THER
PEG_*	CLK_*	*	PEG_2CLK	PEG3_*	CLK_*	*	PEG3_2CLK

DIGITAL VIDEO SIGNAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3X_DIELECTRIC	?	DP_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
DP_20THER	*	=4X_DIELECTRIC	?	DP_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
HDMICKL_2CLK	*	=7X_DIELECTRIC	?	HDMICKL_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
HDMICKL_2DP	*	=4X_DIELECTRIC	?	HDMICKL_2DP	TOP,BOTTOM	=6X_DIELECTRIC	?
HDMICKL_20THER	*	=7X_DIELECTRIC	?	HDMICKL_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	*	*	DP_20THER
HDMI_CLK	CLK_*	*	HDMICKL_2CLK
HDMI_CLK	DISPLAYPORT	*	HDMICKL_2DP
HDMI_CLK	*	*	HDMICKL_20THER

DisplayPort/TMDs intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.

DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.

SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.

MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DMI_S2N	CPU_85D	DMI_S2N	DMI_S2N P<3:0>	5 12 87
DMI_S2N	CPU_85D	DMI_S2N	DMI_S2N N<3:0>	5 12 87
DMI_N2S	CPU_85D	DMI_N2S	DMI_N2S P<3:0>	5 12 87
DMI_N2S	CPU_85D	DMI_N2S	DMI_N2S N<3:0>	5 12 87
FDI_INT	CPU_50S	CPU_AGTI	FDI_INT	5 12
FDI_CSXNC	CPU_50S	CPU_AGTI	FDI_CSXNC	5 12
DMI_CLK	CPU_85D	CLK_DMT	DMI_CLK100M CPU P	6 11
DMI_CLK	CPU_85D	CLK_DMT	DMI_CLK100M CPU N	6 11
CPU_CLK135_PLI	CPU_85D	CLK_PCTF	CPU_CLK135M DPLLREF N	6 11
CPU_CLK135_PLI	CPU_85D	CLK_PCTF	CPU_CLK135M DPLLREF P	6 11
CPU_CLK135_PLI	CPU_85D	CLK_PCTF	CPU_CLK135M DPLLSS N	6 11
CPU_CLK135_PLI	CPU_85D	CLK_PCTF	CPU_CLK135M DPLLSS P	6 11
CPU_EDP_COMP	CPU_27P4S	CPU_COMP	CPU_EDP_RCOMP	5
CPU_PEG_COMP	CPU_27P4S	CPU_COMP	CPU_PEG_RCOMP	5
CPU_CFG	CPU_45S	CPU_ITP	CPU_CFG<19..0>	6 18 86

XDP_CLK_PCH	CLK_PCTF_85D	CLK_PCTF	NC ITPXDP CLK100MP	11 87
XDP_CLK_PCH	CLK_PCTF_85D	CLK_PCTF	NC ITPXDP CLK100MN	11 87
XDP_TDI	CPU_45S	CPU_ITP	XDP_CPU_TDI	6 18 86
XDP_TDO	CPU_45S	CPU_ITP	XDP_CPU_TDO	6 18 86
XDP_TMS	CPU_45S	CPU_ITP	XDP_CPU_TMS	6 18 86
XDP_TCK	CPU_45S	CPU_ITP	XDP_CPU_TCK	6 18 86
XDP_TRST_L	CPU_45S	CPU_ITP	XDP_CRUPCH_TRST_L	6 18 86
XDP_BPM	CPU_45S	CPU_ITP	XDP_BPM L<3..0>	6 18
XDP_BPM_L	CPU_45S	CPU_ITP	XDP_BPM L<7..4>	6 18
XDP_DBRESET_L	CPU_45S	CPU_ITP	XDP_DBRESET_L	6 18 19
XDP_PRDY_L	CPU_45S	CPU_ITP	XDP_CPU_PRDY_L	6 18 86
XDP_PREQ_L	CPU_45S	CPU_ITP	XDP_CPU_PREQ_L	6 18 86
CPU_CATERR_L	CPU_45S	CPU_AGTI	CPU_CATERR_L	6 40
CPU_PECT	CPU_45S	CPU_VTD	CPU_PECT	6 14 41
CPU_PROCHOT_L	CPU_45S	CPU_AGTI	CPU_PROCHOT_L	6 40 41 88

CPU_PWRGD	CPU_45S	CPU_AGTI	CPU_PWRGD	6 14 18
PM_THRMTRIP_L	CPU_45S	CPU_AGTI	PM_THRMTRIP_L	6 14 41
PM_MEM_PWRGD	CPU_45S	CPU_AGTI	PM_MEM_PWRGD	6 12 21
PM_SYNC	CPU_45S	CPU_AGTI	PM_SYNC	6 12
CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2..0>	6
CPU_VID	CPU_45S	CPU_VTD	CPU_VIDSOUT	8 58
CPU_VTD	CPU_45S	CPU_VTD	CPU_VIDSCCLK	8 58
CPU_VTD	CPU_45S	CPU_VTD	CPU_VIDALERT_L	8 58
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	8 58
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	9 58
CPU_MEM_VREF		MEM_12MIL	CPU_DIMMA_VREFDQ	7 22
CPU_MEM_VREF		MEM_12MIL	CPU_DIMMB_VREFDQ	7 22
CPU_MEM_VREF		MEM_PWR	PP0V75_S3_MEM_VREFDQ_A	22 23 24 85 92
CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFDQ_B	22 25 26 85
CPU_MEM_VREF		MEM_PWR	PP0V75_S3_MEM_VREFCA	22 23 24 25 26 85 89 92
CPU_MEM_VREF		MEM_PWR	PP0V75_S3_MEM_VREFCA	22 23 24 25 26 85 89 92

PCIE_D2R_GPU	PEG_80D	PEG3_D2R	PEG_D2R C P<7..0>	68 70
PCIE_D2R_GPU	PEG_80D	PEG3_D2R	PEG_D2R C N<7..0>	68 70
PCIE_D2R_GPU	PEG_80D	PEG3_D2R	PEG_D2R P<7..0>	70 85
PCIE_D2R_GPU	PEG_80D	PEG3_D2R	PEG_D2R N<7..0>	70 85
PCIE_R2D_GPU	PEG_80D	PEG3_R2D	PEG_R2D C P<7..0>	68 70 85
PCIE_R2D_GPU	PEG_80D	PEG3_R2D	PEG_R2D C N<7..0>	68 70 85
PCIE_R2D_GPU	PEG_80D	PEG3_R2D	PEG_R2D P<7..0>	70
PCIE_R2D_GPU	PEG_80D	PEG3_R2D	PEG_R2D N<7..0>	70

PCIE_D2R_SSD	CPU_85D	PEG3_D2R	PCIE_SSD D2R P<3..0>	5 34 85
PCIE_D2R_SSD	CPU_85D	PEG3_D2R	PCIE_SSD D2R N<3..0>	5 34 85
PCIE_R2D_SSD	CPU_85D	PEG3_R2D	PCIE_SSD R2D C P<3..0>	5 34 85
PCIE_R2D_SSD	CPU_85D	PEG3_R2D	PCIE_SSD R2D C N<3..0>	5 34 85
PCIE_R2D_SSD	CPU_85D	PEG3_R2D	PCIE_SSD R2D P<3..0>	34
PCIE_R2D_SSD	CPU_85D	PEG3_R2D	PCIE_SSD R2D N<3..0>	34

PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT D2R P<3..0>	5 28 85
PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT D2R N<3..0>	5 28 85
PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT D2R C P<3..0>	28 87
PEG_D2R_TBT	CPU_85D	PEG_D2R	PCIE_TBT D2R C N<3..0>	28 87
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT R2D P<3..0>	28 87
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT R2D N<3..0>	28 87
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT R2D C P<3..0>	5 28 85
PEG_R2D_TBT	CPU_85D	PEG_R2D	PCIE_TBT R2D C N<3..0>	5 28 85

DP AUX NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DP_TG_AUX	DP_85D	DISPLAYPORT	DPA_IG_AUX_CH_P	12 83 85
DP_TG_AUX	DP_85D	DISPLAYPORT	DPA_IG_AUX_CH_N	12 83 85
DP_TG_AUX	DP_85D	DISPLAYPORT	DPB_IG_AUX_CH_P	12 83 85
DP_TG_AUX	DP_85D	DISPLAYPORT	DPB_IG_AUX_CH_N	12 83 85

DP / HDMI NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C P<3..0>	28 76 97
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C N<3..0>	28 76 97
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	28 97
DP_TBT_ML0	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	28 97
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C P<3..0>	28 76 97
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C N<3..0>	28 76 97
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	28 97
DP_TBT_ML1	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	28 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_P	28 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_N	28 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_P	28 83 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_N	28 83 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_P	28 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_N	28 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_P	28 83 97
TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_N	28 83 97

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SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?
SATA_20THER	*	=4X_DIELECTRIC	?
SATA_RCOMP	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
SATA_RCOMP	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	=SAME	*	SATA_2SAME
SATA_R2D	SATA_D2R	*	SATA_TXRX
SATA_*	*	*	SATA_20THER

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?
USB_RBIAS	*	=6X_DIELECTRIC	?
BT_WAKE	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAS	TOP,BOTTOM	=10X_DIELECTRIC	?
BT_WAKE	TOP,BOTTOM	=6X_DIELECTRIC	?

USB 3.0 INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?
USB3_20THER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB3_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	=SAME	*	USB3_2SAME
USB3_R2D	USB3_D2R	*	USB3_TXRX
USB3_*	*	*	USB3_20THER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_SLOW	*	=4X_DIELECTRIC	?
CLK_25M	*	=5X_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.
NOTE: Latest Intel DG calls out 50ohms SE for sys clocks

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	SATA_85D	SATA_R2D	NC SATA A R2D_CP	11 87
	SATA_85D	SATA_R2D	NC SATA A R2D_CN	11 87
	SATA_85D	SATA_D2R	NC SATA A D2RP	11 87
	SATA_85D	SATA_D2R	NC SATA A D2RN	11 87
	SATA_85D	SATA_R2D	NC SATA B R2D_CP	11 87
	SATA_85D	SATA_R2D	NC SATA B R2D_CN	11 87
	SATA_85D	SATA_D2R	NC SATA B D2RP	11 87
	SATA_85D	SATA_D2R	NC SATA B D2RN	11 87
	SATA_45SE	SATA_RCOMP	PCH SATA RCOMP	11
	USB_85D	USB	USB EXTA P	13 37
	USB_85D	USB	USB EXTA N	13 37
	USB_85D	USB	USB EXTA MUXED P	37
	USB_85D	USB	USB EXTA MUXED N	37
	USB_85D	USB	USB LT1 P	37
	USB_85D	USB	USB LT1 N	37
	USB_85D	USB	NC USB EXTCP	13 87
	USB_85D	USB	NC USB EXTCN	13 87
	USB_85D	USB	NC USB SDP	13 87
	USB_85D	USB	NC USB SDN	13 87
	CPU_45S	CPU_ITP	SMC DEBUGPRT RX L	37 40 41
	CPU_45S	CPU_ITP	SMC DEBUGPRT TX L	37 40 41
	USB_85D	USB	NC USB SMCP	87
	USB_85D	USB	NC USB SMCN	87
	USB_85D	USB	NC USB 6P	13 87
	USB_85D	USB	NC USB 6N	13 87
	USB_85D	USB	NC USB 7P	13 87
	USB_85D	USB	NC USB 7N	13 87
	USB_85D	USB	USB EXTB P	13 81 86
	USB_85D	USB	USB EXTB N	13 81 86
	USB_85D	USB	NC USB EXTDP	13 87
	USB_85D	USB	NC USB EXTDN	13 87
	USB_85D	USB	USB BT P	13 33
	USB_85D	USB	USB BT N	13 33
	USB_85D	USB	USB BT CONN P	33 86
	USB_85D	USB	USB BT CONN N	33 86
	USB_85D	USB	NC USB IRP	13 87
	USB_85D	USB	NC USB IRN	13 87
	USB_85D	USB	USB TPAD P	13 38 86
	USB_85D	USB	USB TPAD N	13 38 86
	USB_85D	USB	USB TPAD R P	
	USB_85D	USB	USB TPAD R N	
	PCH_USB_RBIAS	USB_RBIAS	PCH USB RBIAS	13
	USB_85D	USB3_D2R	USB3 EXTA D2R P	13 37
	USB_85D	USB3_D2R	USB3 EXTA D2R N	13 37
	USB_85D	USB3_D2R	USB3 EXTA D2R C P	
	USB_85D	USB3_D2R	USB3 EXTA D2R C N	
	USB_85D	USB3_R2D	USB3 EXTA R2D P	37
	USB_85D	USB3_R2D	USB3 EXTA R2D N	37
	USB_85D	USB3_R2D	USB3 EXTA R2D C P	13 37
	USB_85D	USB3_R2D	USB3 EXTA R2D C N	13 37
	USB_85D	USB3_D2R	USB3 EXTB D2R P	13 81 86
	USB_85D	USB3_D2R	USB3 EXTB D2R N	13 81 86
	USB_85D	USB3_D2R	USB3 EXTB D2R C P	
	USB_85D	USB3_D2R	USB3 EXTB D2R C N	
	USB_85D	USB3_R2D	USB3 EXTB R2D P	81 86
	USB_85D	USB3_R2D	USB3 EXTB R2D N	81 86
	USB_85D	USB3_R2D	USB3 EXTB R2D C P	13 81
	USB_85D	USB3_R2D	USB3 EXTB R2D C N	13 81
	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RP	13 87
	USB_85D	USB3_D2R	NC_USB3_EXTC_D2RN	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CP	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTC_R2D_CN	13 87
	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RP	13 87
	USB_85D	USB3_D2R	NC_USB3_EXTD_D2RN	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CP	13 87
	USB_85D	USB3_R2D	NC_USB3_EXTD_R2D_CN	13 87

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTC	11 19
	SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB	11 19
	SYSCLK_CLK25M_SB_R	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB_R	11 19
	SYSCLK_CLK25M_CAMERA	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	19 36
	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	19 28
	SYSCLK_CLK25M_TBT_R	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	28

SYNC MASTER=SIDLE_J45

SYNC DATE=12/10/2012

PCH Constraints 1

Apple Inc.

<SCH_NUM>

1

<E4LABEL>

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?
SPI3X	*	=3x_DIELECTRIC	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_SE	*	=2x_DIELECTRIC	?	PCH_SE	TOP,BOTTOM	=3x_DIELECTRIC	?

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_8SD	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_8SD	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	*	=2X_DIELECTRIC	?	PCIE_2SAME	TOP, BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?	PCIE_TXRX	TOP, BOTTOM	=10X_DIELECTRIC	?
PCIE_20THER	*	=4X_DIELECTRIC	?	PCIE_20THER	TOP, BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?	PCIE_2CLK	TOP, BOTTOM	=10X_DIELECTRIC	?
PCIECLK_20THER	*	=7X_DIELECTRIC	?	PCIECLK_20THER	TOP, BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_R2D	PCIE_D2R	*	PCIE_TXRX
PCIE_*	*	*	PCIE_20THER
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_20THER

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		SPACING	
PHYSICAL		SPACING			
	LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	13 40 82
	LPC_FRAME_I	LPC_45S	LPC	LPC_FRAME_L	13 40 82
	SMBUS_PCH_CLK	SMB_45S	SMB	SMBUS_PCH_CLK	13 18 43 81 85 86
	SMBUS_PCH_DATA	SMB_45S	SMB	SMBUS_PCH_DATA	13 18 43 81 85 86
	SMBUS_PCH_0_CLK	SMB_45S	SMB	SML_PCH_0_CLK	13 43
	SMBUS_PCH_0_DATA	SMB_45S	SMB	SML_PCH_0_DATA	13 43
	SMBUS_PCH_1_CLK	SMB_45S	SMB	SML_PCH_1_CLK	13 43
	SMBUS_PCH_1_DATA	SMB_45S	SMB	SML_PCH_1_DATA	13 43
	HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	11 52
		HDA_45S	HDA	HDA_BIT_CLK_R	11
	HDA_SYNC	HDA_45S	HDA	HDA_SYNC	11 52
		HDA_45S	HDA	HDA_SYNC_R	11
		HDA_45S	HDA	HDA_RST_R_L	11
	HDA_RST_I	HDA_45S	HDA	HDA_RST_I	11 52
	HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	11 52
	HDA_SDIN0_R	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	52
	HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	11 52
		HDA_45S	HDA	HDA_SDOUT_R	11 19
	USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_P	13 20 81 86
	USB3_SD_R2D	USB3_85D	USB3_R2D	USB3_SD_R2D_C_N	13 20 81 86
	USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_P	13 20 81 86
	USB3_SD_D2R	USB3_85D	USB3_D2R	USB3_SD_D2R_N	13 20 81 86
	PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_P	33 86
	PCIE_AP_R2D	PCIE_85D	PCIE_R2D	PCIE_AP_R2D_N	33 86
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_P	33 33
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_C_N	33 33
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_P	
		PCIE_85D	PCIE_R2D	PCIE_AP_R2D_PI_N	
	PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_P	33 20 33
	PCIE_AP_D2R	PCIE_85D	PCIE_D2R	PCIE_AP_D2R_N	33 20 33
		PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_P	86
		PCIE_85D	PCIE_D2R	PCIE_AP_D2R_PI_N	86
	PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_P	35 36
	PCIE_CAMERA_R2D	PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_N	35 36
		PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_P	33 36
		PCIE_85D	PCIE_R2D	PCIE_CAMERA_R2D_C_N	33 36
	PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_P	35 20 36
	PCIE_CAMERA_D2R	PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_N	35 20 36
		PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_P	35 36
		PCIE_85D	PCIE_D2R	PCIE_CAMERA_D2R_C_N	35 36
		CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC_R	11 19
	PCH_LPC_CLK0	CLK_LPC_45S	CLK_LPC	LPC_CLK33M_SMC	19 40
		CLK_LPC_45S	CLK_LPC	NC_LPC_CLK33M_LPCPLUS_R	11 87
	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIEIN	11 19
	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK14P3M_REFCLK	11
	PCIE_CLK100M	CPU_45S	CLK_PCIE	PCH_CLK33M_PCIEOUT	11 19
	PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_P	11
	PCIE_CLK100M_PCH	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_PCH_N	11
	PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P	11 28
	PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N	11 28
	PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_P	11
	PCIE_CLK100M_DOT	CLK_PCIE_85D	CLK_PCIE	PCH_CLK96M_DOT_N	11
	PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_P	11
	PCIE_CLK100M_SATA	PCIE_85D	CLK_PCIE	PCH_CLK100M_SATA_N	11
	PCIE_CLK100M_ENET	PCIE_85D	CLK_PCIE	PCIE_CLK100M_SD_P	
	PCIE_CLK100M_ENET	PCIE_85D	CLK_PCIE	PCIE_CLK100M_SD_N	
	PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P	11 33
	PCIE_CLK100M_AP	PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N	11 33
		PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	35 86
		PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	35 86
	PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	11 36
	PCIE_CLK100M_S2	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	11 36
		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	35 36
		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	35 36
	PCIE_CLK100M_FW	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P	11 34
		CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N	11 34
	PCIE_CLK100M_GPU	CLK_PCIE_85D	CLK_PCIE	PEG_CLK100M_P	11 70 85
		CLK_PCIE_85D	CLK_PCIE	PEG_CLK100M_N	11 70 85

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		PHYSICAL		NET_TYPE	SPACING	
H62B	PCH_PM_NET	PCH_45S	PCH_5E	PCH_INTRUDER_L		11
H62C	PCH_PM_NET	PCH_45S	PCH_5E	PCH_INTVRMEN_L		11
H63A	PCH_PM_NET	PCH_45S	PCH_5E	PCH_DSWVRMEN		12
H63B	PCH_PM_NET	PCH_45S	PCH_5E	PCH_SRTCRST_L		11
H61D	PCH_PM_NET	PCH_45S	PCH_5E	PM_RSMRST_L		12 67 86
H61D	PCH_PM_NET	PCH_45S	PCH_5E	PM_SYSRST_L		12 19 40 86
H631	PCH_PM_NET	PCH_45S	PCH_5E	PM_PCH_PWROK		12 19 86 91
H631	PCH_PM_NET	PCH_45S	PCH_5E	PM_PCH_PWROK		12 19 86 91
H62D	PCH_PM_NET	PCH_45S	PCH_5E	PM_DSW_PWRGD		12 40 86
H62E	PCH_PM_NET	PCH_45S	PCH_5E	PM_PCH_SYS_PWROK		12 18 19 40 86
H62D	PCH_PM_NET	PCH_45S	PCH_5E	PM_PWRBTN_L		12 18 40
H62D	PCH_PM_NET	PCH_45S	PCH_5E	PM_THRMTRIP_L_R		14 41 42
H62E	PCH_PCTF_WAKE	PCH_45S	PCH_5E	PCIE_WAKE_L		12 13 35 86
H62E	PCH_PM_NET	PCH_45S	PCH_5E	PCH_RCIN_L		14
H61B	SPT_MLB	SPT_45S	SPT3X	SPI_ALT_CLK		
H61B	SPT_MLB	SPT_45S	SPT	SPI_CLK		50
H61B	SPT_MLB	SPT_45S	SPT	SPI_CLK_R		13 50
H61B	SPT_MLB	SPT_45S	SPT3X	SPI_MLB_CLK		50
H61B	SPT_MLB	SPT_45S	SPT3X	SPI_SMC_CLK		40 50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_ALT_CS_L		
H61D	SPT_MLB	SPT_45S	SPT	SPI_CS0_L		50
H61D	SPT_MLB	SPT_45S	SPT	SPI_CS0_R_L		13 50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_MLB_CS_L		50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_SMC_CS_L		40 50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_ALT_I01_MISO		
H61D	SPT_MLB	SPT_45S	SPT	SPI_MISO		13 50
H61D	SPT_MLB	SPT_45S	SPT	SPI_MISO_R		50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_MLB_I01_MISO		50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_SMC_MISO		40 50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_ALT_I00_MOSI		
H61D	SPT_MLB	SPT_45S	SPT	SPI_MOSI		50
H61D	SPT_MLB	SPT_45S	SPT	SPI_MOSI_R		13 50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_MLB_I00_MOSI		50
H61D	SPT_MLB	SPT_45S	SPT3X	SPI_SMC_MOSI		40 50
H61D	SPT_MLB_I02	SPT_45S	SPT3X	SPI_I0<2>		13 50
H61D	SPT_MLB_I02	SPT_45S	SPT3X	SPI_MLB_I02_WP_L		50
H61D	SPT_MLB_I02	SPT_45S	SPT3X	SPI_ALT_I02_WP_L		
H61D	SPT_MLB_I03	SPT_45S	SPT3X	SPI_I0<3>		13 50
H61D	SPT_MLB_I03	SPT_45S	SPT3X	SPI_MLB_I03_HOLD_L		50
H61D	SPT_MLB_I03	SPT_45S	SPT3X	SPI_ALT_I03_HOLD_L		
H61D	SPT_TPAD	SPT_45S	SPT	TPAD_SPI_SCLK		38 85
H61D	SPT_TPAD_CS	SPT_45S	SPT	TPAD_SPI_CS_L		38 85
H61D	SPT_TPAD	SPT_45S	SPT	TPAD_SPI_MISO		38 85
H61D	SPT_TPAD	SPT_45S	SPT	TPAD_SPI_MOSI		38 85

SYNC MASTER=CLEAN X305 PEG		SYNC DATE=02/18/2014	
PAGE TITLE			
PCH Constraints 2			
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTRL	*	=2x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_20THERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_20THER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTRL2CTRL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_20THERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	*	*	MEM_20THER
MEM_*_DQS_*	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS2OWNDATA

DDR3 (Memory Down):

DQ signals should be matched within 0.508mm of associated DQS pair
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to cClock matching should be within [CLK-139.73mm] and [CLK-30.48mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-2.54mm] to [CLK+0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-2.54mm] to [CLK+2.54mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to first DRAM device is 139.7mm max, to last DRAM device is 194.31mm max.

SOURCE: Double checked with Doc#486985 Chief River SFF Platform DG: Memory Down
SOURCE: Need to re-confirm CRW DG for memory down (Intel not yet provided)

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

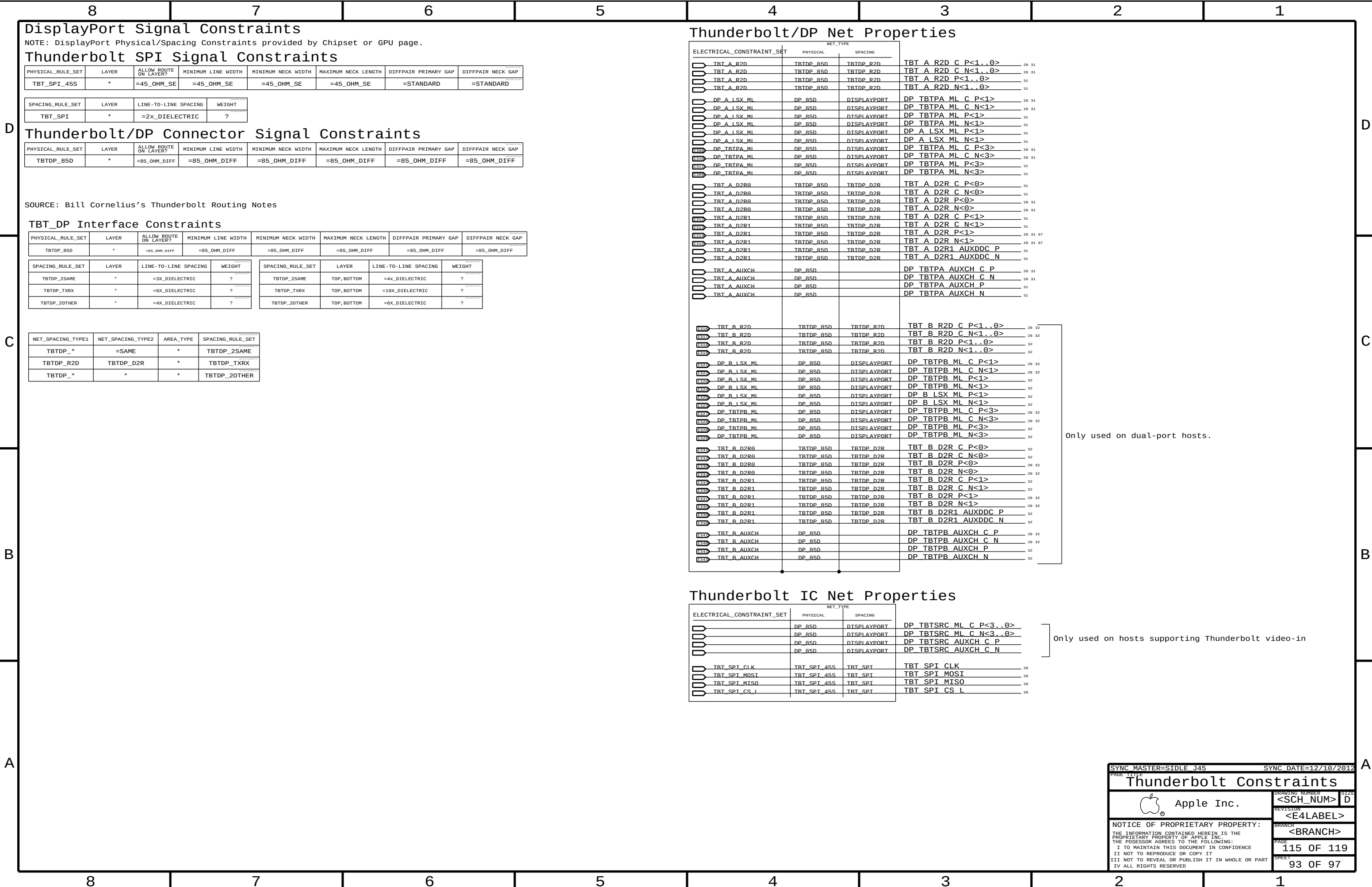
Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
MEM_A_CLK0	MEM_72D	MEM_CLK	MEM A CLK P<0> 7 23 27
MEM_A_CLK0	MEM_72D	MEM_CLK	MEM A CLK N<0> 7 23 27
MEM_A_CLK1	MEM_72D	MEM_CLK	MEM A CLK P<1> 7 24 27
MEM_A_CLK1	MEM_72D	MEM_CLK	MEM A CLK N<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A CKE<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A CKE<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A CS L<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A CS L<1> 7 24 27
MEM_A_CNTRL0	MEM_40S	MEM_CTRL	MEM A ODT<0> 7 23 27
MEM_A_CNTRL1	MEM_40S	MEM_CTRL	MEM A ODT<1> 7 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0> 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0> 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L 7 23 24 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L 7 23 24 27
MEM_A_DATA_0	MEM_45S	MEM_A_DATA_0	MEM A DQ<7..0> 7 23 24
MEM_A_DATA_1	MEM_45S	MEM_A_DATA_1	MEM A DQ<15..8> 7 23 24
MEM_A_DATA_2	MEM_45S	MEM_A_DATA_2	MEM A DQ<23..16> 7 23 24
MEM_A_DATA_3	MEM_45S	MEM_A_DATA_3	MEM A DQ<31..24> 7 23 24
MEM_A_DATA_4	MEM_45S	MEM_A_DATA_4	MEM A DQ<39..32> 7 23 24
MEM_A_DATA_5	MEM_45S	MEM_A_DATA_5	MEM A DQ<47..40> 7 23 24
MEM_A_DATA_6	MEM_45S	MEM_A_DATA_6	MEM A DQ<55..48> 7 23 24
MEM_A_DATA_7	MEM_45S	MEM_A_DATA_7	MEM A DQ<63..56> 7 23 24
MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM A DQS P<0> 7 23 24
MEM_A_DQS0	MEM_85D	MEM_A_DQS_0	MEM A DQS N<0> 7 23 24
MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM A DQS P<1> 7 23 24
MEM_A_DQS1	MEM_85D	MEM_A_DQS_1	MEM A DQS N<1> 7 23 24
MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM A DQS P<2> 7 23 24
MEM_A_DQS2	MEM_85D	MEM_A_DQS_2	MEM A DQS N<2> 7 23 24
MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM A DQS P<3> 7 23 24
MEM_A_DQS3	MEM_85D	MEM_A_DQS_3	MEM A DQS N<3> 7 23 24
MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM A DQS P<4> 7 23 24
MEM_A_DQS4	MEM_85D	MEM_A_DQS_4	MEM A DQS N<4> 7 23 24
MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM A DQS P<5> 7 23 24
MEM_A_DQS5	MEM_85D	MEM_A_DQS_5	MEM A DQS N<5> 7 23 24
MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM A DQS P<6> 7 23 24
MEM_A_DQS6	MEM_85D	MEM_A_DQS_6	MEM A DQS N<6> 7 23 24
MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM A DQS P<7> 7 23 24
MEM_A_DQS7	MEM_85D	MEM_A_DQS_7	MEM A DQS N<7> 7 23 24
MEM_B_CLK0	MEM_72D	MEM_CLK	MEM B CLK P<0> 7 25 27
MEM_B_CLK0	MEM_72D	MEM_CLK	MEM B CLK N<0> 7 25 27
MEM_B_CLK1	MEM_72D	MEM_CLK	MEM B CLK P<1> 7 26 27
MEM_B_CLK1	MEM_72D	MEM_CLK	MEM B CLK N<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B CKE<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B CKE<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B CS L<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B CS L<1> 7 26 27
MEM_B_CNTRL0	MEM_40S	MEM_CTRL	MEM B ODT<0> 7 25 27
MEM_B_CNTRL1	MEM_40S	MEM_CTRL	MEM B ODT<1> 7 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0> 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0> 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L 7 25 26 27
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L 7 25 26 27
MEM_B_DATA_0	MEM_45S	MEM_B_DATA_0	MEM B DQ<7..0> 7 25 26
MEM_B_DATA_1	MEM_45S	MEM_B_DATA_1	MEM B DQ<15..8> 7 25 26
MEM_B_DATA_2	MEM_45S	MEM_B_DATA_2	MEM B DQ<23..16> 7 25 26
MEM_B_DATA_3	MEM_45S	MEM_B_DATA_3	MEM B DQ<31..24> 7 25 26
MEM_B_DATA_4	MEM_45S	MEM_B_DATA_4	MEM B DQ<39..32> 7 25 26
MEM_B_DATA_5	MEM_45S	MEM_B_DATA_5	MEM B DQ<47..40> 7 25 26
MEM_B_DATA_6	MEM_45S	MEM_B_DATA_6	MEM B DQ<55..48> 7 25 26
MEM_B_DATA_7	MEM_45S	MEM_B_DATA_7	MEM B DQ<63..56> 7 25 26
MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM B DQS P<0> 7 25 26
MEM_B_DQS0	MEM_85D	MEM_B_DQS_0	MEM B DQS N<0> 7 25 26
MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM B DQS P<1> 7 25 26
MEM_B_DQS1	MEM_85D	MEM_B_DQS_1	MEM B DQS N<1> 7 25 26
MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM B DQS P<2> 7 25 26
MEM_B_DQS2	MEM_85D	MEM_B_DQS_2	MEM B DQS N<2> 7 25 26
MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM B DQS P<3> 7 25 26
MEM_B_DQS3	MEM_85D	MEM_B_DQS_3	MEM B DQS N<3> 7 25 26
MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM B DQS P<4> 7 25 26
MEM_B_DQS4	MEM_85D	MEM_B_DQS_4	MEM B DQS N<4> 7 25 26
MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM B DQS P<5> 7 25 26
MEM_B_DQS5	MEM_85D	MEM_B_DQS_5	MEM B DQS N<5> 7 25 26
MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM B DQS P<6> 7 25 26
MEM_B_DQS6	MEM_85D	MEM_B_DQS_6	MEM B DQS N<6> 7 25 26
MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM B DQS P<7> 7 25 26
MEM_B_DQS7	MEM_85D	MEM_B_DQS_7	MEM B DQS N<7> 7 25 26
		MEM_PWR	PP0V75_S3 MEM VREFD0 A 22 23 24 25 85 89
		MEM_PWR	PP0V75_S3 MEM VREFCA 22 23 24 25 26 85 89
		MEM_PWR	PP1V35_S3 MEM 22 23 24 25 26 27 45 84

SYNC MASTER=SIDLE J45		SYNC DATE=12/10/2012	
PAGE TITLE			
Memory Constraints			
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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICKL_20THER	*	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICKL_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS20WNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_20THERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER
S2_MEM_DQS*	*	*	S2MEM_20THER
S2_MEM_CMD	*	*	S2MEM_20THER
S2_MEM_CTRL	*	*	S2MEM_20THER
S2_MEM_CLK	*	*	S2MEM_20THER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

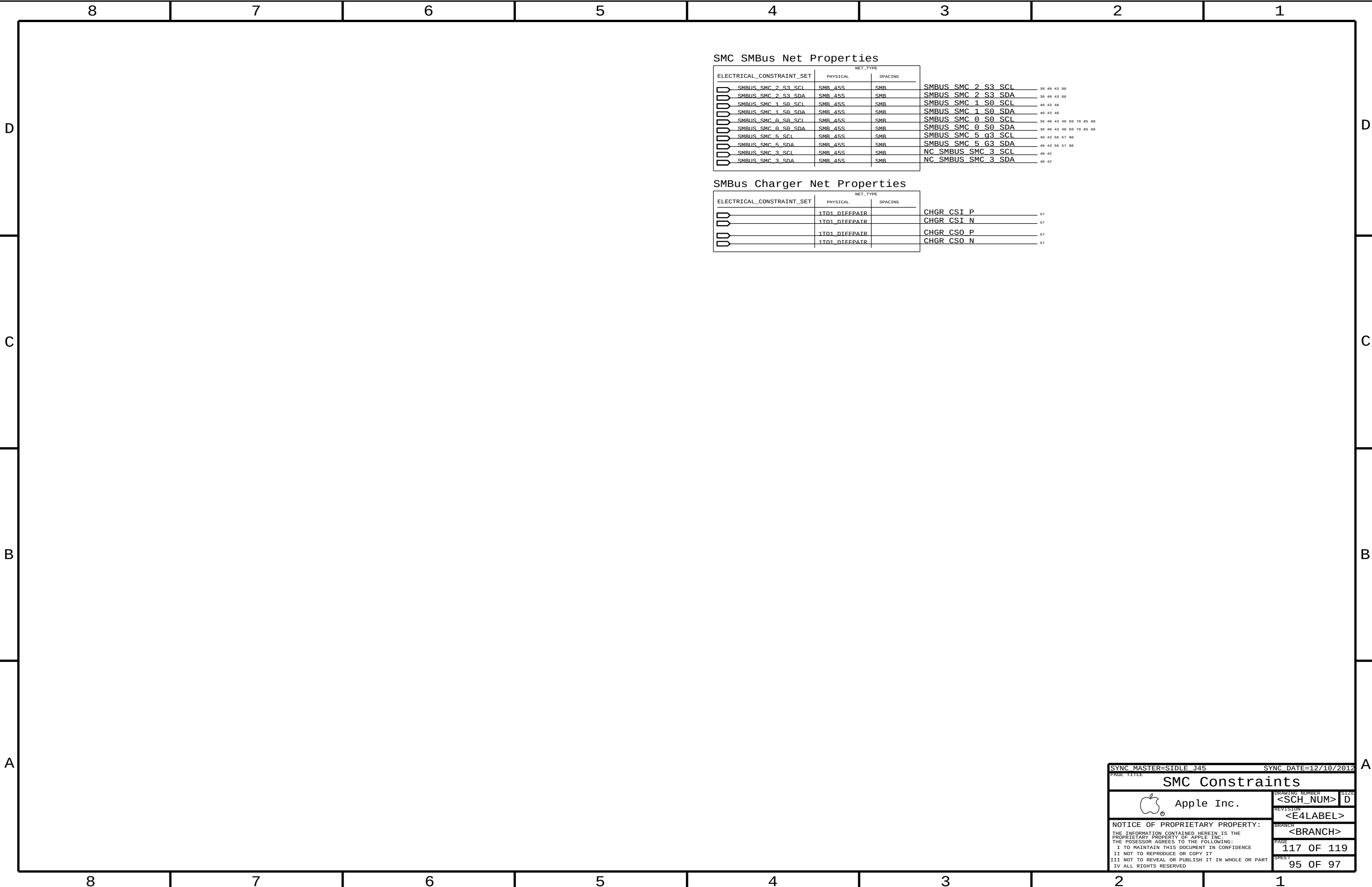
Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
S2_MEM_CLK	S2_MFM_85D	S2_MEM_CLK	MEM_CAM_CLK_P
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N
S2_MEM_CTRL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE
S2_MEM_CTRL	S2_MFM_45S	S2_MEM_CTRL	MEM_CAM_CS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>
S2_MEM_DATA_0	S2_MFM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>
S2_MEM_DATA_1	S2_MFM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>
S2_MEM_DATA_0	S2_MFM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N
		S2_MEM_PWR	PP1V35_CAM
		S2_MEM_PWR	PP0V675_CAM_VREF
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ

SYNC MASTER=SIDLE J45		SYNC DATE=12/10/2012	
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Camera Constraints			
 Apple Inc.	DRAWING NUMBER		SIZE
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SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
 SMBUS_SMC_2_S3_SCL	SMB_45S	SMR	SMBUS_SMC_2_S3_SCL	38 40 43 86
 SMBUS_SMC_2_S3_SDA	SMB_45S	SMR	SMBUS_SMC_2_S3_SDA	38 40 43 86
 SMBUS_SMC_1_S0_SCL	SMB_45S	SMR	SMBUS_SMC_1_S0_SCL	40 43 48
 SMBUS_SMC_1_S0_SDA	SMB_45S	SMR	SMBUS_SMC_1_S0_SDA	40 43 48
 SMBUS_SMC_0_S0_SCL	SMB_45S	SMR	SMBUS_SMC_0_S0_SCL	36 40 43 48 60 76 85 86
 SMBUS_SMC_0_S0_SDA	SMB_45S	SMR	SMBUS_SMC_0_S0_SDA	36 40 43 48 60 76 85 86
 SMBUS_SMC_5_SCL	SMB_45S	SMR	SMBUS_SMC_5_S3_SCL	40 43 56 57 86
 SMBUS_SMC_5_SDA	SMB_45S	SMR	SMBUS_SMC_5_G3_SDA	40 43 56 57 86
 SMBUS_SMC_3_SCL	SMB_45S	SMR	NC SMBUS_SMC_3_SCL	40 42
 SMBUS_SMC_3_SDA	SMB_45S	SMR	NC SMBUS_SMC_3_SDA	40 42

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
	1T01_DIEFPATR		CHGR_CSI_P	57
	1T01_DIEFPATR		CHGR_CSI_N	57
	1T01_DIEFPATR		CHGR_CS0_P	57
	1T01_DIEFPATR		CHGR_CS0_N	57

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_50S	"	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_50S	"	=1:1_DIFFPAIR	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	"	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	"	=1:1_DIFFPAIR	0.1 MM	0.1 MM	16 MM	0.1 MM	0.1 MM
THERM_45S_CPUVISMS1	"	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.2 MM	0.2 MM
THERM_1T01_45S	"	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
SENSE_1T01_45S	"	=1:1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P29M
GND	PCIE_*	*	GND_P29M
GND	SATA_*	*	GND_P29M
USB	GND	*	GND_P29M
CLK_PCIE	SB_POWER	*	PWR_P29M
SB_POWER	SATA_*	*	PWR_P29M
USB	SB_POWER	*	PWR_P29M

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.1 MM	500 MIL		
CPU_27P4S	BOTTOM			0.23 MM	100 MIL		
USB3_85D	TOP			0.1 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
1T01_DIFFPAIR	*	1:1_DIFFPAIR

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_GPUVR	*	Y	0.300 MM	0.200 MM	3.000 MM	0.400 MM	0.200 MM

AMD Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
1E6A	SENSE DIFFPAIR	THERM 1701 45S	THERM	GPUFB CS P 47 73
1E6B		THERM 1701 45S	THERM	GPUFB CS N 47 73
1E6C	SENSE DIFFPAIR	THERM 1701 45S	THERM	GPU TDIODE P 48 76
1E6D		THERM 1701 45S	THERM	GPU TDIODE N 48 76
1E6E	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPU VDDCISENSE P
1E6F	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPU VDDCISENSE N
1E6G	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPU VDDCI_SENSE_XW_P 89
1E6H	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPU VDDCI_SENSE_XW_N 89
1E6I	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPUVCORE SENSE P 79
1E6J	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPUVCORE SENSE N 79
1E6K	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_PP8V95_S0GPU_R_P 47
1E6L	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_PP8V95_S0GPU_R_N 47
1E6M	SENSE DIFFPAIR	THERM 1701 45S	SENSE	VSNS_GPU_0V95_XW_P 73
1E6N		THERM 1701 45S	SENSE	VSNS_GPU_0V95_XW_N 73
1E6O	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	VDDCIS0_CS_R_P 47
1E6P		SENSE 1701 45S	SENSE	VDDCIS0_CS_R_N 47
1E6Q	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	GPUFB CS_R_P 47
1E6R		SENSE 1701 45S	SENSE	GPUFB CS_R_N 47
1E6S	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	VSNS_GPU_VDDC_P 71 79
1E6T	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	VSNS_GPU_VDDC_N 71 79
1E6U	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	VSNS_GPU_VDDI_P 71 88
1E6V	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	VSNS_GPU_VDDI_N 71 88
1E6W	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_1V8_GPU_R_P
1E6X	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_1V8_GPU_R_N
1E6Y	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_1V8_GPU_P
1E6Z	SENSE DIFFPAIR	SENSE 1701 45S	SENSE	ISNS_1V8_GPU_N
1E7A	SENSE DIFFPAIR	THERM 1701 45S	SENSE	VSNS_GPU_FB_XW_P 73
1E7B		THERM 1701 45S	SENSE	VSNS_GPU_FB_XW_N 73

X425G Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
	PHYSICAL		SPACING		
ISNS_CPUDDR_P ISNS_CPUDDR_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_CPUDDR_P	46
ISNS_CPUDDR_N ISNS_CPU_DDR_R_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_CPUDDR_N	46
ISNS_CPU_DDR_R_P ISNS_CPU_DDR_R_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_CPU_DDR_R_P	46
ISNS_CPU_DDR_R_N CPUTHMSNS_D2_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_CPU_DDR_R_N	46
CPUTHMSNS_D2_P CPUTHMSNS_D2_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	CPUTHMSNS_D2_P	48
CPUTHMSNS_D2_N ISNS_LCD_PANEL_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	CPUTHMSNS_D2_N	48
ISNS_LCD_PANEL_P ISNS_LCD_PANEL_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCD_PANEL_P	46
ISNS_LCD_PANEL_N DDR3THMSNS_D1_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCD_PANEL_N	46
DDR3THMSNS_D1_P DDR3THMSNS_D1_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	DDR3THMSNS_D1_P	48
DDR3THMSNS_D1_N FNTHMSNS_D_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	DDR3THMSNS_D1_N	48
FNTHMSNS_D_P FNTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	FNTHMSNS_D_P	48
ISNS_1V35_MEM_P ISNS_1V35_MEM_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_1V35_MEM_P	45
ISNS_1V35_MEM_N ISNS_1V35_MEM_R_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_1V35_MEM_N	45
ISNS_1V35_MEM_R_P ISNS_1V35_MEM_R_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_1V35_MEM_R_P	45
ISNS_AIRPORT_P ISNS_AIRPORT_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_AIRPORT_P	45
ISNS_AIRPORT_N ISNS_AIRPORT_R_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_AIRPORT_N	45
ISNS_AIRPORT_R_P ISNS_AIRPORT_R_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_AIRPORT_R_P	46
ISNS_AIRPORT_R_N ISNS_LCDBKLT_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_AIRPORT_R_N	46
ISNS_LCDBKLT_N ISNS_LCDBKLT_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCDBKLT_N	63
ISNS_LCDBKLT_P ISNS_LCD_PANEL_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCDBKLT_P	63
ISNS_LCD_PANEL_N ISNS_LCD_PANEL_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCD_PANEL_N	46
ISNS_PCH_R_P ISNS_PCH_R_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_LCD_PANEL_P	46
ISNS_PCH_R_N ISNS_TPAD_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PCH_R_P	45
ISNS_TPAD_P ISNS_TPAD_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PCH_R_N	45
ISNS_HS_OTHER5V_P ISNS_HS_OTHER5V_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_TPAD_P	45
ISNS_HS_OTHER5V_N ISNS_HS_OTHER3V3_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_OTHER5V_P	44
ISNS_HS_OTHER3V3_P ISNS_HS_OTHER3V3_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_OTHER5V_N	44
ISNS_HS_OTHER3V3_N ISNS_HS_COMPUTING_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_OTHER3V3_P	44
ISNS_HS_COMPUTING_P ISNS_HS_COMPUTING_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_OTHER3V3_N	44
ISNS_HS_COMPUTING_N CPUVR_ISNS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_COMPUTING_P	44
CPUVR_ISNS_P CPUVR_ISNS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	CPUVR_ISNS_P	45
CPUVR_ISNS_N P1V05_GPU_PEX_IOVDD_SNS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	CPUVR_ISNS_N	45
P1V05_GPU_PEX_IOVDD_SNS_P P1V05_GPU_PEX_IOVDD_SNS_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05_GPU_PEX_IOVDD_SNS_P	45
P1V05_GPU_PEX_IOVDD_SNS_N DIFFERENTIAL_PAIR	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05_GPU_PEX_IOVDD_SNS_N	45
DIFFERENTIAL_PAIR CPUVR_ISNS1_P	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	DIFFERENTIAL_PAIR	55
CPUVR_ISNS1_P CPUVR_ISNS1_N	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	CPUVR_ISNS1_P	45
CPUVR_ISNS1_N CPUVR_ISNS2_P	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	CPUVR_ISNS1_N	45
CPUVR_ISNS2_P CPUVR_ISNS2_N	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	CPUVR_ISNS2_P	45
CPUVR_ISNS2_N CPUVR_ISNS3_P	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	CPUVR_ISNS2_N	45
CPUVR_ISNS3_P CPUVR_ISNS3_N	SENSE_DIFFPAIR	THERM_45S_CPUVIRISNS1	THERM	CPUVR_ISNS3_P	45
ISNS_TBT_N ISNS_TBT_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	CPUVR_ISNS3_N	45
ISNS_TBT_P ISNS_TBT_R_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_TBT_N	45
ISNS_TBT_R_N ISNS_TBT_R_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_TBT_P	45
ISNS_TBT_R_P ISNS_SSD_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_TBT_R_P	45
ISNS_SSD_P ISNS_SSD_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_SSD_P	45
ISNS_SSD_N ISNS_SSD_R_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_SSD_N	45
ISNS_SSD_R_P ISNS_SSD_R_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_SSD_R_P	45
ISNS_SSD_R_N P1V05S0_CS_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	ISNS_SSD_R_N	45
P1V05S0_CS_P P1V05S0_CS_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05S0_CS_P	45
P1V05S0_CS_N DIFFERENTIAL_PAIR	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05S0_CS_N	45
DIFFERENTIAL_PAIR P1V05S0_SENSE_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	DIFFERENTIAL_PAIR	62
P1V05S0_SENSE_P P1V05S0_SENSE_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05S0_SENSE_P	62
P1V05S0_SENSE_N TBT_THERMDP	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	P1V05S0_SENSE_N	28
TBT_THERMDP TBT_THERMDN	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	TBT_THERMDP	48
CHGR_CSI_R_P CHGR_CSI_R_N	AUDIODIFF	AUDIODIFF	AUDIO	TBT_THERMDN	48
CHGR_CSI_R_N CHGR_CS0_R_P	AUDIODIFF	AUDIODIFF	AUDIO	CHGR_CSI_R_P	57
CHGR_CS0_R_P CHGR_CS0_R_N	AUDIODIFF	AUDIODIFF	AUDIO	CHGR_CSI_R_N	57
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	CHGR_CS0_R_P	57
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	47
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_P	47
ISNS_PP0V95_S0GPU_P ISNS_PP0V95_S0GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_HS_GPU_N	47
ISNS_PP0V95_S0GPU_N VDDCI90_CS_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_P	47
VDDCI90_CS_P VDDCI90_CS_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	ISNS_PP0V95_S0GPU_N	47
VDDCI90_CS_N GPUTHMSNS_D_P	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	VDDCI90_CS_P	48
GPUTHMSNS_D_P GPUTHMSNS_D_N	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	VDDCI90_CS_N	48
GPUTHMSNS_D_N ISNS_HS_GPU_P	SENSE_DIFFPAIR	THERM_1T01_45S	THERM	GPUTHMSNS_D_P	48
ISNS_HS_GPU_P ISNS_HS_GPU_N	SENSE_DIFFPAIR	SENSE_1T01_45S	SENSE	GPUTHMSNS_D_N	48
ISNS_HS_GPU_N ISNS_PP0V95_S0GPU_P	SENSE_DIFFPAIR	SENSE_1T01_45S			

X425G Specific Net Properties

[illegible]

SYNC MASTER=SIDLE J45		SYNC DATE=12/10/2012	
PAGE TITLE		PAGE	
Project Specific Constraints			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
		REVISION <E4LABEL>	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH <BRANCH>	
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GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=3x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=7x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR5_CMD	TOP, BOTTOM	=4x_DIELECTRIC	?
GDDR5_DATA	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR5_EDC	TOP, BOTTOM	=7x_DIELECTRIC	?

GDDR5_CMD spacing can be relaxed to 2x per AMD recommendation for x32_4.5G config.

Breakout Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR5_*	*	BGA	GDDR5_BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_BGA	*	=1.3x_DIELECTRIC	?

GDDR5 FB A Net Properties

ELECTRICAL_CONSTRAINT_SET		PHYSICAL	SPACING	
FE00	FB A0 CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK P
FE00	FB A0 CLK	GDDR5_80D	GDDR5_CLK	FB A0 CLK N
FE00	FB A1 CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK P
FE00	FB A1 CLK	GDDR5_80D	GDDR5_CLK	FB A1 CLK N
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 A<8...0>
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 A<8...0>
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 ABI L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 ABI L
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 RAS L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 RAS L
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 CAS L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 CAS L
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 WE L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 WE L
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 CKE L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 CKE L
FE00	FB A0 CMD	GDDR5_45SE	GDDR5_CMD	FB A0 CS L
FE00	FB A1 CMD	GDDR5_45SE	GDDR5_CMD	FB A1 CS L
FE00	FB A0 EDC0	GDDR5_45SE	GDDR5_FDC	FB A0 EDC<0>
FE00	FB A0 EDC1	GDDR5_45SE	GDDR5_FDC	FB A0 EDC<1>
FE00	FB A0 EDC2	GDDR5_45SE	GDDR5_FDC	FB A0 EDC<2>
FE00	FB A0 EDC3	GDDR5_45SE	GDDR5_FDC	FB A0 EDC<3>
FE00	FB A1 EDC0	GDDR5_45SE	GDDR5_FDC	FB A1 EDC<0>
FE00	FB A1 EDC1	GDDR5_45SE	GDDR5_FDC	FB A1 EDC<1>
FE00	FB A1 EDC2	GDDR5_45SE	GDDR5_FDC	FB A1 EDC<2>
FE00	FB A1 EDC3	GDDR5_45SE	GDDR5_FDC	FB A1 EDC<3>
FE00	FB A0 DBI I 0	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<0>
FE00	FB A0 DBI I 1	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<1>
FE00	FB A0 DBI I 2	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<2>
FE00	FB A0 DBI I 3	GDDR5_45SE	GDDR5_DATA	FB A0 DBI L<3>
FE00	FB A1 DBI I 0	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<0>
FE00	FB A1 DBI I 1	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<1>
FE00	FB A1 DBI I 2	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<2>
FE00	FB A1 DBI I 3	GDDR5_45SE	GDDR5_DATA	FB A1 DBI L<3>
FE00	FB A0 WCLK0	GDDR5_80D	GDDR5_CLK	FB A0 WCLK P<0>
FE00	FB A0 WCLK0	GDDR5_80D	GDDR5_CLK	FB A0 WCLK N<0>
FE00	FB A0 WCLK1	GDDR5_80D	GDDR5_CLK	FB A0 WCLK P<1>
FE00	FB A0 WCLK1	GDDR5_80D	GDDR5_CLK	FB A0 WCLK N<1>
FE00	FB A1 WCLK0	GDDR5_80D	GDDR5_CLK	FB A1 WCLK P<0>
FE00	FB A1 WCLK0	GDDR5_80D	GDDR5_CLK	FB A1 WCLK N<0>
FE00	FB A1 WCLK1	GDDR5_80D	GDDR5_CLK	FB A1 WCLK P<1>
FE00	FB A1 WCLK1	GDDR5_80D	GDDR5_CLK	FB A1 WCLK N<1>
FE00	FB A0 DQ BYTE0	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<7...0>
FE00	FB A0 DQ BYTE1	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<15...8>
FE00	FB A0 DQ BYTE2	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<23...16>
FE00	FB A0 DQ BYTE3	GDDR5_45SE	GDDR5_DATA	FB A0 DQ<31...24>
FE00	FB A1 DQ BYTE0	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<7...0>
FE00	FB A1 DQ BYTE1	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<15...8>
FE00	FB A1 DQ BYTE2	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<23...16>
FE00	FB A1 DQ BYTE3	GDDR5_45SE	GDDR5_DATA	FB A1 DQ<31...24>
FE00	FB A0 CMD_R	GDDR5_45SE	GDDR5_CMD	FB A0 RESET L
FE00	FB A1 CMD_R	GDDR5_45SE	GDDR5_CMD	FB A1 RESET L

GDDR5 FB B Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
NEW	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK P
NEW	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB B0 CLK N
NEW	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK P
NEW	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB B1 CLK N
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 A<8...0>
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 A<8...0>
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 ABI L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 ABI L
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 RAS L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 RAS L
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 CAS L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 CAS L
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 WE L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 WE L
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 CKE L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 CKE L
NEW	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB B0 CS L
NEW	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB B1 CS L
NEW	FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<0>
NEW	FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<1>
NEW	FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<2>
NEW	FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB B0 EDC<3>
NEW	FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<0>
NEW	FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<1>
NEW	FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<2>
NEW	FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB B1 EDC<3>
NEW	FB_B0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<0>
NEW	FB_B0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<1>
NEW	FB_B0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<2>
NEW	FB_B0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB B0 DBI L<3>
NEW	FB_B1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<0>
NEW	FB_B1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<1>
NEW	FB_B1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<2>
NEW	FB_B1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB B1 DBI L<3>
NEW	FB_B0_WCLK0	GDDR5_80D	GDDR5_CLK	FB B0 WCLK P<0>
NEW	FB_B0_WCLK0	GDDR5_80D	GDDR5_CLK	FB B0 WCLK N<0>
NEW	FB_B0_WCLK1	GDDR5_80D	GDDR5_CLK	FB B0 WCLK P<1>
NEW	FB_B0_WCLK1	GDDR5_80D	GDDR5_CLK	FB B0 WCLK N<1>
NEW	FB_B1_WCLK0	GDDR5_80D	GDDR5_CLK	FB B1 WCLK P<0>
NEW	FB_B1_WCLK0	GDDR5_80D	GDDR5_CLK	FB B1 WCLK N<0>
NEW	FB_B1_WCLK1	GDDR5_80D	GDDR5_CLK	FB B1 WCLK P<1>
NEW	FB_B1_WCLK1	GDDR5_80D	GDDR5_CLK	FB B1 WCLK N<1>
NEW	FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<7...0>
NEW	FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<15...8>
NEW	FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<23...16>
NEW	FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B0 DQ<31...24>
NEW	FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<7...0>
NEW	FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<15...8>
NEW	FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<23...16>
NEW	FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB B1 DQ<31...24>
NEW	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B0 RESET L
NEW	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB B1 RESET L

MUXGFX & DP AUX MUX NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET		PHYSICAL	NET TYPE	SPEC TAG	
MEM0	DP_85D	DISPLAYPORT	DP INT ML C P<3..0>	69	82
MEM1	DP_85D	DISPLAYPORT	DP INT ML C N<3..0>	69	82
MEM2	DP_85D	DISPLAYPORT	DP INT ML F P<3..0>	69	
MEM3	DP_85D	DISPLAYPORT	DP INT ML F N<3..0>	69	
MEM4	DP_INT_MI	DP_85D	DP INT ML P<3..0>	69	85
MEM5	DP_INT_MI	DP_85D	DP INT ML N<3..0>	69	85
MEM6	DP_85D	DISPLAYPORT	DP INT AUXCH C P	69	82
MEM7	DP_85D	DISPLAYPORT	DP INT AUXCH C N	69	82
MEM8	DP_INT_AUXCH	DP_85D	DP INT AUX P		
MEM9	DP_INT_AUXCH	DP_85D	DP INT AUX N	69	85
MEM0	DP_85D	DISPLAYPORT	DP INT EG AUX P	76	77
MEM1	DP_85D	DISPLAYPORT	DP INT EG AUX N	76	77
MEM2	DP_85D	DISPLAYPORT	DP INT EG ML P<3..0>	76	82
MEM3	DP_85D	DISPLAYPORT	DP INT EG ML N<3..0>	76	82
MEM4	DP_85D	DISPLAYPORT	DP INT IG AUX P	5	82
MEM5	DP_85D	DISPLAYPORT	DP INT IG AUX N	5	82
MEM6	DP_INT_MI	DP_85D	DP INT IG ML P<3..0>	5	82
MEM7	DP_85D	DISPLAYPORT	DP INT IG ML N<3..0>	5	82
MEM0	DP_EG_AUX	DP_85D	DP TBTSNK0 EG AUXCH P	76	77
MEM1	DP_EG_AUX	DP_85D	DP TBTSNK0 EG AUXCH N	76	77
MEM2	DP_EG_AUX	DP_85D	DP TBTSNK0 EG AUXCH P	76	77
MEM3	DP_EG_AUX	DP_85D	DP TBTSNK1 EG AUXCH N	76	77
MEM0	TBTSNK_AUXCH	DP_85D	DP TBTSNK0 AUXCH P	28	89
MEM1	TBTSNK_AUXCH	DP_85D	DP TBTSNK0 AUXCH N	28	89
MEM2	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH C P	28	83
MEM3	DP_85D	DISPLAYPORT	DP TBTSNK0 AUXCH C N	28	83
MEM4	TBTSNK_AUXCH	DP_85D	DP TBTSNK1 AUXCH P	28	89
MEM5	TBTSNK_AUXCH	DP_85D	DP TBTSNK1 AUXCH N	28	89
MEM6	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH C P	28	83
MEM7	DP_85D	DISPLAYPORT	DP TBTSNK1 AUXCH C N	28	83
MEM8	DP_TBT_MI	DP_85D	DP TBTSNK0 ML P<3..0>	28	89
MEM9	DP_TBT_MI	DP_85D	DP TBTSNK0 ML N<3..0>		
MEM0	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C P<3..0>	28	89
MEM1	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C N<3..0>	28	89
MEM2	DP_TBT_MI	DP_85D	DP TBTSNK1 ML P<3..0>	28	89
MEM3	DP_TBT_MI	DP_85D	DP TBTSNK1 ML N<3..0>		
MEM4	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C P<3..0>	28	89
MEM5	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C N<3..0>	28	89

Kepler Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE	SPACING
H63D GPU_CLK_TEST RC P	1-1 DIEPAIR		72
H63D GPU_CLK_TEST RC N	1-1 DIEPAIR		72
H63D GPU_CLK_TEST P	1-1 DIEPAIR		72
H63D GPU_CLK_TEST N	1-1 DIEPAIR		72
HDMI_DATA DP_85D DISPLAYPORT HDMI EG DATA P<2..0>	DP_85D	DISPLAYPORT	76 E1 R
HDMI_DATA DP_85D DISPLAYPORT HDMI EG DATA N<2..0>	DP_85D	DISPLAYPORT	76 E1 R
HDMI_CLK DP_85D HDMI_CLK HDMI EG CLK P	DP_85D	HDMI_CLK	76 E1 R
HDMI_CLK DP_85D HDMI_CLK HDMI EG CLK N	DP_85D	HDMI_CLK	76 E1 R

SYNC MASTER=J45G_AMD		SYNC DATE=97/01/20	
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